



Phinma University of Iloilo

College of Information Technology Education

ProfitLens

A Web-Based Platform for Smart Profit Monitoring and Business Insights
for Auto Parts Sales

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The rapid advancement of information technology has fundamentally transformed how businesses operate, manage their finances, and make strategic decisions. In the contemporary business landscape, digital tools and automated systems have become indispensable for organizations seeking to maintain competitive advantage, improve operational efficiency, and achieve sustainable growth. This digital transformation is particularly relevant for small and medium enterprises (SMEs), which constitute the backbone of economies worldwide and face unique challenges in managing their financial operations effectively.

Small and medium enterprises represent approximately 99% of all businesses globally and contribute significantly to employment generation, economic output, and innovation across diverse industries. In the Philippines specifically, SMEs account for over 99% of all registered businesses and employ roughly 63% of the national workforce, making them critical drivers of economic development and social stability. Despite their substantial contribution to the national economy, many Philippine SMEs continue to struggle with fundamental business management practices, particularly in the areas of financial tracking, expense monitoring, and profit analysis.

The automotive parts retail industry presents a compelling case study for examining these challenges. As vehicle ownership continues to increase across the Philippines, driven by urbanization, economic growth, and expanding transportation networks, the demand for auto parts and related services has correspondingly grown. The automotive aftermarket industry in the Philippines encompasses a vast network of retailers, wholesalers, and service providers who collectively ensure that vehicle owners have access to the replacement parts, maintenance supplies, and technical expertise needed to keep their vehicles operational. Within this ecosystem, small auto parts retailers play a particularly important role, serving as accessible points of purchase for individual consumers and small repair shops who require immediate availability of specific parts and consumables.

However, the operational realities of running a small auto parts retail business present numerous financial management challenges. These challenges are compounded by several factors characteristic of the industry: the diversity of product inventory (ranging from high-value components to low-cost consumables), the variability in demand patterns (influenced by vehicle models, seasonal factors, and economic conditions), the complexity of pricing structures (involving cost-plus pricing, bulk discounts, and competitive adjustments), and the need for accurate stock management to prevent both stockouts and overstocking conditions.

The reference business for this study, a local auto parts store owned by French Joy Salvador in Iloilo City, exemplifies these challenges in a concrete operational context. During the preliminary observation and requirements gathering phases of this research, the research team identified numerous instances where the absence of automated financial tracking systems led to operational inefficiencies, financial inaccuracies, and missed business opportunities. The store's reliance on manual record-keeping methods, while understandable given the scale of operations and resource constraints, created systematic vulnerabilities that undermined the owner's ability to manage the business effectively.

Manual financial record-keeping in small businesses typically involves handwritten sales logs, paper-based expense receipts, basic spreadsheet calculations, and end-of-month reconciliation processes that are both time-consuming and error-prone. These methods, while simple to implement and requiring no technical infrastructure, suffer from several inherent limitations that become increasingly problematic as business complexity grows. Human calculation errors, incomplete data capture, inconsistent categorization of transactions, difficulty in retrieving historical information, and the absence of real-time financial visibility collectively create an environment where business owners operate with incomplete and potentially inaccurate financial information.

The consequences of inadequate financial tracking extend beyond mere inconvenience. When business owners cannot accurately determine their actual profit margins, identify which products are most profitable, recognize spending patterns, or detect when expenses are exceeding budgetary allocations, they are effectively making business decisions based on incomplete or erroneous information. This informational deficit can lead to suboptimal pricing strategies, inefficient inventory management, missed cost-reduction opportunities, and ultimately, compromised business viability.

Recognizing these challenges, the researchers conceived ProfitLens as a purpose-built solution designed to address the specific financial management needs of small auto parts retailers. The platform was conceived as a web-based application that would leverage readily available technologies to deliver automated financial tracking, real-time profit calculation, comprehensive expense monitoring, and intuitive visual reporting capabilities. By consolidating these functions within a single, accessible platform, ProfitLens aims to eliminate the informational barriers that prevent small business owners from making informed, data-driven decisions.

The development of ProfitLens was guided by several core design principles that reflect both the theoretical foundations of information systems design and the practical

requirements of the target user population. First, the system was designed to minimize the technical expertise required for effective use, ensuring that business owners and staff with limited computer literacy could operate the platform productively. Second, the system prioritized data accuracy and integrity through automated calculations and validated data entry processes. Third, the platform emphasized real-time information delivery, providing users with current financial data rather than requiring end-of-period reconciliation. Fourth, the system incorporated visual reporting mechanisms that transformed raw financial data into intuitive charts and graphs accessible to non-technical users.

The choice of web-based architecture was deliberate and reflects several strategic considerations. Web-based applications can be accessed from any device with a modern web browser, eliminating the need for software installation on individual computers and enabling access from multiple locations. This architectural decision also facilitates centralized data storage, ensuring that all financial records are maintained in a single, secure database that can be backed up and protected against data loss. Furthermore, the PHP-MySQL technology stack employed in ProfitLens represents a mature, well-documented, and widely-supported technology combination that offers robust functionality, strong security features, and excellent compatibility with the XAMPP development environment commonly used in educational and small business contexts.

The significance of this study extends beyond the immediate operational benefits to the reference business. By documenting the complete process of identifying business requirements, designing a technical solution, implementing the system using established development methodologies, and evaluating the outcomes, this research contributes to the broader body of knowledge regarding technology adoption and digital transformation in small businesses within the Philippine context. The findings and methodology presented in this documentation can serve as a reference model for similar initiatives targeting other small business sectors facing analogous financial management challenges.

1.2 Statement of the Problem

The central operational challenge addressed by ProfitLens can be comprehensively understood through examination of the specific deficiencies in the current financial management practices employed by the reference business. The auto parts store operated by French Joy Salvador in Iloilo City lacked any automated system for tracking, aggregating, visualizing, or reporting financial data. All financial management was conducted through manual methods that were inadequate in precision, transparency, and actionability, creating a cascade of operational inefficiencies that collectively undermined the owner's ability to manage the business effectively.

Direct Observation: Shadowing daily store operations, point-of-sale interactions, and end-of-day reconciliation routines.

Structured Interviews: Engaging in deep-dive discussions with the owner and operational staff to uncover pain points and process gaps.

Documentary Analysis: Auditing historical financial records, paper ledgers, supplier invoices, and physical receipts.

Benchmarking: Evaluating current store workflows against established industry best practices in small-and-medium enterprise (SME) financial management.

The research team identified ten distinct but interconnected problems through a combination of direct observation of store operations, structured interviews with the owner and staff, analysis of existing financial documents and records, and comparison with established best practices in small business financial management. These problems define the comprehensive problem space that ProfitLens was engineered to resolve.

Problem 1: Profit Monitoring Difficulty

The fundamental challenge of accurately determining business profitability represented the most critical financial management deficiency identified during the requirements gathering process. Calculating monthly net profit required the manual aggregation of all individual sales transactions across the entire month, followed by a manual subtraction of all recorded expenses. This process was performed at the end of each month, typically by the owner herself, and regularly required several hours of concentrated effort to complete.

The manual nature of this calculation introduced multiple points of potential error that collectively undermined the reliability of the resulting profit figure. Individual sales entries could be missed during the aggregation process, particularly during busy periods when transaction volumes were high. Transcription errors could propagate through multi-step calculations, where a single incorrect figure could cascade into incorrect totals. The absence of any validation mechanism meant that erroneous totals could be accepted without detection, as there was no independent means of verifying the accuracy of manual calculations.

The implications of this problem extended beyond mere calculation accuracy. The time-consuming nature of manual profit calculation meant that the owner received profitability information only at the end of each month, precluding any possibility of real-time or intra-month financial monitoring. The lack of granular profit analysis, broken down by product categories, time periods, or other relevant dimensions, prevented the owner from understanding the relative contribution of different business elements to overall financial performance. Without this detailed insight, decisions regarding product selection, pricing strategies, and resource allocation were necessarily based on incomplete information.

The owner acknowledged during interviews that the monthly profit figure produced through manual calculation could not be relied upon with full confidence. This admission highlights the fundamental inadequacy of manual profit monitoring for supporting informed business decision-making. When business owners cannot trust their own

profitability calculations, every subsequent business decision is potentially compromised by inaccurate foundational data.

ProfitLens addresses this problem through the implementation of automated net profit calculation that occurs in real time whenever sales and expense records are entered into the system. The computation is performed using a clearly defined mathematical formula applied consistently by the PHP backend, ensuring mathematical accuracy and eliminating the manual aggregation step entirely. The formula $\text{Net Profit} = \text{Total Sales Revenue} - \text{Total Expenses}$ is applied automatically, with each transaction update immediately reflected in the current profit calculation.

Problem 2: Absence of Structured Financial Reports

The absence of any formal mechanism for producing structured financial reports represented a significant limitation in the reference business's financial management capabilities. No standardized process existed for generating monthly, quarterly, or annual financial summaries that could provide the owner with comprehensive visibility into business performance over defined time periods.

Historical financial data was stored in physical documents and informal spreadsheets that were not organized in a way that would support retrospective reporting. Sales records were maintained in handwritten logs, expense receipts were stored in folders or envelopes, and any calculations were performed on an ad hoc basis using basic spreadsheet applications or manual arithmetic. This fragmented approach to data storage made it virtually impossible to generate comprehensive financial reports without extensive manual effort to reconstruct and consolidate historical information.

When the owner needed to review past performance for business planning purposes, the process required manual reconstruction of relevant figures from archived receipts, handwritten notes, and informal records. For example, if the owner wanted to assess whether a particular month's revenue was sufficient to sustain a planned purchase of additional inventory, she was required to manually gather and calculate all relevant financial data from physical sources. This reconstruction process was impractical for any period longer than a few months and became effectively impossible for periods where physical records had been lost, damaged, or misplaced.

The absence of structured financial reporting had several negative consequences for business operations. First, it prevented the owner from identifying trends in business performance over time, making it impossible to determine whether the business was improving, declining, or remaining stable. Second, it eliminated the ability to compare performance across different time periods, such as comparing current month performance to the same month in previous years. Third, it precluded the development of financial projections or budgets based on historical performance data. Fourth, it made it difficult to provide financial information to external parties, such as banks or suppliers, who might require evidence of business performance for credit or partnership decisions.

ProfitLens resolves this problem through the maintenance of all financial data in a structured MySQL database and the provision of a dedicated Financial Reports module. This module generates comprehensive profit-and-loss summaries, monthly expense

breakdowns, and a full January-to-December grouped bar chart comparing revenue against expenses. All reports are available on demand within the system, with no manual reconstruction required, and can be generated for any historical period for which data has been entered into the system.

Problem 3: No Trend Analysis Capability

The absence of trend analysis capabilities prevented the reference business from developing the kind of forward-looking, evidence-based business strategies that characterize successful enterprises. Without a centralized data store and automated reporting tools, it was impossible for the store owner to visualize performance trends, identify seasonal demand patterns, compare monthly or annual results, or make data-driven predictions about future revenue.

The importance of trend analysis for business success cannot be overstated. Understanding historical performance patterns enables business owners to anticipate future demand, optimize inventory levels, allocate resources effectively, and develop realistic growth targets. Without these insights, business decisions are necessarily reactive rather than proactive, addressing problems as they arise rather than preventing them through advance planning.

The store had no way of determining, for instance, whether June was consistently its strongest month, whether demand for oil-based consumables peaked before the rainy season, or whether year-over-year revenue was growing, declining, or stable. These are the kinds of insights that enable proactive business decision-making: adjusting inventory orders in advance of high-demand periods, negotiating better purchase terms with suppliers during low-revenue months, or setting realistic revenue targets based on historical performance patterns.

The absence of trend analysis capability was particularly problematic for the reference business given the seasonal nature of automotive demand in the iloiloCity region. Vehicle maintenance patterns, consumer purchasing behavior, and economic conditions all fluctuate throughout the year, creating predictable variations in demand for auto parts and services. Without the ability to identify and analyze these patterns, the business was unable to optimize its operations to capitalize on peak demand periods or mitigate the impact of slower periods.

ProfitLens addresses this limitation through its Revenue Analysis module, which presents monthly revenue trends in both bar and line chart formats, supports year-over-year comparison across any two calendar years, and identifies the top five products by revenue and units sold. All visualizations are computed dynamically from the database and rendered visually in the browser without requiring any external chart library, ensuring that the system remains self-contained and does not depend on third-party service availability.

Problem 4: Undetected Overspending

Operating expenses at the auto parts store encompassed a diverse range of cost categories, including utilities, office supplies, advertising, salaries and wages, rent, and

various miscellaneous operational costs. The complexity and variety of these expenses made it difficult to maintain accurate awareness of total expenditure levels without systematic tracking and monitoring.

Without automated monitoring, the cumulative total of these expenses frequently exceeded the budget available for a given month without the owner becoming aware of the overrun until she conducted her end-of-month manual reconciliation. This temporal lag between the occurrence of overspending and its detection meant that corrective action could not be taken in time to prevent the financial impact from materializing.

The consequences of undetected overspending can be severe for small businesses operating with limited financial reserves. When expenses exceed revenue for extended periods without detection, the resulting cash flow deficits can accumulate to threatening levels before the problem is identified. By the time manual reconciliation reveals the extent of overspending, opportunities for corrective action may have passed, and the business may face cash flow crises that threaten its operational viability.

Furthermore, the absence of categorized expense tracking made it impossible to identify which specific cost categories were consuming disproportionate shares of revenue. Without this granular visibility, the owner had no basis for targeted cost reduction efforts and could only resort to generalized expense cutting that might negatively impact essential business operations.

ProfitLens addresses this through its Expense Tracking module, which maintains a real-time running total of all recorded expenses for the current month, categorizes expenses for granular visibility, and provides visual indicators that highlight high-spend categories relative to overall revenue. This gives the owner and staff an always-current view of the expense position without requiring any manual computation, enabling timely identification and correction of overspending conditions.

Problem 5: Inaccurate Inventory Records

Operating a retail business requires a precise understanding of physical stock levels to support purchasing decisions, sales activities, and financial planning. Without real-time tracking mechanisms, a significant discrepancy frequently develops between the amount of stock a business believes it has according to its records and the actual inventory sitting on the warehouse shelves. This discrepancy, commonly referred to as inventory inaccuracy, represents a fundamental operational challenge that can have far-reaching consequences for business performance.

The causes of inventory inaccuracy are varied and include data entry errors, theft or shrinkage, damage or spoilage, misplaced items, and delays in updating records to reflect actual stock movements. Without systematic tracking and reconciliation procedures, these sources of inaccuracy accumulate over time, creating increasingly large gaps between recorded and actual inventory levels.

The consequences of inaccurate inventory records are significant and multifaceted. When managers make purchasing decisions based on inflated inventory records, they may fail to reorder products that are actually running low, leading to stockouts and lost

sales opportunities. Conversely, when records understate actual inventory levels, unnecessary reorders may be placed, resulting in excess inventory that ties up working capital and incurs holding costs. In both cases, the lack of accurate inventory information undermines operational efficiency and financial performance.

ProfitLens addresses this through its stock management functionality, which maintains real-time stock quantities for all products and automatically updates these quantities whenever sales transactions are recorded. This ensures that the system's inventory records accurately reflect the quantities that have been sold, providing a reliable basis for purchasing and restocking decisions.

Problem 6: Stockouts and Overstocking

Balancing optimal stock levels represents a constant operational hurdle for retail businesses, and the auto parts store was no exception. When demand spikes unexpectedly, a lack of inventory results in immediate stockouts, leading to missed sales opportunities and frustrated customers who may turn to competitors for their immediate needs. Conversely, the fear of running out of popular items often prompts excessive ordering, resulting in overstocking that ties up vital cash flow and inflates warehouse holding costs for items that may take months to sell.

The challenge of inventory optimization is particularly acute in the auto parts industry due to the diversity of products carried, the unpredictability of demand for specific items, and the long tail of slow-moving products that must be maintained for customer convenience. Without systematic analysis of historical demand patterns and current inventory positions, stocking decisions are necessarily based on intuition and experience rather than data-driven optimization.

Stockouts have both immediate and long-term consequences for retail businesses. In the immediate term, stockouts result in lost sales and disappointed customers. In the longer term, repeated stockouts can damage customer relationships, reduce customer loyalty, and drive customers to establish purchasing relationships with competitors. Overstocking, while avoiding the immediate negative consequences of stockouts, creates its own set of problems including increased storage costs, risk of obsolescence, reduced working capital availability, and potential write-offs for items that become unsalable.

ProfitLens addresses this challenge through its inventory management functionality, which tracks current stock levels and provides visibility into inventory positions for all products. By maintaining accurate, real-time inventory data, the system enables business owners to make informed purchasing decisions that balance the competing objectives of avoiding stockouts and minimizing excess inventory.

Problem 7: Demand Forecasting Errors

Predicting future customer purchasing patterns is inherently challenging, particularly for businesses with diverse product portfolios and variable demand patterns. Without systematic data collection and analysis capabilities, demand forecasting at the

reference business relied on the owner's personal experience and intuition, supplemented by basic historical observations that lacked statistical rigor.

Miscalculating future market demand creates a cascade of operational problems that affect multiple aspects of business performance. When demand is overestimated, the business accumulates excess inventory that ties up working capital, incurs storage costs, and risks becoming obsolete or unsalable. When demand is underestimated, the business experiences stockouts that result in lost sales, customer dissatisfaction, and potential damage to business relationships.

The limitations of intuition-based demand forecasting are particularly pronounced in the auto parts industry, where demand patterns are influenced by multiple factors including vehicle registration trends, seasonal weather conditions, economic activity levels, and the specific maintenance requirements of different vehicle models. Without systematic data collection and analysis, these influencing factors cannot be quantified or incorporated into forecasting models, resulting in predictions that are inherently less accurate than data-driven alternatives.

While ProfitLens in its current implementation does not include sophisticated predictive analytics capabilities, the systematic collection and storage of sales data creates a foundation upon which future demand forecasting functionality could be built. The historical sales data maintained in the system's database provides the raw material for statistical analysis of demand patterns, seasonal trends, and product-specific performance that could support more accurate demand predictions.

Problem 8: Supply Chain Disruptions

A retail business's internal operations are heavily dependent on its external supply chain relationships with vendors, distributors, and manufacturers. Unexpected delays, factory closures, quality issues, or shipping bottlenecks can create severe disruptions that stall the arrival of essential inventory and directly impact product availability and customer satisfaction.

The auto parts supply chain is particularly complex, involving multiple tiers of suppliers, distributors, and retailers, with products originating from diverse manufacturing locations both domestically and internationally. This complexity creates numerous potential points of failure that can disrupt the flow of inventory to retail locations.

Without systematic supply chain monitoring and management capabilities, businesses are often caught unprepared by supply disruptions, lacking the early warning systems and contingency plans needed to mitigate the impact on operations. When a critical product becomes unavailable due to supply chain issues, the business may be forced to turn away customers, lose sales opportunities, and potentially damage long-term customer relationships.

While comprehensive supply chain management is beyond the scope of ProfitLens's current implementation, the system's inventory tracking capabilities provide visibility into stock levels that can help business owners anticipate potential supply needs and plan purchasing activities accordingly. The data maintained by the system regarding sales

patterns and inventory turnover rates can inform more strategic supplier relationship management.

Problem 9: Inventory Shrinkage

Inventory shrinkage, defined as the loss of inventory between the point of receipt from suppliers and the point of sale to customers, represents a significant but often underestimated source of financial loss for retail businesses. This shrinkage is driven by a combination of factors including shoplifting, employee theft, product damage, spoilage, administrative errors, and vendor fraud.

Without systematic inventory tracking and reconciliation procedures, shrinkage losses can accumulate to substantial levels before they are detected. The reference business, relying on manual inventory records and periodic physical counts, had limited visibility into the magnitude and sources of inventory shrinkage, making it impossible to implement targeted prevention strategies.

The financial impact of inventory shrinkage extends beyond the direct cost of lost merchandise. Shrinkage also affects the accuracy of inventory records, which in turn undermines purchasing decisions, financial reporting, and tax calculations. Furthermore, shrinkage reduces the gross margin available to cover operating expenses and generate profit, directly impacting business profitability.

While ProfitLens does not include dedicated shrinkage detection or prevention functionality, the system's inventory tracking capabilities provide a foundation for identifying discrepancies between expected and actual inventory levels. By maintaining accurate records of all sales transactions and their associated stock deductions, the system creates an audit trail that can be compared against physical inventory counts to identify potential shrinkage issues.

Problem 10: Lack of Visibility

Managing inventory across multiple storage locations, including sales floors, backrooms, and warehouse areas, often results in data silos that prevent managers from developing a comprehensive view of overall inventory positions. Without a centralized system for aggregating inventory information from all locations, managers lack visibility into exactly where specific items are located, how quickly they are moving, and what quantities are available across the entire business.

This lack of visibility forces staff to make operational decisions based on incomplete information, leading to inefficiencies such as unnecessary inter-location transfers, double-handling of goods, and missed opportunities to optimize inventory placement based on demand patterns. When managers cannot see the complete inventory picture, they cannot make the kind of informed decisions that optimize both customer service and operational efficiency.

The consequences of poor inventory visibility are particularly pronounced in businesses with diverse product portfolios, where the sheer number of individual items makes it difficult to maintain accurate mental models of inventory positions without systematic

tracking support. In such environments, the risk of making suboptimal decisions based on incomplete information is significantly elevated.

ProfitLens addresses this through its centralized data architecture, which aggregates all sales, expense, and inventory information into a single database that can be accessed through the system's various modules and dashboards. This centralized approach ensures that all users have access to consistent, up-to-date information regardless of their physical location within the business.

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The consequences of poor inventory visibility are particularly pronounced in businesses with diverse product portfolios, where the sheer number of individual items makes it difficult to maintain accurate mental models of inventory positions without systematic tracking support. In such environments, the risk of making suboptimal decisions based on incomplete information is significantly elevated. Staff may spend valuable time searching for products that are already available but stored in different locations, while managers may unknowingly reorder products that remain in stock, resulting in unnecessary procurement costs and inefficient inventory utilization.

Limited visibility also affects the quality of business planning and decision-making. Without consolidated and real-time inventory information, managers are unable to accurately evaluate stock availability, identify fast-moving and slow-moving products, monitor inventory turnover, or recognize purchasing trends. This lack of insight reduces the organization's ability to forecast demand, prepare for seasonal fluctuations, and respond promptly to changes in customer purchasing behavior. Consequently, businesses may experience higher inventory carrying costs, increased stock obsolescence, and reduced customer satisfaction due to delayed order fulfillment or unavailable products.

Furthermore, fragmented inventory information makes it difficult to perform routine inventory audits and reconcile physical stock with recorded quantities. Discrepancies between actual inventory and recorded data may remain undetected for extended

periods, increasing the likelihood of inventory shrinkage, misplaced products, recording errors, or unauthorized stock movements. The absence of centralized visibility also limits management's ability to monitor inventory performance and enforce accountability across different operational areas.

ProfitLens addresses this challenge through its centralized data architecture, which consolidates sales, expense, product, and inventory information into a single database that can be accessed through the system's integrated modules and reporting dashboard. By maintaining a unified repository of business data, the system ensures that all authorized users work with consistent, accurate, and up-to-date information regardless of their physical location within the business. This centralized approach minimizes data duplication, improves information reliability, and enables faster retrieval of inventory records whenever needed.

In addition, the system provides managers with organized reports and real-time inventory updates that support informed operational and strategic decision-making. By presenting a comprehensive view of inventory levels alongside sales performance and financial information, ProfitLens enables businesses to identify inventory shortages, monitor stock movement, evaluate product demand, and make timely purchasing decisions. The improved visibility ultimately enhances operational efficiency, reduces unnecessary inventory costs, strengthens inventory control, and supports better customer service by ensuring that product availability can be monitored and managed more effectively.

1.3 Objectives of the Study

1.3.1 General Objective

The overarching objective of this study is to design, develop, and deploy ProfitLens, a smart, web-based profit monitoring platform that delivers automated business insights, accurate financial tracking, and proactive alerts for overspending in an auto parts sales and services environment. The platform is intended to replace the manual financial management practices currently in use at the reference business and to provide its users with a reliable, easy-to-use, and visually informative system that supports daily operational decision-making and longer-term financial planning.

This general objective encompasses several interconnected goals that collectively address the financial management challenges identified in the problem statement. First, the study aims to demonstrate that web-based technologies can be effectively applied to create affordable, purpose-built financial monitoring solutions for small businesses. Second, the study seeks to validate the effectiveness of automated financial tracking in improving the accuracy, timeliness, and accessibility of financial information for business decision-making. Third, the study intends to produce a thoroughly documented system that can serve as a reference model for similar initiatives targeting other small business contexts.

The achievement of this general objective requires the successful completion of several interdependent tasks, including the identification and analysis of business requirements, the design of system architecture and user interfaces, the implementation of functional modules using appropriate technologies, the testing and validation of system functionality, and the deployment of the system in the reference business environment. Each of these tasks contributes to the overall objective while also generating knowledge and insights that are valuable beyond the immediate project context.

1.3.1 General Objective

The overarching objective of this study is to design, develop, implement, and evaluate **ProfitLens**, a smart, web-based profit monitoring platform that delivers automated business insights, accurate financial tracking, and proactive alerts for overspending in an auto parts sales and services environment. The platform is intended to replace the manual financial management practices currently employed by the reference business and provide users with a reliable, user-friendly, and visually informative system that supports both daily operational decision-making and long-term financial planning.

Specifically, the system seeks to improve the efficiency and accuracy of recording sales and expenses by replacing paper-based and spreadsheet-dependent processes with a centralized digital platform. Through automation, real-time financial monitoring, and organized reporting, ProfitLens enables business owners and staff to maintain accurate financial records while reducing the time, effort, and errors associated with manual bookkeeping.

This general objective encompasses several interconnected goals that collectively address the financial management challenges identified in the problem statement. First, the study aims to demonstrate that modern web-based technologies can be effectively utilized to develop affordable, scalable, and purpose-built financial monitoring solutions for small and medium-sized enterprises. Second, the study seeks to validate the effectiveness of automated financial tracking in improving the accuracy, consistency, timeliness, and accessibility of financial information used for business decision-making. Third, the study aims to evaluate the usability and practicality of the proposed system in supporting the day-to-day operations of an auto parts sales and services business.

In addition, the study intends to enhance financial transparency by providing business owners with real-time visibility into sales, expenses, profit trends, and operational performance through interactive dashboards and analytical reports. The availability of timely and organized financial information enables managers to identify spending patterns, monitor business performance, detect potential financial issues, and make informed strategic decisions based on reliable data rather than estimates or manual calculations.

Another important objective is to promote the digital transformation of small businesses by demonstrating how information systems can streamline routine financial processes, reduce administrative workload, and improve overall operational efficiency. By integrating essential financial monitoring functions into a single web-based platform,

ProfitLens minimizes redundant tasks, supports standardized data management, and improves collaboration among authorized users within the organization.

Furthermore, the study aims to develop a secure and maintainable information system that incorporates appropriate authentication mechanisms, role-based access control, and centralized data storage to protect sensitive business information while ensuring data integrity and accessibility. These features contribute to the reliability and sustainability of the proposed solution, making it suitable for continuous use in a real business environment.

The achievement of this general objective requires the successful completion of several interdependent tasks, including the identification and analysis of business requirements, the design of system architecture and user interfaces, the development of a secure database and functional modules, the implementation of reporting and analytics features, comprehensive system testing and validation, user acceptance evaluation, and the deployment of the completed system within the reference business environment. Each of these activities contributes to the realization of the study's primary objective while generating practical knowledge regarding the design, implementation, and adoption of financial monitoring systems for small businesses.

Ultimately, this study seeks not only to provide an effective financial monitoring solution for the partner business but also to contribute to the growing body of knowledge on information systems development for small enterprises. By demonstrating the practical application of web technologies in improving financial management, ProfitLens serves as a model for future researchers and developers seeking to create accessible, cost-effective, and technology-driven solutions for similar business environments.

1.3.2 Specific Objectives

In support of the general objective, this study aims to accomplish the following specific objectives:

Objective 1: Automated Profit Tracking Module

To provide a comprehensive module for tracking monthly profit by recording individual sales transactions, each defined by a selected product, an entered quantity, and a specified sale date, against a live-updated register of product costs and operating expenses. The module must automatically compute the total value of each sale from the product's selling price, deduct the sold quantity from the corresponding stock record, and present a continuously updated net profit figure computed as the difference between the period's total sales revenue and total recorded expenses.

This objective addresses the fundamental need for accurate, real-time profit visibility that was identified as a critical deficiency in the reference business's current operations. The module must implement the mathematical formula $\text{Net Profit} = \text{Total Sales Revenue} - \text{Total Expenses}$ with complete accuracy, ensuring that every transaction is properly captured, calculated, and reflected in the overall profit figure. The automated nature of

this calculation eliminates the manual aggregation process that was identified as a primary source of errors and delays in the existing system.

The implementation of this objective requires careful attention to several technical considerations, including the accurate calculation of individual sale totals, the proper handling of stock deductions to maintain inventory accuracy, the correct aggregation of sales data across different time periods, and the accurate subtraction of expenses from revenue to determine net profit. Each of these technical requirements must be implemented with precision to ensure that the resulting profit figures are reliable and accurate.

Objective 2: Financial Reporting Module

To implement a financial reporting module that displays monthly, quarterly, and annual profit-and-loss summaries within the system's web interface. The module must present a full January-to-December comparison of revenue and expenses in both tabular and graphical formats, support the selection of specific calendar years for historical review, and make all reported data available without requiring the export of raw records to an external application for processing.

This objective addresses the need for structured financial reporting that was identified as a significant deficiency in the reference business's current operations. The module must generate comprehensive financial reports that provide the business owner with complete visibility into financial performance across different time periods and dimensions. The reports must be generated automatically from the data stored in the system's database, eliminating the manual reconstruction process that was identified as impractical and error-prone.

The implementation of this objective requires the development of database queries that can aggregate financial data across different time periods, the creation of tabular report formats that present information in a clear and organized manner, and the implementation of graphical visualizations that transform raw data into intuitive visual representations. The module must also support year selection functionality that enables users to review historical performance data for any year for which data has been entered into the system.

Objective 3: Revenue Analysis and Visualization

To implement automated revenue insights that display bar graphs comparing monthly revenue versus expenses across the full calendar year (January through December), supplemented by line chart representations of revenue trends and year-over-year comparative visualizations. These visual outputs must be computed dynamically from the database without relying on any external charting library, ensuring that the system remains self-contained and does not depend on third-party service availability.

This objective addresses the need for trend analysis and visual reporting that was identified as a critical deficiency in the reference business's current operations. The module must generate visual representations of financial data that enable the business owner to identify patterns, compare performance across time periods, and develop

insights that support strategic decision-making. The requirement for self-contained chart generation ensures that the system can operate independently of external services, providing reliable access to visual reporting capabilities regardless of internet connectivity or third-party service availability.

The implementation of this objective requires the development of SVG chart generation capabilities using PHP and JavaScript, the implementation of data aggregation algorithms that compile raw transaction data into monthly and annual summaries, and the creation of interactive visualizations that enable users to explore financial data from multiple perspectives. The charts must be rendered directly in the user's web browser without requiring any external plugins or libraries.

Objective 4: Expense Tracking and Analysis

To develop an expense tracking module with category-based expense breakdowns covering categories including Office Supplies, Advertising, Utilities, and Salaries and Wages, and visual indicators that highlight high-spend areas relative to total revenue for the current period. The module must support the addition, editing, and deletion of individual expense records, maintain accurate running totals, and provide the owner and staff with immediate visual feedback when expense levels in any category represent a disproportionately high share of the period's revenue.

This objective addresses the need for comprehensive expense monitoring and analysis that was identified as a critical deficiency in the reference business's current operations. The module must enable systematic capture and categorization of all business expenses, providing the business owner with granular visibility into spending patterns across different cost categories. The visual indicator functionality must alert users to potential overspending conditions before they escalate into serious financial problems.

The implementation of this objective requires the development of a categorized expense entry interface, the implementation of real-time expense totaling and aggregation, the creation of visual indicators that compare category-level expenses against revenue thresholds, and the provision of CRUD (Create, Read, Update, Delete) functionality for expense records. The module must maintain accurate running totals that are immediately updated whenever expense records are added, modified, or deleted.

1.4 Scope and Delimitation

1.4.1 Functional Scope

ProfitLens covers the following functional areas, all of which are fully implemented in the delivered system:

User Authentication

Secure login and registration for admin-level users, including password hashing using PHP's `password_hash()` function and session-based access management. The authentication system ensures that only authorized users can access the system's financial data and functionality, protecting sensitive business information from unauthorized access. Session management maintains user authentication state across

multiple page requests, eliminating the need for users to re-authenticate during normal system use.

The authentication system implements several security best practices, including the use of bcrypt password hashing through PHP's password_hash() function, which provides strong protection against brute-force and dictionary attacks. Session identifiers are regenerated upon successful authentication to prevent session fixation attacks, and session data is validated on every protected page request to ensure continued authorization.

Product Management

The ability to add, edit, and delete products with associated attributes including name, category, selling price, cost price, and current stock quantity. The product management module provides the foundation for all sales-related functionality by maintaining the product catalog that users select from when recording sales transactions.

Each product record contains comprehensive attribute information that supports both operational and analytical functions. The product name and category attributes enable product identification and grouping. The selling price and cost price attributes support profit margin calculations and pricing analysis. The current stock quantity attribute enables real-time inventory tracking and supports the stock deduction functionality that occurs automatically when sales are recorded. The **Product Management** module provides a centralized facility for maintaining the complete product catalog used by the business. It enables authorized users to add, edit, and delete product records while managing essential product attributes such as the product name, category, selling price, cost price, and current stock quantity. As one of the core components of ProfitLens, this module serves as the foundation for sales recording, inventory monitoring, revenue analysis, and profit computation by ensuring that all product information is accurate, organized, and readily available throughout the system.

Rather than maintaining separate product lists in paper records or spreadsheets, ProfitLens stores all product information in a centralized database. This centralized approach ensures consistency across all system modules, eliminates duplicate product records, and simplifies the management of inventory information. Since every sales transaction references the product database, maintaining accurate product information directly contributes to the reliability of financial reports and business analytics.

Each product record contains comprehensive attribute information that supports both operational activities and financial analysis. The **product name** serves as the primary identifier, allowing users to easily locate and select products during sales recording. The **product category** groups similar products together, making it easier to organize inventory, generate category-based reports, and analyze product performance across different segments of the business.

The **selling price** attribute represents the amount charged to customers for each unit sold and is automatically retrieved during sales transactions. This ensures consistent pricing throughout the system while eliminating the need for users to manually enter

prices for every transaction. Automated price retrieval reduces pricing inconsistencies, minimizes data entry errors, and accelerates the sales recording process.

The **cost price** attribute records the acquisition or purchase cost of each product. This information is essential for calculating gross profit, determining profit margins, and evaluating the profitability of individual products. By maintaining both selling and cost prices within each product record, ProfitLens enables more accurate financial analysis and allows business owners to assess whether current pricing strategies generate sufficient returns to cover operating expenses and support business growth.

Another critical attribute is the **current stock quantity**, which represents the number of units currently available for sale. This value is automatically updated whenever a sales transaction is successfully recorded, ensuring that inventory levels remain synchronized with actual business activities. Automatic stock adjustments eliminate the need for manual inventory updates, reducing administrative effort while improving inventory accuracy.

The Product Management module also supports inventory integrity by preventing inconsistencies between product records and sales transactions. Since all sales are linked directly to products stored in the centralized database, users can only record transactions for products that have been properly registered within the system. This controlled process reduces the likelihood of invalid product entries, duplicate records, or inaccurate inventory reporting.

The module includes comprehensive **Create, Read, Update, and Delete (CRUD)** functionality, allowing administrators to manage product information efficiently. New products can be added as the business expands its inventory, while existing product information may be updated to reflect changes in pricing, product categories, or stock quantities. Obsolete or discontinued products can also be removed when they are no longer required, helping maintain an organized and relevant product catalog.

Editing functionality provides flexibility for maintaining accurate product information over time. If selling prices change due to market conditions, supplier cost adjustments, or promotional strategies, administrators can update the corresponding product records without affecting the integrity of previously recorded sales transactions. Likewise, stock quantity adjustments may be performed when necessary to reflect inventory reconciliation or corrections identified during physical inventory counts.

The searchable product catalog significantly improves user efficiency during transaction processing. As the number of available products increases, manually locating specific items becomes increasingly time-consuming. ProfitLens addresses this challenge by providing searchable product selection, enabling users to quickly identify products using keywords or partial names. This feature reduces transaction processing time and improves the overall user experience, particularly for businesses with extensive product inventories.

Beyond supporting daily operations, the Product Management module also contributes to business analysis and strategic decision-making. Product information serves as the basis for identifying best-selling items, evaluating product profitability, monitoring inventory turnover, and analyzing revenue generated by different product categories. These insights help business owners make informed decisions regarding inventory replenishment, pricing adjustments, promotional activities, and product assortment planning.

The centralized product database further enhances data consistency throughout the system. Because all modules—including Sales Recording, Revenue Analysis, Inventory Monitoring, Financial Dashboard, and Profit Calculation—retrieve product information from the same source, changes made within the Product Management module are automatically reflected across the application. This integrated design eliminates redundant data maintenance and ensures that users always work with current and consistent product information.

From a usability perspective, the Product Management interface is designed to be intuitive and easy to navigate. Clearly labeled input fields, organized layouts, and built-in validation help users enter complete and accurate product information while minimizing data entry errors. Validation mechanisms ensure that essential fields, such as product name, selling price, cost price, and stock quantity, contain valid values before records are saved to the database, thereby improving overall data quality.

The module also promotes operational efficiency by reducing the administrative burden associated with inventory maintenance. Business owners no longer need to manually update multiple records whenever products are added or modified, as all relevant information is maintained within a single centralized system. This streamlined approach saves time, improves record accuracy, and simplifies day-to-day inventory management activities.

Although the current implementation focuses on essential product management functions, the module has been developed using a modular architecture that supports future enhancements. Potential improvements include barcode generation and scanning, supplier associations, product image management, batch and serial number tracking, warranty information, minimum stock thresholds, automated reorder notifications, product variants, category hierarchies, and advanced inventory analytics. These enhancements could further strengthen inventory management capabilities while preserving compatibility with the existing financial monitoring features.

Overall, the **Product Management** module serves as the operational backbone of ProfitLens by maintaining accurate, organized, and centralized product information that supports every major function within the system. Through comprehensive product records, automated inventory synchronization, flexible record management, and seamless integration with sales, reporting, and financial analysis modules, the feature enables efficient inventory administration, improves data consistency, supports accurate profit calculations, and provides the reliable product information necessary for effective financial monitoring and informed business decision-making.

Sales Recording

A form-based interface for entering sales transactions, with automatic total calculation, stock deduction, and validation to prevent recording of sales that would result in negative stock. The sales recording module streamlines the transaction entry process while ensuring data accuracy and inventory integrity.

The sales recording interface implements several validation and automation features that enhance data quality and user efficiency. The searchable product dropdown enables quick product selection even when the product catalog contains many items. Automatic total calculation eliminates manual arithmetic errors and ensures consistent pricing. Stock deduction validation prevents the recording of sales that would result in negative inventory levels, maintaining the integrity of inventory records. The **Sales Recording** module provides a structured, form-based interface for recording daily sales transactions in a consistent, efficient, and accurate manner. It enables authorized users to select products, specify quantities sold, and automatically calculate transaction totals while simultaneously updating inventory records. Through built-in validation mechanisms and automated processing, the module minimizes manual effort, reduces data entry errors, and ensures that all recorded sales contribute accurately to the system's financial reports and profit calculations.

Instead of relying on handwritten sales logs or spreadsheet-based records, ProfitLens centralizes all sales transactions within a single database. Every completed sale is immediately stored and becomes available for financial reporting, revenue analysis, inventory monitoring, and profit computation. This centralized approach improves data consistency, eliminates duplicate record-keeping, and provides business owners with immediate access to updated sales information.

The sales recording interface is designed with usability and efficiency in mind. Users complete transactions through clearly labeled input fields and guided forms that simplify the recording process while minimizing the likelihood of incomplete or inaccurate entries. The intuitive layout enables staff members to record sales quickly with minimal training, making the module suitable for daily operational use in small business environments.

One of the module's key features is its **searchable product dropdown**, which allows users to rapidly locate products by typing part of the product name rather than scrolling through an extensive list. As the business expands its product catalog, this feature significantly reduces the time required to identify the correct item and improves overall transaction efficiency. Quick product selection also minimizes the risk of selecting incorrect products, contributing to more accurate sales records.

The module automatically retrieves product information, including the unit selling price, directly from the product database. Once users specify the quantity being sold, the system performs real-time calculations to determine the total transaction amount. This automation eliminates the need for manual arithmetic, reducing computational errors

and ensuring consistent pricing across all recorded sales. Automatic calculations also accelerate transaction processing, allowing staff to complete sales more efficiently.

To preserve inventory integrity, the system automatically deducts the quantity sold from the available stock immediately after a successful transaction is recorded. This real-time inventory synchronization ensures that stock levels remain current and accurately reflect the actual quantity of products available for future sales. Because inventory updates occur automatically, users are not required to perform separate inventory adjustments following each transaction, reducing administrative workload and preventing inconsistencies between sales and stock records.

An important validation feature of the module prevents the recording of sales that would result in negative inventory levels. Before a transaction is finalized, the system verifies that sufficient stock is available for the requested quantity. If the available inventory is insufficient, the transaction is rejected, and an appropriate validation message is displayed to the user. This safeguard prevents overselling, protects the accuracy of inventory records, and ensures that reported stock quantities remain consistent with actual product availability.

In addition to inventory validation, the Sales Recording module performs input validation to ensure that all required transaction details are properly completed before submission. Mandatory fields such as product selection, quantity, and transaction date are verified, while invalid values such as negative quantities or empty fields are rejected. These validation mechanisms improve overall data quality and reduce the likelihood of inaccurate financial records entering the system.

Every recorded sales transaction contributes directly to the financial monitoring capabilities of ProfitLens. Sales information is automatically incorporated into revenue reports, profit calculations, dashboard summaries, and analytical modules without requiring duplicate data entry or manual report preparation. This seamless integration ensures that financial reports always reflect the latest business activities and provides management with real-time visibility into business performance.

The module also supports historical transaction management by maintaining a complete record of previously entered sales. Authorized users can review, edit, or delete sales transactions when corrections are necessary, ensuring that financial data remains accurate over time. These maintenance capabilities enable businesses to address data entry mistakes promptly while preserving the integrity and consistency of the overall financial database.

Furthermore, the Sales Recording module serves as a valuable source of operational insights. By maintaining comprehensive sales records, the system enables business owners to identify high-demand products, evaluate sales trends, monitor transaction volumes, and assess overall business performance. These insights support inventory planning, purchasing decisions, pricing strategies, and promotional activities that contribute to improved profitability and customer service.

The module also improves accountability by ensuring that every sales transaction is systematically documented within the centralized database. Compared with manual record-keeping methods, this structured approach reduces the possibility of lost records, duplicate entries, and undocumented transactions while providing a reliable audit trail for financial review and business analysis.

From a usability perspective, the interface emphasizes speed, clarity, and simplicity. Clearly organized forms, responsive validation messages, automatic computations, and intuitive navigation reduce the learning curve for new users while improving productivity for experienced staff. These design characteristics enable the system to integrate smoothly into daily business operations with minimal disruption.

The modular architecture of the Sales Recording feature also supports future enhancements. Potential improvements include barcode scanning for faster product selection, customer transaction histories, invoice generation, discount and promotional pricing management, multiple payment methods, sales returns processing, and integration with electronic payment systems or point-of-sale hardware. These enhancements could expand the functionality of the module while preserving its core purpose of accurate and efficient sales recording.

Overall, the **Sales Recording** module serves as one of the core functional components of ProfitLens by providing a reliable, automated, and user-friendly mechanism for recording business transactions. Through guided data entry, automatic total calculation, real-time inventory updates, comprehensive validation, and seamless integration with financial reporting, the module enhances data accuracy, improves operational efficiency, protects inventory integrity, and provides the accurate sales information necessary for effective profit monitoring and informed business decision-making.

Expense Tracking

A categorized expense logging system supporting addition, editing, and deletion of records, with real-time totaling and category-level breakdowns. The expense tracking module enables systematic capture and categorization of all business expenses, providing the foundation for expense analysis and monitoring.

The expense tracking interface supports multiple expense categories including Office Supplies, Advertising, Utilities, and Salaries and Wages, enabling granular categorization of expenses for analysis purposes. Real-time totaling provides immediate visibility into current period expenses, while the editing and deletion functionality ensures that expense records can be maintained accurately over time. The **Expense Tracking** module provides a structured and centralized system for recording, organizing, and monitoring business expenditures. It enables authorized users to add, edit, and delete expense records while automatically categorizing each transaction according to its corresponding expense type. Through real-time total computation and category-level breakdowns, the module provides business owners with accurate and

up-to-date information regarding operating expenses, forming the foundation for effective financial monitoring and profitability analysis.

Rather than relying on handwritten expense logs or scattered spreadsheet entries, ProfitLens stores all expense transactions within a centralized database. This approach improves the consistency, accessibility, and reliability of financial records while reducing the likelihood of missing, duplicated, or incorrectly recorded expense entries. Each recorded expense is immediately reflected in the system's financial summaries and reports, ensuring that business owners always have access to current expense information.

The expense tracking interface supports multiple predefined expense categories, including **Office Supplies**, **Advertising**, **Utilities**, **Salaries and Wages**, and other operational expenditures relevant to the business. Categorizing expenses enables users to analyze spending patterns more effectively by identifying which areas consume the largest portion of the operating budget. This organized classification also simplifies financial reporting by grouping similar expenditures into meaningful categories that support business analysis and planning.

The module allows users to record essential details for each expense transaction, including the expense category, amount, date, and a brief description of the expenditure. Capturing this information creates a complete financial record that can be reviewed for auditing, reporting, and managerial decision-making purposes. The structured input process also reduces inconsistencies that commonly occur when expense records are maintained manually.

One of the module's key features is its **real-time expense totaling** capability. As new expenses are entered, modified, or removed, the system automatically recalculates total expenditures without requiring manual computation. This automation eliminates arithmetic errors, reduces administrative workload, and ensures that financial summaries always reflect the most current business data. Users can immediately determine the total amount spent during a selected reporting period, enabling continuous monitoring of operational costs.

The editing functionality provides flexibility for correcting data entry errors or updating expense information whenever necessary. If incorrect amounts, categories, or descriptions are entered, authorized users can modify the existing records without creating duplicate transactions. Similarly, the deletion feature allows invalid or mistakenly entered expense records to be removed while maintaining the overall accuracy of the financial database. These maintenance capabilities help preserve the integrity and reliability of financial information over time.

The Expense Tracking module also contributes directly to accurate profit computation. Since expenses are automatically incorporated into the system's financial calculations, business owners receive a more precise representation of net profit after operational costs have been deducted from recorded revenue. This integration eliminates the need

to manually consolidate expense information from multiple sources, thereby improving both efficiency and financial accuracy.

Beyond recording expenses, the module supports expense analysis by presenting categorized expenditure summaries that allow users to compare spending across different operational areas. Business owners can identify which expense categories account for the largest share of business costs, evaluate whether spending aligns with budget expectations, and recognize opportunities for cost reduction. Such insights enable more informed financial decisions and encourage responsible allocation of business resources.

Historical expense records maintained within the system also support long-term financial evaluation. Users can review previous expense transactions to identify recurring expenditures, monitor changes in operational costs over time, and compare spending across different reporting periods. This historical information assists management in preparing budgets, forecasting future expenses, and evaluating the effectiveness of cost-control initiatives implemented by the business.

The module further enhances financial transparency by ensuring that all recorded expenditures are systematically documented and easily retrievable. Instead of searching through paper receipts or manually reviewing spreadsheets, users can quickly locate specific expense records using the system's organized database structure. This accessibility simplifies internal reviews, financial reconciliation, and report preparation while reducing the time required to retrieve historical financial information.

From a usability perspective, the Expense Tracking interface emphasizes simplicity and efficiency. Clearly labeled input fields, intuitive navigation, and organized expense categories enable users to record transactions with minimal effort and training. Automatic validation of required fields helps prevent incomplete or inaccurate entries, contributing to higher data quality and improved reliability of financial reports.

The modular design of the Expense Tracking feature also supports future enhancement. Additional capabilities such as receipt image uploads, recurring expense scheduling, expense approval workflows, budget allocation monitoring, automated overspending alerts, supplier-specific expense tracking, and advanced filtering options may be incorporated in future versions of ProfitLens. These enhancements would further strengthen the system's financial management capabilities while preserving the existing foundation for accurate expense recording and analysis.

Overall, the Expense Tracking module serves as a critical component of ProfitLens by providing a reliable and organized mechanism for recording, categorizing, and analyzing business expenditures. Through automated expense computation, categorized financial records, real-time updates, and comprehensive reporting support, the module enables business owners to monitor operating costs effectively, maintain accurate financial records, identify spending trends, and make informed decisions that contribute to improved financial control, operational efficiency, and long-term business sustainability.

Revenue Analysis

A dedicated module presenting monthly revenue trends in bar and line chart formats, year-over-year comparisons, and a ranked list of top-performing products. The revenue analysis module transforms raw sales data into visual representations that enable trend identification and performance comparison.

The revenue analysis module implements SVG-based chart generation that creates visual representations directly in the user's web browser without requiring external charting libraries. The module supports multiple visualization types including bar charts for monthly comparisons, line charts for trend visualization, and ranked listings for product performance analysis. The **Revenue Analysis** module provides a dedicated environment for examining the business's sales performance through interactive visualizations and summarized financial data. It presents monthly revenue trends using bar and line charts, year-over-year revenue comparisons, and a ranked list of top-performing products based on recorded sales transactions. By transforming raw sales records into meaningful graphical representations, the module enables business owners and authorized users to identify trends, evaluate business performance, and make informed operational and strategic decisions.

Rather than requiring users to manually calculate revenue totals or compare transaction records across different periods, the module automatically aggregates sales data stored in the centralized database. This automation reduces the time required to prepare financial analyses while minimizing the possibility of calculation errors associated with manual reporting. Users can immediately view updated revenue information whenever new sales transactions are recorded, ensuring that business performance indicators remain current and reliable.

The module implements **SVG (Scalable Vector Graphics)**-based chart generation, allowing visual reports to be rendered directly within the user's web browser without relying on external charting libraries or third-party visualization services. This implementation reduces software dependencies, improves system portability, and ensures consistent chart rendering across supported web browsers. Because SVG graphics are resolution-independent, charts remain clear and readable regardless of screen size or display resolution, contributing to a responsive and user-friendly interface.

Multiple visualization formats are incorporated to present revenue information from different analytical perspectives. Bar charts provide clear comparisons of monthly revenue figures, allowing users to identify periods of high and low sales performance at a glance. Line charts illustrate revenue trends over time, making it easier to recognize long-term growth patterns, seasonal fluctuations, and recurring changes in business activity. These graphical representations simplify the interpretation of financial data and enable users to identify trends that might not be immediately apparent from numerical reports alone.

The year-over-year comparison feature further enhances financial analysis by allowing users to evaluate revenue performance across different calendar years. This comparison assists business owners in measuring long-term business growth, determining whether sales performance has improved or declined, and assessing the effectiveness of business strategies implemented during previous periods. Historical comparisons also support strategic planning by providing evidence of recurring seasonal demand and long-term market behavior.

In addition to graphical analysis, the module generates a ranked list of top-performing products based on recorded sales data. Products may be ranked according to sales quantity, revenue generated, or overall contribution to business income, enabling managers to identify which inventory items consistently perform well in the marketplace. This information supports more informed purchasing decisions, inventory planning, pricing strategies, and promotional activities by highlighting products that contribute significantly to business profitability.

The Revenue Analysis module also assists in identifying slow-moving or underperforming products by comparing sales performance across the product catalog. Although its primary purpose is to highlight high-performing products, users can infer which products require additional marketing efforts, pricing adjustments, or inventory review by analyzing comparative sales results. Such insights contribute to more efficient inventory management and better allocation of business resources.

Furthermore, the module supports evidence-based decision-making by providing timely and organized financial information. Instead of relying solely on intuition or isolated transaction records, business owners can use summarized revenue reports to evaluate sales performance, monitor business growth, assess the success of promotional campaigns, and determine whether operational objectives are being achieved. These analytical capabilities enable more strategic management practices and reduce uncertainty in financial planning.

Because the module retrieves information directly from the centralized database, all visualizations reflect real-time business data. Revenue charts and ranked product listings are automatically updated as new sales transactions are entered into the system, eliminating the need for manual report generation or periodic data consolidation. This real-time capability ensures that decision-makers always have access to accurate and current financial information.

The module's user interface emphasizes simplicity, readability, and accessibility. Charts are presented using clearly labeled axes, organized layouts, and intuitive visual elements that can be interpreted even by users with limited financial or technical expertise. This design philosophy makes complex revenue information easier to understand while reducing the learning curve associated with using analytical reporting tools.

In addition, the Revenue Analysis module complements the Financial Dashboard by providing more detailed insights into sales performance. While the dashboard offers a

concise overview of key financial indicators, the Revenue Analysis module enables users to explore revenue patterns in greater depth through historical comparisons, graphical trend analysis, and product performance evaluation. Together, these modules provide both high-level summaries and detailed analytical capabilities that support comprehensive financial monitoring.

The modular implementation of the Revenue Analysis feature also provides opportunities for future enhancement. Subsequent versions of ProfitLens may incorporate advanced analytical capabilities such as revenue forecasting, moving averages, growth projections, customizable reporting periods, category-based sales analysis, branch comparisons, interactive chart filtering, exportable visual reports, and predictive analytics using historical sales data. These enhancements could further strengthen the system's ability to support strategic business planning while building upon the analytical foundation established in the current implementation.

Overall, the Revenue Analysis module serves as a vital decision-support component of ProfitLens by converting recorded sales transactions into meaningful visual and statistical information. Through automated revenue aggregation, SVG-based chart generation, historical trend visualization, year-over-year performance comparison, and product ranking, the module enables business owners to monitor financial performance efficiently, identify emerging sales patterns, evaluate business growth, and make well-informed decisions that contribute to improved profitability and long-term business success.

Profit and Loss Reporting

A comprehensive reporting module providing a January-to-December grouped bar chart of revenue versus expenses, a tabular profit-and-loss summary, and a monthly expense breakdown report with year selection capability. The profit and loss reporting module provides the most comprehensive financial overview available in the system, combining graphical and tabular representations of financial performance.

The reporting module aggregates data across all twelve months of the calendar year, providing a complete annual financial picture that enables pattern identification and year-over-year comparison. The year selection capability enables historical review of financial performance for any year for which data has been entered into the system. The **Profit and Loss Reporting** module serves as the primary financial reporting component of ProfitLens by providing a comprehensive overview of the business's financial performance over a selected reporting period. The module presents a **January-to-December grouped bar chart** comparing monthly revenue and expenses, a detailed **Profit and Loss (P&L) summary table**, and a **monthly expense breakdown report** with year selection capability. By combining graphical visualizations with structured tabular reports, the module enables business owners and authorized users to evaluate profitability, monitor operational costs, and assess overall business performance using accurate and up-to-date financial information.

Unlike manual financial reporting methods that require users to consolidate sales and expense records from multiple sources, ProfitLens automatically retrieves and processes financial data stored within its centralized database. Revenue, expenses, and profit values are calculated dynamically based on recorded transactions, eliminating manual computations and significantly reducing the risk of reporting errors. This automated approach ensures that financial reports remain accurate, consistent, and readily available whenever required.

One of the module's key features is the **January-to-December grouped bar chart**, which visually compares monthly revenue against corresponding expenses throughout the calendar year. Displaying both financial indicators side by side allows users to immediately evaluate the relationship between business income and operating costs for each month. By presenting comparative financial information in a graphical format, the system enables users to recognize periods of strong profitability, increased expenditures, or declining financial performance without manually analyzing extensive numerical data.

The graphical representation also assists in identifying seasonal business patterns. Users can observe recurring periods of increased revenue, seasonal declines in sales, or months characterized by unusually high operating expenses. These insights are valuable for planning inventory purchases, budgeting operational costs, scheduling promotional activities, and preparing for predictable fluctuations in business demand throughout the year.

Complementing the graphical presentation is a detailed **Profit and Loss summary table** that provides a structured breakdown of monthly revenue, total expenses, and resulting net profit or loss. The tabular report presents financial information in a precise numerical format, allowing users to review exact financial figures alongside the visual trends presented in the accompanying charts. This combination of graphical and tabular reporting accommodates different user preferences while supporting more comprehensive financial analysis.

The Profit and Loss summary enables business owners to evaluate whether monthly revenues are sufficient to cover operating expenses and generate sustainable profits. By presenting net financial results for each reporting period, the module assists management in determining the overall financial health of the business and identifying months requiring closer operational review. Negative profit values immediately indicate periods where expenses exceeded revenues, while positive results demonstrate profitable business performance.

Another important feature of the reporting module is the **monthly expense breakdown report**, which categorizes operating expenditures according to predefined expense classifications. Rather than displaying only total monthly expenses, the report identifies how much was spent on categories such as office supplies, advertising, utilities, salaries and wages, and other business expenses. This detailed categorization enables users to determine which operational areas contribute most significantly to overall expenditures and identify opportunities for cost optimization.

The **year selection capability** further enhances the module's analytical functionality by allowing users to generate reports for any calendar year stored within the system's database. Instead of limiting financial analysis to the current reporting period, business owners can review historical financial performance, compare annual results, and evaluate long-term business growth. Historical reporting supports strategic planning by enabling users to identify recurring financial trends, assess the effectiveness of previous business decisions, and establish realistic financial goals based on historical performance.

Year-over-year comparisons also provide valuable insights into business development. Users can determine whether revenue has increased, expenses have remained under control, or profitability has improved over successive years. These comparisons assist management in evaluating the success of implemented business strategies, pricing adjustments, inventory management practices, and operational improvements over extended periods.

Because all financial reports are generated directly from the centralized database, the information presented remains synchronized with the latest sales and expense transactions. Whenever new financial data is entered into the system, corresponding reports are automatically updated without requiring manual recalculation or report preparation. This real-time reporting capability ensures that business owners always have access to current financial information for operational monitoring and decision-making.

The reporting module also contributes significantly to financial transparency and accountability. By maintaining organized records of revenues, expenses, and profits across multiple reporting periods, the system enables users to review historical financial performance whenever necessary. These reports can support internal financial reviews, budgeting activities, performance evaluations, and discussions regarding future business strategies.

From a usability perspective, the module emphasizes clarity, accessibility, and ease of interpretation. Charts utilize clearly labeled axes, organized legends, and proportional graphical elements that facilitate quick understanding of financial performance. Likewise, tabular reports present financial information using structured layouts that are easy to read and interpret, even for users with limited accounting or financial management experience. The intuitive design reduces the time required to analyze business performance while improving the overall user experience.

The integration of graphical and tabular reporting also enhances decision-making by providing multiple perspectives on the same financial information. While charts emphasize trends, comparisons, and overall performance patterns, detailed tables provide the precise numerical values necessary for budgeting, financial verification, and operational planning. Together, these complementary reporting formats allow users to gain both high-level and detailed insights into the financial condition of the business.

The modular design of the Profit and Loss Reporting feature also supports future enhancements. Subsequent versions of ProfitLens may incorporate downloadable PDF and Excel reports, customizable reporting periods, comparative quarterly analyses, gross profit and operating margin calculations, break-even analysis, cash flow summaries, forecasted financial projections, customizable dashboards, and interactive chart filtering. Integration with advanced business intelligence and data visualization tools may also be considered to provide more sophisticated financial analytics while maintaining compatibility with the existing reporting framework.

Overall, the **Profit and Loss Reporting** module serves as the most comprehensive financial analysis component of ProfitLens by consolidating revenue, expenses, and profitability into meaningful graphical and tabular reports. Through automated financial aggregation, annual trend visualization, categorized expense analysis, historical year selection, and real-time report generation, the module empowers business owners to evaluate operational performance, identify financial trends, monitor profitability, support budgeting activities, and make informed strategic decisions that contribute to the long-term sustainability and growth of the business.

Financial Dashboard

A centralized landing page presenting key performance indicators including net profit for the current month, growth percentage relative to the prior month, a mini bar chart of the current year's revenue distribution, the best-selling product, and visual summaries from the expense and revenue modules. The financial dashboard provides at-a-glance visibility into the most important financial metrics, enabling quick assessment of business performance.

The dashboard design prioritizes information density and visual clarity, presenting multiple performance indicators in a single view that can be quickly scanned and understood. The growth percentage indicator provides immediate feedback on whether performance is improving or declining relative to the previous period, enabling rapid identification of trends that may require attention. A centralized landing page presents key performance indicators (KPIs), including the current month's net profit, the percentage of growth or decline compared with the previous month, a mini bar chart illustrating the current year's monthly revenue distribution, the best-selling product, and visual summaries of sales and expense data. The financial dashboard provides users with an at-a-glance overview of the business's financial performance, enabling business owners and authorized staff to quickly assess the organization's overall financial condition without manually reviewing individual transaction records or multiple reports.

The dashboard is designed to prioritize information density while maintaining visual clarity and ease of interpretation. Essential financial metrics are organized into intuitive dashboard cards, charts, and summary panels, allowing users to obtain meaningful insights within a single interface. By consolidating important financial information into one screen, the system minimizes the time required to locate critical business data and improves the overall efficiency of financial monitoring activities.

One of the dashboard's primary features is the display of the current month's net profit, which is automatically calculated using recorded sales and expense transactions. This real-time computation eliminates the need for manual calculations and ensures that business owners always have access to updated profitability information. Since the displayed figures are generated directly from the centralized database, users can rely on the accuracy and consistency of the financial information presented.

The monthly growth percentage indicator provides immediate insight into business performance by comparing the current month's results with those of the previous month. Positive growth values indicate improving financial performance, while negative values highlight potential declines that may require managerial attention. This comparative analysis enables business owners to identify emerging trends, evaluate the effectiveness of operational decisions, and respond promptly to changes in sales performance or operating expenses.

To further enhance data visualization, the dashboard includes a compact bar chart displaying the distribution of revenue across the current calendar year. Rather than presenting numerical values alone, the graphical representation enables users to recognize seasonal fluctuations, recurring sales patterns, and periods of exceptional business performance more easily. Visualizing historical revenue trends assists management in identifying peak selling periods, evaluating business growth over time, and planning future sales strategies based on historical performance.

The dashboard also identifies the best-selling product based on recorded sales transactions. This feature provides valuable insight into customer demand by highlighting the product that has generated the highest sales volume or revenue during the selected reporting period. Knowing which products perform best enables business owners to make informed inventory decisions, prioritize high-demand items, optimize purchasing strategies, and develop promotional campaigns that capitalize on customer preferences.

In addition, visual summaries generated from the sales and expense modules provide users with a concise breakdown of business income and operational expenditures. These summaries allow managers to compare revenue against expenses, monitor spending behavior, and quickly determine whether business operations remain financially sustainable. The use of charts and graphical indicators simplifies complex financial information, making it easier for users with limited accounting knowledge to understand the financial status of the business.

The dashboard is designed to display real-time information by automatically updating whenever new sales or expense transactions are recorded. This ensures that all key performance indicators accurately reflect the latest business activities without requiring manual report generation or data refresh procedures. The availability of current financial information enables management to make timely operational decisions based on the most recent business performance.

Furthermore, the centralized dashboard reduces the need to navigate between multiple modules to obtain essential financial information. Instead of opening separate sales reports, expense reports, inventory records, and profit summaries, users can access the most important performance indicators from a single interface immediately after logging into the system. This streamlined workflow improves user productivity, reduces navigation time, and enhances the overall usability of the application.

The dashboard also supports better strategic planning by providing historical and comparative financial information that assists in evaluating business performance over time. Managers can monitor changes in profitability, identify unusual fluctuations in revenue or expenses, assess the effectiveness of cost-control measures, and determine whether business objectives are being achieved. These analytical capabilities transform raw financial data into actionable business insights that support evidence-based decision-making.

From a usability perspective, the dashboard employs a clean and organized interface with clearly labeled metrics, responsive chart components, and logically grouped information. The visual hierarchy ensures that the most critical financial indicators receive immediate attention while maintaining an uncluttered appearance that is easy to understand for both experienced users and newly trained staff. This intuitive presentation reduces cognitive load and allows users to interpret financial information quickly and accurately.

The dashboard also serves as a centralized monitoring tool that promotes proactive financial management. Rather than identifying financial issues only after generating detailed reports, business owners can observe changes in profit, revenue, expenses, and sales performance as they occur. Early visibility into these indicators enables timely corrective actions, such as controlling unnecessary expenditures, adjusting purchasing decisions, or implementing sales strategies to improve profitability.

Overall, the Financial Dashboard serves as the primary decision-support interface of ProfitLens by consolidating critical financial information into a comprehensive, visually informative, and easy-to-use display. Through real-time performance indicators, automated profit calculations, interactive charts, and summarized financial data, the dashboard enables business owners and authorized users to monitor operational performance efficiently, identify emerging trends, and make informed decisions that contribute to improved financial management, operational efficiency, and long-term business sustainability.

Role-Based Access Control

Differentiated access levels for admin users (full system access) and standard users (restricted access), enforced through PHP session validation on every protected page. The role-based access control system ensures that users can only access functionality appropriate to their authorization level, protecting sensitive financial data and system configuration from unauthorized modification.

The admin role provides complete access to all system modules including the ability to add, edit, and delete products, sales, and expenses. The standard user role provides read-only access to financial data and reports without the ability to modify records. This differentiation enables appropriate access control while still allowing staff members to view financial information relevant to their responsibilities. ProfitLens implements a **Role-Based Access Control (RBAC)** mechanism that regulates user access according to predefined authorization levels. The system distinguishes between **administrator** and **standard user** accounts, with each role assigned a specific set of permissions based on operational responsibilities. Access restrictions are enforced through PHP session validation on every protected page, ensuring that only authenticated users with the appropriate privileges can access system resources. This security mechanism safeguards sensitive financial information and prevents unauthorized users from viewing or modifying confidential business data.

The implementation of role-based access control strengthens system security by ensuring that users are granted only the permissions necessary to perform their assigned tasks. Rather than allowing unrestricted access to all system functions, ProfitLens follows the principle of **least privilege**, whereby users receive the minimum level of access required to fulfill their responsibilities. This approach reduces the risk of accidental data modification, unauthorized transactions, and intentional misuse of financial records while maintaining efficient business operations.

The **administrator role** is granted complete access to all modules and system functionalities. Administrators can manage user accounts, add, edit, and delete product records, record and modify sales and expense transactions, generate financial reports, monitor inventory, and configure system settings. They are also responsible for maintaining data accuracy, resolving inconsistencies, overseeing user activities, and ensuring that the system operates correctly. Because administrators have unrestricted access to critical business information, their accounts are protected through secure authentication and continuous session validation.

The **standard user role** is designed for staff members whose responsibilities involve viewing business information without making permanent modifications to financial records. Standard users are permitted to access dashboards, financial reports, inventory information, and other authorized modules in a read-only capacity. They cannot create, edit, or delete products, sales, expenses, or user accounts, thereby protecting the integrity of business data while still providing employees with the information necessary to perform their daily responsibilities. This separation of responsibilities minimizes the likelihood of unauthorized or accidental changes to financial records.

Access control decisions are enforced throughout the application using PHP session management. Whenever a user successfully logs into the system, a secure session is established containing authentication and authorization information, including the user's assigned role. Each protected page validates the active session before granting access. If a user attempts to access a page without proper authentication or without sufficient privileges, the system automatically redirects the user to the login page or displays an

access restriction message. This mechanism prevents unauthorized access through manually entered URLs or bookmarked administrative pages.

In addition to improving security, role-based access control contributes to better accountability within the organization. Since each authorized user logs into the system using individual credentials, actions performed within the application can be associated with specific user accounts. This separation of user identities promotes responsible system usage and simplifies administrative oversight by ensuring that access privileges are clearly defined according to organizational roles.

The implementation of RBAC also protects the integrity and consistency of financial information. Because only administrators are authorized to modify products, sales records, expense entries, and other critical data, the possibility of accidental deletion, duplicate entries, or unauthorized alterations is significantly reduced. This controlled environment helps maintain reliable financial records that accurately represent the business's operational activities and support dependable financial reporting.

Furthermore, role-based access control simplifies system administration by allowing permissions to be managed through predefined user roles instead of assigning individual permissions to every user. As new employees are added to the organization, administrators can assign the appropriate role during account creation, ensuring that users immediately receive the correct level of access. This standardized permission structure reduces administrative effort, improves consistency, and facilitates easier management of user accounts as the business grows.

The RBAC framework also supports compliance with fundamental information security principles, including confidentiality, integrity, and availability. Confidentiality is maintained by restricting access to sensitive financial information to authorized personnel only. Integrity is preserved by limiting data modification privileges to trusted administrators, thereby preventing unauthorized or unintended alterations. Availability is ensured by allowing authorized users to access the information necessary for their responsibilities while preventing inappropriate access that could compromise system operations.

Although the current implementation includes only two user roles—administrator and standard user—the system architecture has been developed with scalability in mind. Future versions of ProfitLens may introduce additional roles such as inventory manager, accountant, cashier, supervisor, or auditor, each with customized permissions tailored to specific organizational responsibilities. This extensible design allows the access control framework to evolve alongside the operational needs of expanding businesses.

Overall, the Role-Based Access Control feature serves as a fundamental security component of ProfitLens by ensuring that users interact with the system according to their assigned responsibilities. Through secure authentication, session validation, predefined user permissions, and controlled access to sensitive financial information, the system protects business data from unauthorized access while promoting

accountability, maintaining data integrity, and supporting the secure and efficient operation of the financial monitoring platform.

1.4.2 Delimitations

The following functional areas are explicitly outside the scope of ProfitLens as designed and implemented. These delimitations reflect deliberate decisions made to keep the system focused, maintainable, and appropriate for the scale and technical environment of the reference business:

Point-of-Sale Hardware Integration

ProfitLens does not interface with barcode scanners, receipt printers, cash drawers, or any other physical POS hardware. Sales are entered manually through the web interface. This delimitation reflects the resource constraints of the reference business, which does not currently employ POS hardware, and the desire to minimize system complexity and deployment requirements. ProfitLens does not interface with physical Point-of-Sale (POS) hardware such as barcode scanners, receipt printers, cash drawers, customer display units, card payment terminals, weighing scales, or other peripheral devices commonly used in retail environments. All sales transactions are entered manually through the system's web-based interface, where authorized users select products, specify quantities, and complete transaction details using standard input forms. The system operates independently of specialized retail hardware and relies solely on conventional computer hardware and web browser functionality.

This delimitation was established to align the proposed system with the operational environment and technological capabilities of the reference auto parts sales and services business. At the time of the study, the business primarily relied on manual sales recording and did not utilize dedicated POS hardware in its daily operations. Consequently, integrating support for specialized devices would not directly address the immediate needs of the business and would introduce unnecessary development complexity.

Supporting POS hardware integration would require the implementation of additional software drivers, device communication protocols, hardware-specific application programming interfaces (APIs), and compatibility testing for multiple manufacturers and equipment models. Because different hardware vendors employ proprietary communication standards and device configurations, ensuring reliable compatibility across various POS devices would require extensive development, testing, and maintenance beyond the intended scope of this undergraduate capstone project.

Furthermore, acquiring and maintaining POS hardware represents a significant financial investment for many small businesses. The purchase of barcode scanners, receipt

printers, cash drawers, payment terminals, and related accessories increases implementation costs and may require additional technical support for installation, calibration, and maintenance. Since one of the objectives of ProfitLens is to provide an affordable and accessible financial monitoring solution for small businesses, excluding hardware dependencies allows the system to remain practical and economically feasible for organizations with limited financial and technological resources.

The manual sales entry process implemented in ProfitLens remains sufficient for the transaction volume and operational requirements of the reference business. Through intuitive data entry forms, searchable product selection, automatic computation of totals, and real-time inventory updates, users can efficiently record transactions without the need for specialized retail equipment. These features simplify transaction recording while maintaining the accuracy required for reliable financial monitoring and profit analysis.

The absence of barcode scanning functionality does not prevent accurate product identification because the system provides searchable product lists and organized product information that allow users to quickly locate and select the appropriate items. Likewise, although receipt printing is not supported, sales transactions are securely stored within the centralized database and can be reviewed later through transaction histories and financial reports. This approach satisfies the business's operational requirements while avoiding the additional complexity associated with printer configuration and receipt formatting.

Similarly, the system does not integrate with electronic payment terminals or digital payment gateways. Payment processing continues to follow the business's existing procedures, with ProfitLens serving only as a financial recording and monitoring platform. This separation ensures that the system remains focused on managing business financial data rather than processing customer payments or supporting payment authorization services.

Excluding POS hardware integration also contributes to easier system deployment and maintenance. Because the application requires only a web server, database, and standard web browser, installation can be completed without configuring hardware drivers or troubleshooting peripheral device compatibility issues. This simplified deployment process reduces technical barriers for small businesses and enables faster implementation with minimal infrastructure requirements.

Although hardware integration is beyond the scope of the current implementation, the modular architecture of ProfitLens provides opportunities for future enhancement. Subsequent versions of the system may incorporate barcode scanning for faster product identification, receipt printing for completed transactions, cash drawer control, electronic payment terminal integration, QR code support, and other POS hardware capabilities through appropriate software interfaces and device communication protocols. These enhancements could further improve transaction efficiency and customer service while building upon the financial monitoring functions already implemented.

Ultimately, this delimitation enables the study to concentrate on its primary objective of developing a reliable, cost-effective, and user-friendly financial monitoring platform that addresses the operational needs of the reference auto parts sales and services business. By excluding specialized POS hardware integration, the research maintains a realistic development scope while delivering a practical solution that effectively supports sales recording, inventory updating, expense tracking, profit computation, and financial reporting without requiring additional hardware investments or complex technical infrastructure.

Payroll Processing

While salaries and wages are trackable as an expense category, ProfitLens does not compute individual employee payroll, manage attendance records, or generate payslips. This delimitation recognizes that payroll processing involves complex regulatory requirements and specialized functionality that are beyond the scope of a profit monitoring platform. While salaries, wages, and other employee compensation can be recorded and categorized as business expenses, ProfitLens does not provide a comprehensive payroll processing module. The system does not compute individual employee salaries, overtime pay, holiday pay, night differential, commissions, bonuses, deductions, or net pay. Likewise, it does not manage employee attendance, work schedules, leave records, timesheets, or biometric attendance data, nor does it generate payslips, payroll registers, or payroll summaries. Instead, payroll-related expenditures are recorded solely as expense entries to ensure that labor costs are reflected accurately in the business's financial records and profit computations.

This delimitation was established to maintain the study's focus on financial monitoring and profit analysis rather than human resource and payroll administration. Although payroll expenses significantly affect business profitability, payroll management represents a distinct operational domain that requires specialized processes, regulations, and system functionalities. Incorporating a complete payroll system would considerably expand the scope of the project by requiring additional modules for employee management, attendance monitoring, compensation computation, benefits administration, and payroll reporting, which extend beyond the primary objectives of this research.

Furthermore, payroll processing in the Philippines is governed by various labor laws and government regulations that require accurate computation of mandatory employee benefits and statutory deductions. A comprehensive payroll system must account for contributions to the Social Security System (SSS), Philippine Health Insurance Corporation (PhilHealth), Home Development Mutual Fund (Pag-IBIG), and applicable withholding taxes administered by the Bureau of Internal Revenue (BIR). It must also comply with regulations concerning minimum wage rates, overtime compensation, holiday pay, premium pay, service incentive leave, and other labor standards. Ensuring full compliance with these evolving legal requirements would require continuous system updates and regulatory monitoring, making payroll management unsuitable for inclusion within the scope of the present study.

In addition, payroll processing requires the collection and storage of sensitive employee information, including personal identification details, employment records, salary rates, government identification numbers, bank account information, attendance history, and other confidential employment data. Managing such information introduces significant privacy, security, and legal responsibilities that necessitate enhanced access controls, encryption mechanisms, audit trails, and compliance with the Data Privacy Act of 2012 (Republic Act No. 10173). These additional requirements would substantially increase system complexity and development effort.

Despite the absence of payroll processing functionality, ProfitLens remains capable of supporting financial monitoring by allowing payroll-related expenditures to be recorded as operating expenses. Business owners can categorize salary and wage payments appropriately within the expense management module, enabling labor costs to be incorporated into overall expense reports, profit calculations, and financial analyses. This ensures that employee compensation is accurately reflected in the financial performance of the business without requiring detailed payroll administration.

The simplified approach also contributes to improved usability. Users are not required to maintain employee attendance records, define compensation structures, configure deduction rules, or perform complex payroll calculations within the system. Instead, they can focus on recording completed payroll payments as expense transactions, allowing the platform to remain intuitive and aligned with its primary objective of financial monitoring. This streamlined design minimizes administrative workload while providing sufficient financial information for business decision-making.

Although comprehensive payroll management is outside the scope of the current implementation, the modular architecture of ProfitLens allows for future enhancement. Subsequent versions of the system may incorporate employee management, attendance tracking, automated payroll computation, statutory deduction calculations, electronic payslip generation, leave management, and payroll reporting. Integration with biometric attendance devices or cloud-based human resource management systems may also be considered to provide a more comprehensive workforce management solution while maintaining compatibility with the existing financial monitoring features.

Ultimately, this delimitation enables the study to concentrate on developing a reliable and efficient profit monitoring platform tailored to the operational needs of a small auto parts sales and services business. By excluding specialized payroll processing functions, the research maintains a realistic development scope while ensuring that the implemented system effectively supports financial recording, expense management, profit computation, and business performance monitoring without the additional complexity associated with payroll administration and regulatory compliance.

Tax Filing and Compliance

The system does not generate **Bureau of Internal Revenue (BIR)**-compliant tax forms or reports, compute Value-Added Tax (VAT), withholding taxes, or other statutory tax obligations, nor does it integrate with government tax filing or electronic tax reporting

platforms. ProfitLens is designed primarily as a financial monitoring and profit tracking system rather than a comprehensive accounting or tax management solution.

This delimitation reflects the specialized and highly regulated nature of tax compliance in the Philippines, where tax computations, reporting requirements, and filing procedures are governed by laws and regulations that are subject to periodic revisions by the Bureau of Internal Revenue. Developing and maintaining a fully compliant tax management module would require continuous updates to accommodate changes in tax policies, rates, reporting standards, and electronic filing requirements, which are beyond the scope of the present study.

Instead, the system focuses on organizing and maintaining accurate records of sales, expenses, and profit-related transactions that can serve as supporting financial information for tax preparation. Business owners may use the reports generated by ProfitLens as reference materials when preparing financial statements or submitting records to accountants, bookkeepers, or tax professionals responsible for ensuring compliance with applicable tax regulations.

Furthermore, the system is not intended to replace professional accounting services or certified tax preparation software. Users remain responsible for ensuring that all tax computations, filings, and regulatory submissions comply with the requirements of the Bureau of Internal Revenue and other relevant government agencies. This limitation allows the study to concentrate on its primary objective of improving financial monitoring, business performance analysis, and operational decision-making while maintaining a manageable project scope consistent with the objectives of an undergraduate capstone project.

External Accounting System Integration

ProfitLens does not connect to or synchronize data with any third-party accounting application such as QuickBooks, Xero, or any cloud-based ERP system. This delimitation keeps the system self-contained and eliminates dependencies on external services or subscriptions. ProfitLens does not connect to, exchange data with, or synchronize financial records with any third-party accounting software or enterprise resource planning (ERP) platforms such as QuickBooks, Xero, SAP Business One, Oracle NetSuite, or other cloud-based accounting and financial management systems. The proposed system functions as a standalone web-based financial monitoring platform specifically developed to address the operational needs of the reference auto parts sales and services business. Consequently, all sales, expense, product, and profit data are stored and managed exclusively within the system's own database.

This delimitation was established to maintain a manageable development scope that aligns with the objectives, available resources, and timeframe of the study. Integrating external accounting platforms would require the implementation of application programming interfaces (APIs), authentication protocols, data synchronization mechanisms, compatibility testing, and ongoing maintenance to accommodate updates

introduced by third-party software providers. These additional technical requirements extend beyond the intended scope of this capstone project and would significantly increase system complexity.

Moreover, third-party accounting systems often require paid subscriptions, licensing agreements, internet-based services, and compliance with proprietary integration standards. Such dependencies could limit the accessibility and affordability of the proposed system for small businesses that may not have the financial resources to subscribe to commercial accounting platforms. By operating independently, ProfitLens remains a cost-effective solution that can be deployed without requiring additional software subscriptions or specialized technical infrastructure.

The absence of external accounting integration does not diminish the usefulness of the system for its intended purpose. ProfitLens provides comprehensive recording and monitoring of sales, expenses, profits, and related financial activities within a centralized database. The system generates organized financial reports that business owners can use for operational analysis, financial monitoring, and decision-making. These reports may also serve as reference documents that can be manually transferred into external accounting software when more advanced accounting or auditing functions are required.

Furthermore, maintaining a self-contained architecture simplifies system installation, configuration, maintenance, and troubleshooting. Because the platform does not rely on external services, users are less likely to experience disruptions caused by API changes, service outages, software version incompatibilities, or subscription expirations. This independence contributes to greater system reliability and allows businesses to continue using the platform without interruption or dependence on third-party providers.

The modular design of ProfitLens also allows for future expansion should integration become necessary. Although external accounting connectivity is not included in the current implementation, the system architecture has been developed in a manner that could accommodate future API-based integrations with accounting platforms, tax management systems, or enterprise resource planning solutions if the business requirements evolve. Such enhancements are recommended for future studies and subsequent versions of the system but remain outside the scope of the present research.

Ultimately, this delimitation enables the study to focus on its primary objective of providing an accessible, efficient, and user-friendly financial monitoring system tailored to the operational needs of small auto parts sales and services businesses. By prioritizing core financial recording and profit monitoring functions over complex third-party integrations, ProfitLens delivers a practical solution that effectively addresses the financial management challenges identified in the study while remaining technically feasible within the scope of an undergraduate capstone project. Additionally, external accounting platforms are designed primarily to support comprehensive bookkeeping, financial statement preparation, payroll management, tax compliance, accounts payable and receivable, and other advanced accounting functions. ProfitLens, on the other

hand, was intentionally developed as a specialized financial monitoring and profit analysis platform that focuses on the day-to-day recording of sales and expenses, real-time profit computation, and business performance visualization. By limiting the scope to these essential functions, the system provides a simpler and more intuitive user experience that is appropriate for small business owners who may not possess formal accounting knowledge or professional bookkeeping experience.

Another reason for excluding third-party accounting integration is to maintain data consistency and system stability throughout the implementation and evaluation phases of the study. Synchronizing information between multiple software platforms introduces additional challenges such as data mapping, conflict resolution, duplicate records, synchronization delays, and dependency on external application programming interfaces (APIs). Addressing these issues would require extensive testing under various operating conditions and would divert the research from its primary objective of developing and evaluating a profit monitoring solution specifically tailored to the operational requirements of the reference business.

Furthermore, different accounting software applications utilize varying database structures, financial reporting standards, and application interfaces. Developing universal compatibility with multiple accounting platforms would require separate integration modules for each software provider, significantly increasing development time, implementation costs, and long-term maintenance requirements. Since these accounting systems regularly introduce software updates and API modifications, maintaining compatibility would also require continuous monitoring and redevelopment beyond the duration of this research project.

The standalone nature of ProfitLens also enhances data ownership and operational independence. Because all financial records are stored within the system's centralized database, the business retains full control over its operational information without relying on external cloud services or third-party vendors. This approach minimizes the risks associated with vendor lock-in, subscription cancellations, service interruptions, or changes in third-party pricing policies. It also enables the system to continue functioning as intended even in environments where internet connectivity is unstable, provided that the web application remains accessible within the business's local infrastructure.

Despite the absence of automated accounting software integration, the reports generated by ProfitLens remain valuable sources of financial information for accountants, auditors, and financial consultants. Business owners may export or manually reference sales summaries, expense records, and profit reports when preparing financial statements, conducting business performance evaluations, or completing accounting procedures using separate accounting applications. As such, ProfitLens complements rather than replaces comprehensive accounting systems by serving as an operational financial monitoring tool that provides organized and accurate financial data for further processing when necessary.

Future researchers may consider extending the capabilities of ProfitLens by implementing secure API-based integration with widely used accounting platforms,

enabling automated data synchronization and reducing the need for manual data transfer. Such enhancements could support a more comprehensive financial management ecosystem while preserving the core monitoring and reporting functionalities developed in this study. However, these advanced integration features require additional technical expertise, security considerations, licensing compliance, and extensive interoperability testing, making them appropriate topics for future system enhancements rather than the present implementation.

Multi-Branch Management

The system is designed specifically for a single-location business and does not support the management of multiple branches or business locations under a unified platform. ProfitLens maintains a centralized database intended for one store only and does not provide features such as consolidated financial reporting, branch-specific user management, inter-branch inventory transfers, or comparative performance analysis across multiple locations. All sales, expense, product, and financial records are recorded and managed within the operational context of the reference business.

This delimitation reflects the current organizational structure and operational requirements of the reference auto parts sales and services business, which operates from a single physical location. Since the business does not maintain multiple branches or warehouses that require centralized coordination, the inclusion of multi-branch management features would not contribute to the immediate objectives of the study. Instead, the research focuses on addressing the existing financial monitoring challenges encountered within the current business environment.

Supporting multiple branches would require a substantially more sophisticated system architecture. Such functionality would involve implementing branch-specific databases or identifiers, centralized synchronization mechanisms, role-based access for different branch personnel, branch-level financial reporting, consolidated profit analysis, and secure communication between geographically separated locations. These additional requirements would significantly increase the complexity of system design, database management, security implementation, and software testing, making them beyond the intended scope of this undergraduate capstone project.

Furthermore, multi-branch operations introduce additional business processes such as inter-branch inventory transfers, stock balancing, branch-specific pricing, localized promotions, and separate financial accountability for each location. Managing these activities requires advanced inventory control, transaction reconciliation, and reporting mechanisms that are more appropriate for enterprise resource planning (ERP) systems or large-scale retail management solutions than for the financial monitoring platform proposed in this study.

By limiting the system to a single business location, ProfitLens is able to provide a streamlined and user-friendly interface that focuses on the essential financial monitoring functions required by the reference business. This simplified architecture improves system performance, reduces database complexity, minimizes administrative overhead, and allows users to perform routine financial recording tasks efficiently without navigating features that are unnecessary for their operational needs. The focused design also contributes to easier deployment, maintenance, and future system updates.

Although the current implementation does not support multi-branch management, the system has been developed using a modular architecture that allows future expansion. Additional modules for branch management, consolidated financial reporting, branch-specific dashboards, centralized user administration, and inter-branch inventory tracking may be incorporated if the business expands its operations or if future researchers choose to enhance the system. The existing database structure and application framework can serve as a foundation for implementing these capabilities with appropriate modifications.

The absence of multi-branch functionality does not affect the system's ability to achieve its primary objective of providing accurate and efficient financial monitoring for the reference business. Since all business transactions originate from a single operational location, the implemented features are sufficient to record, organize, analyze, and report financial information effectively. Users are able to monitor sales performance, manage expenses, track profits, and generate financial reports without the additional complexity associated with managing multiple business locations.

Ultimately, this delimitation enables the study to concentrate on developing a practical, reliable, and cost-effective financial monitoring solution that directly addresses the operational needs of a single-location auto parts sales and services business. By excluding enterprise-level multi-branch management features, the research maintains a realistic development scope while ensuring that the resulting system remains efficient, maintainable, and aligned with the actual requirements of the partner business.

Customer Relationship Management (CRM)

ProfitLens does not maintain a customer database, track individual customer purchase histories, or support customer communication functions. This delimitation focuses the system on financial monitoring rather than customer management, recognizing that CRM functionality represents a distinct domain with its own specialized requirements. ProfitLens does not include Customer Relationship Management (CRM) functionality such as maintaining a customer database, tracking individual customer purchase histories, managing customer profiles, recording customer preferences, or facilitating customer communication through email, text messaging, or other digital channels. The proposed system is designed primarily as a financial monitoring platform that emphasizes the recording, analysis, and reporting of sales, expenses, and profit rather than the management of customer relationships and marketing activities.

This delimitation was established to ensure that the study remains focused on addressing the financial management challenges identified in the reference business. While customer relationship management is an important aspect of business operations, it represents a separate functional domain that involves its own processes, data requirements, and technological considerations. Incorporating CRM capabilities would significantly broaden the scope of the project by requiring additional modules for customer registration, customer segmentation, loyalty program management, communication tools, customer feedback collection, and marketing campaign administration. These functionalities extend beyond the primary objectives of the present research.

Furthermore, implementing a comprehensive CRM system would require the collection and storage of customer personal information, including names, contact details, purchase histories, and other personally identifiable data. Managing such information introduces additional responsibilities related to data privacy, security, user consent, and compliance with applicable data protection regulations, including the **Data Privacy Act of 2012 (Republic Act No. 10173)**. Addressing these legal, ethical, and technical considerations would require additional security mechanisms and administrative controls that are outside the intended scope of this study.

The absence of CRM functionality does not limit the effectiveness of ProfitLens in achieving its intended purpose. The system records sales transactions based on products sold rather than customer identities, allowing business owners to monitor revenue, expenses, and profitability without maintaining detailed customer records. This approach simplifies the sales recording process, minimizes data entry requirements for staff, and enables faster transaction processing while still providing the financial information necessary for business monitoring and decision-making.

Additionally, excluding customer management features contributes to a simpler and more user-friendly system interface. Staff members can focus on recording financial transactions accurately and efficiently without navigating unnecessary customer-related functions. This streamlined design reduces system complexity, shortens the learning curve for new users, and minimizes the possibility of data entry errors associated with maintaining extensive customer information.

Although ProfitLens does not currently support CRM capabilities, its modular system architecture provides opportunities for future enhancement. Future versions of the system may incorporate customer registration, purchase history tracking, customer loyalty and rewards programs, service reminders, promotional notifications, customer satisfaction surveys, and sales analytics based on customer purchasing behavior. Such enhancements could help businesses strengthen customer relationships, improve customer retention, and develop more effective marketing strategies while complementing the existing financial monitoring functions.

The financial reports generated by ProfitLens may also serve as valuable inputs for future CRM implementations. By analyzing product sales, revenue trends, and purchasing patterns, businesses can identify opportunities for customer-focused

marketing initiatives even without maintaining individual customer profiles. Should the business require advanced customer management capabilities in the future, the existing financial data can complement a dedicated CRM solution or be integrated with an external customer management platform.

Ultimately, this delimitation enables the study to concentrate on its primary objective of developing an efficient, reliable, and user-friendly financial monitoring system tailored to the operational needs of a small auto parts sales and services business. By excluding CRM features, the research maintains a realistic and manageable development scope while delivering a specialized solution that effectively addresses the financial recording, profit monitoring, and reporting requirements of the reference business.

Full Warehouse Management

While product stock quantities are maintained and updated upon each sale, the system does not support purchase order management, supplier tracking, reorder-level alerts, or warehouse location tracking. This delimitation keeps the inventory management functionality focused on the core requirement of maintaining accurate stock levels for profit calculation purposes. While ProfitLens maintains and automatically updates product stock quantities whenever sales transactions are recorded, the system does not provide a comprehensive warehouse management solution. Specifically, it does not support purchase order management, supplier relationship management, goods receiving and inspection, reorder-level alerts, warehouse location mapping, barcode or RFID scanning, batch or lot tracking, expiration date monitoring, inventory cycle counting, or inter-warehouse stock transfers. The inventory functionality is intentionally limited to maintaining accurate stock balances necessary for financial monitoring and profit computation.

This delimitation was established to ensure that the system remains focused on its primary objective of improving financial monitoring and profitability analysis for the reference auto parts sales and services business. Although warehouse management is closely related to inventory control, it encompasses a much broader set of operational processes involving procurement, logistics, supplier coordination, storage optimization, and distribution management. Incorporating these functions would substantially increase the scope, complexity, and development requirements of the project beyond what is appropriate for the objectives of this undergraduate capstone study.

Implementing a complete warehouse management system would require the development of numerous additional modules, including supplier registration and evaluation, purchase requisition processing, purchase order generation, delivery scheduling, receiving verification, stock allocation, warehouse zoning, storage

location mapping, inventory movement tracking, and automated replenishment mechanisms. These capabilities involve sophisticated business workflows and database relationships that are typically found in enterprise inventory management or warehouse management systems rather than in specialized financial monitoring applications.

Furthermore, advanced warehouse management often depends on specialized hardware and supporting technologies such as barcode scanners, RFID readers, handheld inventory terminals, label printers, and mobile warehouse applications. Integrating these technologies would require additional software interfaces, hardware compatibility testing, network infrastructure, and user training, all of which would significantly increase implementation costs and technical complexity. Such requirements are not necessary for addressing the financial monitoring challenges identified in the reference business.

The exclusion of supplier management also reflects the intended focus of the proposed system. ProfitLens does not maintain supplier records, monitor supplier performance, compare supplier pricing, or automate procurement activities. Purchasing decisions and supplier coordination continue to be performed using the business's existing operational procedures. Instead, the system concentrates on accurately recording sales and updating available stock quantities to ensure that inventory values used in financial reports and profit calculations remain reliable.

Likewise, the system does not provide automated reorder-level notifications or inventory forecasting based on historical sales trends. Business owners remain responsible for monitoring stock availability and determining when products should be replenished. Although users can review current stock quantities through the inventory module, the system does not automatically generate purchase recommendations or procurement schedules. This limitation allows the research to prioritize the implementation of accurate financial recording and reporting functionalities while avoiding the additional complexity associated with inventory planning algorithms.

Despite these limitations, the inventory features implemented in ProfitLens are sufficient to support the primary objectives of the study. The system automatically deducts product quantities after each completed sale, maintains updated stock records, and ensures that inventory information remains synchronized with recorded sales transactions. This functionality helps reduce manual inventory updates, minimizes recording errors, and provides business owners with accurate information regarding current stock availability for operational monitoring and financial analysis.

The simplified inventory management approach also contributes to improved usability and maintainability. By focusing on essential inventory functions, the system offers an intuitive interface that can be easily understood by business owners and staff with limited technical expertise. Users can efficiently record transactions and monitor inventory levels without navigating complex warehouse management features that are unnecessary for the day-to-day operations of the reference business. This streamlined design reduces training requirements, improves operational efficiency, and minimizes the likelihood of user errors.

Although comprehensive warehouse management capabilities are outside the scope of the present study, the modular architecture of ProfitLens allows for future enhancement. Subsequent versions of the system may incorporate supplier management, automated purchase order generation, reorder point calculations, demand forecasting, warehouse location tracking, barcode-based inventory management, and advanced stock movement monitoring. These enhancements could transform the platform into a more comprehensive inventory and warehouse management solution while building upon the financial monitoring foundation established by the current implementation.

Ultimately, this delimitation enables the study to remain focused on its core objective of developing an efficient, reliable, and user-friendly financial monitoring system for a small auto parts sales and services business. By limiting inventory management to accurate stock quantity maintenance and automatic inventory updates, ProfitLens delivers the essential functionality required for precise profit computation and financial reporting while maintaining a realistic development scope consistent with the objectives, resources, and timeframe of the research.

Email Notifications

Although a forgot-password page is provided, the system does not send automated email notifications of any kind in its current implementation, as email server configuration is outside the deployment environment's scope. This delimitation reflects the technical requirements and costs associated with email server setup and maintenance. Although ProfitLens includes a **Forgot Password** page to assist users during the account recovery process, the current implementation does not support automated email notifications or email-based communication. The system does not send password reset links, account verification emails, login alerts, transaction confirmations, sales summaries, overspending notifications, report deliveries, or any other system-generated email messages. All system interactions and notifications are performed directly within the web application rather than through external email services.

This delimitation was established because configuring and maintaining an email delivery service falls outside the intended scope and deployment environment of the study.

Implementing email functionality would require the integration of external mail servers or third-party email service providers, configuration of the Simple Mail Transfer Protocol (SMTP), domain verification, authentication mechanisms such as SPF, DKIM, and DMARC, as well as continuous monitoring to ensure reliable message delivery. These additional technical requirements would increase the complexity of system development without directly contributing to the primary objective of improving financial monitoring and profit management.

Furthermore, many email service providers impose usage limitations, subscription fees, or security requirements that may not be practical for a small business or an undergraduate capstone project. Maintaining an email notification service also involves ongoing administrative responsibilities, including managing server credentials, preventing unauthorized access, monitoring delivery failures, addressing spam filtering issues, and updating configuration settings whenever service providers modify their policies. Such operational considerations extend beyond the resources and implementation timeframe allocated for this research.

The exclusion of automated email notifications also simplifies system deployment. Since ProfitLens operates as a self-contained web application, businesses can install and use the system without configuring external email servers, purchasing additional services, or depending on third-party communication platforms. This approach reduces installation complexity, minimizes deployment costs, and allows the system to function effectively in environments with limited technical infrastructure.

Although automated email functionality is not available, the system still provides users with immediate feedback through on-screen notifications, confirmation messages, validation prompts, and status indicators within the application interface. These built-in notifications inform users whether transactions have been successfully recorded, updated, or deleted, ensuring that essential operational feedback is delivered without relying on external communication channels. This design supports efficient system use while maintaining a simple and responsive user experience.

Regarding password recovery, the Forgot Password page serves as an interface for initiating the recovery process; however, it does not automatically transmit password reset links or temporary credentials via email. Password resets are intended to be handled manually by an authorized system administrator or business owner, depending on the organization's established account management procedures. This approach ensures that user accounts can still be recovered while avoiding the additional infrastructure required for automated email delivery.

The absence of email notifications does not affect the system's primary functions of recording sales, managing expenses, monitoring inventory, calculating profits, and generating financial reports. Business owners and staff can continue to access all financial information directly through the web application, where dashboards and reports provide real-time updates regarding business performance. Consequently, the lack of email communication does not diminish the effectiveness of ProfitLens as a financial monitoring platform.

The modular architecture of ProfitLens also provides opportunities for future enhancement. Subsequent versions of the system may incorporate secure email services for password recovery, account verification, scheduled financial reports, low-stock notifications, overspending alerts, monthly profit summaries, and other automated communications. Such enhancements could be implemented through integration with SMTP servers or cloud-based email service providers while incorporating appropriate authentication, encryption, and security measures to ensure reliable and secure message delivery.

Ultimately, this delimitation enables the study to concentrate on developing the core financial monitoring capabilities of ProfitLens while maintaining a practical and manageable implementation scope. By excluding automated email notifications, the research avoids unnecessary technical dependencies and operational costs, resulting in a streamlined, reliable, and cost-effective system that effectively addresses the financial management needs of the reference auto parts sales and services business.

1.5 Significance of the Study

The development and documentation of ProfitLens carries significance that extends beyond the immediate operational benefits to the reference business. The following stakeholder groups stand to benefit from the outcomes of this study:

1.5.1 Business Owners

For the primary user of the system, the small auto parts store owner, ProfitLens provides real-time profit visibility that was previously unavailable without laborious manual calculation. The platform's dashboard presents net profit for the current month alongside a growth indicator showing whether performance is improving or declining relative to the previous period. The full January-to-December revenue versus expense bar chart enables the owner to identify seasonal patterns, assess whether the business is on track relative to prior years, and make informed decisions about inventory investment, staffing levels, and discretionary spending. The categorized expense breakdown gives immediate visibility into which cost categories are consuming the largest proportions of revenue, enabling targeted cost management rather than undifferentiated expense reduction.

Beyond the immediate operational benefits, ProfitLens empowers business owners to transition from intuition-based to data-driven decision-making. This shift represents a fundamental improvement in management capability that can have lasting positive effects on business performance and sustainability. When business owners have access to accurate, timely, and comprehensive financial information, they can identify opportunities for growth, recognize threats to profitability, and allocate resources more effectively than is possible with manual record-keeping methods alone.

The significance of this empowerment is particularly pronounced for micro-enterprises and very small businesses that lack the resources to employ dedicated accounting staff

or engage external financial advisors. For these businesses, ProfitLens effectively provides a level of financial visibility and analytical capability that was previously accessible only to larger organizations with more substantial resources. For the primary users of the system—the owners and managers of small auto parts sales and services businesses—ProfitLens provides a centralized, real-time financial monitoring platform that significantly improves the visibility, accuracy, and accessibility of business information. Instead of relying on handwritten records, spreadsheets, or manual calculations to determine profitability, business owners can immediately access updated financial information through an interactive dashboard that presents key performance indicators, including the current month's net profit, revenue and expense summaries, monthly growth percentage, and the best-selling product. These features allow owners to evaluate the overall health of the business quickly and make informed decisions based on accurate financial data.

The platform's dashboard provides immediate visibility into the financial condition of the business by displaying net profit alongside a growth indicator that compares current performance with the previous reporting period. This comparison enables owners to determine whether profitability is improving, remaining stable, or declining without performing time-consuming manual computations. By identifying financial trends as they emerge, business owners can respond more quickly to operational challenges, adjust business strategies, and implement corrective measures before financial issues become more significant.

ProfitLens also presents a comprehensive January-to-December revenue-versus-expense chart that enables owners to evaluate financial performance across the entire calendar year. The visual comparison between income and expenditures makes it easier to identify seasonal sales patterns, periods of increased operational costs, and months in which profitability exceeded or fell below expectations. Historical financial trends support more effective budgeting, inventory planning, staffing decisions, and marketing initiatives by allowing management to anticipate recurring business conditions rather than reacting only after they occur.

Another significant benefit is the system's categorized expense analysis, which organizes expenditures into meaningful classifications such as office supplies, utilities, advertising, salaries and wages, and other operating costs. This detailed breakdown provides immediate insight into where business resources are being allocated and which expense categories consume the largest share of revenue. Rather than implementing broad cost-cutting measures, business owners can identify specific areas where expenses may be optimized while preserving investments that contribute directly to business growth and customer satisfaction.

Beyond improving day-to-day financial monitoring, ProfitLens empowers business owners to transition from intuition-based management toward evidence-based decision-making. Traditional manual record-keeping often limits financial analysis because business information is fragmented, outdated, or difficult to consolidate. By automatically recording transactions, calculating profits, and generating comprehensive financial reports, the system provides owners with reliable information that supports

more objective and informed business decisions. This transition enables management to evaluate the financial impact of operational changes, pricing adjustments, promotional activities, and inventory investments using measurable data rather than assumptions.

The availability of real-time financial information also enhances strategic planning. Business owners can continuously monitor sales performance, identify high-performing products, evaluate spending patterns, and measure profitability across different reporting periods. These insights support more effective forecasting, financial planning, and resource allocation by helping management determine where additional investment is justified and where operational improvements are necessary. As a result, businesses become better positioned to sustain long-term profitability and adapt to changing market conditions.

ProfitLens further improves operational efficiency by reducing the administrative burden associated with manual bookkeeping. Automated calculations, centralized data storage, and integrated financial reporting eliminate repetitive clerical tasks that previously consumed valuable business time. Instead of manually reconciling sales and expense records, owners can devote more attention to business development activities such as improving customer service, expanding product offerings, strengthening supplier relationships, and exploring new market opportunities. This increased efficiency contributes to higher overall productivity and more effective utilization of managerial resources.

The system also strengthens financial accuracy and record reliability. Because sales, expenses, inventory movements, and profit calculations are processed automatically within a centralized database, the likelihood of arithmetic errors, duplicate records, inconsistent reporting, and lost documentation is significantly reduced. Accurate financial records provide business owners with greater confidence when evaluating business performance, preparing budgets, planning future investments, or reviewing historical financial information.

Another important benefit is the accessibility of business information. Since ProfitLens is implemented as a web-based application, authorized users can access financial reports and dashboards through a standard web browser within the business environment without relying on paper files or standalone spreadsheet documents. Centralized access ensures that financial information remains organized, consistent, and readily available whenever managerial decisions need to be made, thereby improving responsiveness to operational and financial challenges.

The significance of these benefits is particularly pronounced for **micro-enterprises and small businesses** that often operate without dedicated accountants, financial analysts, or professional bookkeeping personnel. Many small business owners are responsible for managing daily operations, supervising employees, purchasing inventory, serving customers, and maintaining financial records simultaneously. ProfitLens reduces this administrative workload by automating essential financial monitoring functions,

effectively providing a level of financial visibility and analytical capability that would otherwise require substantially greater financial and human resources.

Moreover, the system supports stronger financial discipline by encouraging consistent recording of sales and expenses. Because financial reports and profit calculations depend directly on recorded transactions, business owners are encouraged to maintain complete and accurate financial records on a daily basis. This systematic approach promotes better financial accountability, improves record organization, and establishes reliable historical data that can support future business planning, loan applications, investment decisions, or business expansion initiatives.

The insights generated by ProfitLens also contribute to improved risk management. Early identification of declining revenues, rising operating costs, or shrinking profit margins allows business owners to implement timely corrective actions before financial problems become more severe. By monitoring key financial indicators regularly, management can recognize potential operational issues, evaluate their underlying causes, and implement targeted strategies to maintain business stability and profitability.

Furthermore, the comprehensive reports generated by the system provide valuable documentation that supports communication with business partners, suppliers, investors, or financial institutions when financial information is required. Organized revenue reports, expense summaries, and profit analyses demonstrate a higher level of financial management and business professionalism compared with manually maintained records, potentially improving the business's credibility when pursuing financing opportunities or strategic partnerships.

Ultimately, ProfitLens provides business owners with far more than a digital replacement for manual bookkeeping. It serves as a practical decision-support system that transforms routine financial transactions into meaningful business intelligence through automated calculations, centralized record management, interactive dashboards, analytical reports, and real-time performance monitoring. By improving financial visibility, enhancing operational efficiency, strengthening decision-making, reducing administrative effort, and promoting accurate financial management, ProfitLens enables small auto parts sales and services businesses to operate more effectively, improve profitability, and establish a stronger foundation for sustainable long-term growth.

1.5.2 Staff Members

Store staff benefit from a guided, form-based system for recording sales and expenses that replaces the informal and inconsistency-prone manual process. The system's automatic total calculation for each sales entry removes the arithmetic burden from staff and eliminates the most common source of data entry errors. The searchable product dropdown in the sales recording form reduces the time required to locate and select the correct product, particularly as the product catalog grows. Staff can complete routine financial recording tasks more quickly, with greater confidence in the accuracy of the records they are creating.

The standardization of data entry processes enabled by ProfitLens also improves consistency in how financial transactions are recorded across different staff members and time periods. This consistency is important for maintaining data quality over time and ensuring that financial reports accurately reflect actual business operations. Standardized records also make it easier for management to review transactions, identify discrepancies, and maintain reliable documentation for auditing and reporting purposes.

Furthermore, the system's intuitive interface design reduces the training burden associated with onboarding new staff members. The form-based data entry approach, combined with clear labeling and logical navigation, enables new employees to become productive with minimal training investment, reducing the operational disruption typically associated with staff turnover. This allows experienced staff to spend less time supervising routine recording tasks and more time focusing on customer service and other operational responsibilities.

ProfitLens also minimizes duplicate and incomplete entries through built-in input validation and required fields, ensuring that essential transaction details are captured before records are submitted. This feature reduces the need for manual corrections and follow-up, improving the overall reliability of financial data. Additionally, the system provides immediate confirmation when transactions are successfully recorded, giving staff confidence that their work has been properly saved.

Another significant benefit is the improved accessibility of historical transaction records. Instead of manually searching through paper logs or spreadsheets, staff can quickly retrieve previous sales and expense records using the system's search and filtering functions. This capability is particularly useful when responding to customer inquiries, verifying past transactions, or resolving discrepancies in daily financial records.

By automating repetitive tasks and streamlining the recording process, ProfitLens reduces administrative workload and allows staff to complete their responsibilities more efficiently. The reduction in manual effort not only increases productivity but also decreases stress associated with repetitive calculations and record verification. Overall, the system empowers staff with a reliable and user-friendly platform that enhances accuracy, efficiency, consistency, and confidence in carrying out daily financial recording activities.

1.5.3 Researchers and Future Developers

ProfitLens serves as a practical reference model for researchers interested in automated financial monitoring systems designed for small businesses in the Philippine context. The system's architecture, a PHP-MySQL client-server application with inline SVG chart generation, role-based access control implemented through PHP sessions, and a deliberately limited feature scope targeted at a specific operational problem, illustrates design decisions that are transferable to analogous contexts. Future researchers can use the system as a baseline against which more feature-rich or differently architected solutions can be compared, and can build upon the theoretical framework and literature review provided in Chapter 2 as a foundation for further scholarly inquiry.

The documentation of the complete development process, from requirements gathering through system design, implementation, testing, and evaluation, provides a comprehensive case study that can inform future capstone projects and applied research initiatives. The challenges encountered and solutions developed during the project offer practical insights that complement the theoretical knowledge acquired through coursework.

Additionally, the system's use of widely-available, open-source technologies demonstrates that effective financial monitoring solutions can be developed without significant software licensing costs, making the approach accessible to researchers and developers working with limited budgets. This accessibility is particularly relevant in the Philippine educational context, where resource constraints often limit the technologies that can be employed in student projects.

1.5.4 The Auto Parts Sales and Services Industry

More broadly, ProfitLens demonstrates the feasibility of developing cost-effective, fit-for-purpose financial monitoring solutions for small auto parts retailers in the Philippine market. By standardizing the data structures and reporting formats used for financial tracking within a single store, the platform establishes a basis upon which industry-level data collection and benchmarking could eventually be built. If similar systems were adopted across multiple stores, whether through replication of the ProfitLens model or through adaptation to different technology stacks, it would become possible to develop aggregate insights about the financial performance, cost structures, and demand patterns of the Philippine auto parts retail sector that are currently unavailable due to the fragmented and informal nature of financial record-keeping across the industry.

The potential for industry-level insight generation is particularly significant given the importance of the automotive aftermarket to the Philippine economy. As vehicle ownership continues to grow and the vehicle fleet ages, demand for auto parts and maintenance services will continue to increase, making the auto parts retail sector an increasingly important component of the national economy. Improved financial management capabilities across the sector could contribute to better business performance, higher employment, and greater economic contribution.

In addition, the adoption of digital financial monitoring systems such as ProfitLens encourages the modernization of business operations among small and medium-sized auto parts retailers. Many businesses in the sector continue to rely on handwritten records or basic spreadsheets that are prone to errors, data loss, and inconsistencies. By transitioning to a centralized digital platform, retailers can improve the accuracy, accessibility, and security of their financial information while reducing the time spent on administrative tasks. This digital transformation enables business owners to devote more attention to customer service, inventory management, and strategic planning rather than routine record-keeping.

ProfitLens also supports more informed business decision-making by providing timely access to financial reports and performance indicators. Store owners can identify seasonal sales trends, monitor high-performing products, evaluate operating expenses, and assess overall profitability using organized and up-to-date financial data. These insights allow businesses to make evidence-based decisions regarding inventory procurement, pricing strategies, promotional activities, and cost control measures, ultimately improving operational efficiency and competitiveness.

Furthermore, the system promotes better financial discipline within the industry by encouraging consistent recording of sales and expenses. Accurate and complete financial records help businesses prepare budgets, forecast future cash flow, and evaluate long-term financial performance. Such practices strengthen business sustainability by enabling retailers to anticipate financial challenges, allocate resources more effectively, and respond proactively to changes in market demand.

The implementation of systems like ProfitLens may also encourage greater adoption of digital technologies throughout the automotive aftermarket sector. As businesses become more familiar with computerized financial management, they may be more inclined to integrate complementary digital solutions such as inventory management, customer relationship management, supplier coordination, and electronic payment systems. This gradual digitalization can improve operational integration and enhance the overall efficiency of the industry's value chain.

Moreover, standardized financial records generated by systems such as ProfitLens can support stronger relationships between retailers and external stakeholders, including suppliers, financial institutions, investors, and government agencies. Reliable financial documentation facilitates loan applications, investment evaluations, tax preparation, and compliance with regulatory requirements, enabling businesses to access growth opportunities that may otherwise be limited by incomplete or inconsistent financial records.

Ultimately, ProfitLens serves not only as a financial monitoring tool for an individual business but also as a practical example of how affordable digital solutions can contribute to the modernization and professionalization of the Philippine auto parts sales and services industry. As more retailers embrace structured financial management practices, the sector can benefit from improved operational efficiency, enhanced business resilience, increased competitiveness, and stronger support for sustainable economic growth.

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1.6 Definition of Terms

The following terms are used throughout this documentation with the specific meanings defined below. Where terms have both a general meaning and a system-specific technical meaning, the system-specific meaning is provided.

ProfitLens: The web-based platform for smart profit monitoring and business insights that is the subject of this documentation. Built with PHP 8, MySQL, HTML5, CSS3, and JavaScript, and deployed on an Apache/XAMPP local server environment.

Dashboard: The system's primary landing page after login (dashboard.php), which displays real-time financial summaries including net profit for the current month, growth percentage relative to the prior month, the best-selling product, expense breakdown, and revenue trend visualizations.

Sales Module: The system module implemented in profit.php that enables admin users to record sales transactions by selecting a product from a searchable dropdown, entering a sale quantity, and specifying a sale date. The module automatically computes the transaction total from the product's stored selling price and deducts the sold quantity from the product's stock record.

Expense Tracking: The system module implemented in expenses.php that provides functionality for adding, editing, and deleting business expense records, categorized by

type (Office Supplies, Advertising, Utilities, Salaries and Wages, and others). The module maintains real-time monthly and annual expense totals.

Revenue Analysis: The system module implemented in revenue.php that presents visual representations of monthly revenue trends, year-over-year performance comparisons, and a ranked listing of top-performing products by units sold and total revenue.

Financial Reports: The system module implemented in reports.php that presents a comprehensive January-to-December grouped bar chart comparing revenue and expenses, a tabular profit-and-loss summary, and a detailed monthly expense breakdown report. Supports year selection for historical review.

Product Management: The system module implemented in products.php that provides full create, read, update, and delete (CRUD) functionality for product records, each of which carries attributes including name, category, selling price, cost price, and current stock quantity.

Authentication: The process by which the system verifies a user's identity before permitting access to protected pages. ProfitLens implements authentication through a login form (index.php) that validates submitted credentials against hashed passwords stored in the users database table using PHP's password_verify() function.

Session: PHP's built-in mechanism for maintaining state across multiple HTTP requests from the same user, implemented via the \$_SESSION superglobal. ProfitLens stores the authenticated user's user_id, email, and role in the session upon successful login, and checks the session on every protected page to verify continued authorization.

Net Profit: The financial metric computed by the system as: $\text{Net Profit} = \text{Sum of Sales Totals} - \text{Sum of Expense Amounts}$ for a given period. Displayed on the dashboard, profit module, and reports module. A positive value indicates a profitable period; a negative value indicates that expenses exceeded revenue.

formatMoney(): A PHP helper function defined in includes/config.php that formats a numeric amount as a Philippine Peso currency string. Syntax: formatMoney(\$amount) returns a string in the form ₱15,000.00. Used consistently across all modules wherever monetary amounts are displayed to the user.

Prepared Statement: A database query technique in which SQL commands are compiled separately from user-supplied data values, which are then bound to the compiled query as parameters. ProfitLens uses MySQL's prepared statement implementation (via PHP's MySQLi extension) throughout the system to prevent SQL injection attacks.

Admin Role: The user role designation (role = 'admin' in the users table) that grants full access to all system modules, including the ability to add, edit, and delete products, sales, and expenses. Non-admin users who attempt to access admin-protected pages are redirected to an access-denied page by the requireAdmin() function.

XAMPP: A cross-platform local web server environment bundling Apache (web server), MariaDB/MySQL (database server), and PHP. ProfitLens was developed and tested on XAMPP and is designed for deployment in XAMPP or any equivalent Apache/PHP/MySQL environment.

profit_lens Database: The MySQL database that stores all system data. Contains four primary tables: users, products, sales, and expenses. All system modules interact with this database exclusively through the `getDB()` connection function defined in `includes/config.php`.

SME (Small and Medium Enterprise): In the Philippine context, as used in this document, refers to businesses with fewer than 200 employees and annual revenue below the threshold defined by the Philippine Department of Trade and Industry. The auto parts store that serves as the reference business for ProfitLens is classified as a micro-enterprise by this definition.

DFD (Data Flow Diagram): A graphical representation of how data moves through a system, showing sources, destinations, processes, and data stores. ProfitLens is documented using DFD Level 0 (context diagram showing high-level interactions) and DFD Level 1 (detailed process diagram) conventions.

ERD (Entity Relationship Diagram): A visual representation of the relationships between the entities (tables) in a database. The ProfitLens ERD defines the structure of and relationships among the users, products, sales, and expenses tables.

Note on Currency Formatting: All monetary amounts presented in system interfaces and in this documentation are expressed in Philippine Pesos (PHP), formatted using the `formatMoney()` helper function. This formatting is applied consistently across all modules and reports to ensure that all financial figures are immediately recognizable as peso-denominated values.

With the conceptual and definitional groundwork established in this chapter, the following chapter reviews the theoretical and empirical literature that informed the design of ProfitLens, situating the system within the broader scholarly conversation about financial automation, SME digital transformation, and the specific challenges of the Philippine small business context.

1.7 Research Methodology Overview

This section provides a high-level overview of the research methodology employed in this study, with detailed methodology discussion reserved for Chapter 3. The research followed a systematic approach that began with problem identification through observation and interviews at the reference business, followed by theoretical framework development through comprehensive literature review, system design based on identified requirements and established design principles, iterative development using agile methodology, and systematic testing and evaluation.

The research team comprised five BSIT third-year students from PHINMA University of Iloilo, enrolled in the Applied Analytics in Business for IT (BAM) course during the Second Semester of 2025-2026. Each team member contributed specific expertise to the project: Ojeda, Dee Lord served as project lead and primary systems analyst; Abolucion, Marc Emmanuel contributed backend development expertise; Echalar, Wesley focused on database design and implementation; Panizales, Kent Justin specialized in frontend development and user interface design; and Dolorfino, Jhoana Angela managed documentation quality assurance and user acceptance testing coordination.

The study employed a mixed-methods approach combining qualitative data gathered through interviews and observation with quantitative analysis of business operations and system performance metrics. This approach was chosen to ensure that the resulting system would address genuine business needs identified through direct engagement with the target user population, while also meeting technical standards validated through systematic testing.

Data collection occurred over a period of approximately four months, beginning with initial site visits and requirements gathering sessions at the reference business in Iloilo City, continuing through the development phase with regular progress reviews and feedback sessions, and concluding with formal usability testing and system evaluation sessions. The extended timeline allowed for iterative refinement of both the system design and the documentation based on ongoing feedback from the business owner and staff.

1.8 Ethical Considerations

The research was conducted with full informed consent from the business owner and all participating staff members. The reference business owner, French Joy Salvador, provided written consent for the research team to observe business operations, conduct interviews with staff, analyze existing financial records, and deploy the resulting system in the business environment.

All financial data accessed during the research process was treated with strict confidentiality. No individual transaction records, specific revenue figures, or other sensitive financial information were included in this documentation or any public presentation of the research findings. The data presented in this documentation has been generalized to protect the business's competitive interests while still accurately representing the nature and scope of the operational challenges addressed by the system.

The research team ensured that participation in the study was entirely voluntary and that no staff member would face any negative consequences for declining to participate in interviews, observation sessions, or usability testing activities. In practice, all invited participants agreed to take part in the research activities, and the collaborative relationship between the research team and the business staff remained positive throughout the project duration.

The system was designed with data protection principles in mind, including password-based authentication, role-based access control, and session management to prevent unauthorized access to financial data. No personal data from the business owner or staff members was collected or stored by the system beyond the login credentials necessary for system operation.

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The research was conducted with full informed consent from the business owner and all participating staff members. The reference business owner, French Joy Salvador, provided written consent for the research team to observe business operations, conduct interviews with staff, analyze existing financial records, and deploy the resulting system in the business environment. Prior to data collection, the objectives, scope, procedures, expected outcomes, and intended use of the research were clearly explained to all participants to ensure that their participation was based on a thorough understanding of the study.

All financial data accessed during the research process was treated with strict confidentiality. No individual transaction records, specific revenue figures, or other sensitive financial information were included in this documentation or any public presentation of the research findings. The data presented in this documentation has been generalized to protect the business's competitive interests while still accurately representing the nature and scope of the operational challenges addressed by the system. Access to collected information was restricted to the research team, and all research materials were handled responsibly to prevent unauthorized disclosure.

The research team ensured that participation in the study was entirely voluntary and that no staff member would face any negative consequences for declining to participate in interviews, observation sessions, or usability testing activities. Participants were informed that they could withdraw from the study at any stage without penalty or obligation to provide a reason. In practice, all invited participants agreed to take part in the research activities, and the collaborative relationship between the research team and the business staff remained positive throughout the project duration.

The system was designed with data protection principles in mind, including password-based authentication, role-based access control, and session management to prevent unauthorized access to financial data. User permissions were assigned according to designated roles, ensuring that only authorized personnel could access sensitive financial information and perform administrative functions. These security measures help maintain the confidentiality, integrity, and availability of the system's data.

No personal data from the business owner or staff members was collected or stored by the system beyond the login credentials necessary for system operation. The system was intentionally designed to minimize the collection of personally identifiable

information, following the principle of data minimization. Only information necessary for user authentication and system functionality was retained, reducing potential privacy risks.

The research team also adhered to the principles of honesty, integrity, and transparency throughout the development and evaluation of the system. Research findings, system capabilities, and identified limitations were reported accurately without exaggeration or misrepresentation. Any challenges encountered during system development and testing were documented objectively to ensure that the study provides a truthful account of the project's outcomes.

To maintain fairness and prevent bias, feedback from the business owner and participating staff members was collected and considered during system evaluation. Suggestions and concerns raised by users were treated objectively and incorporated into system improvements whenever technically feasible. This participatory approach ensured that the developed system reflected the actual operational needs of the business rather than solely the assumptions of the researchers.

Furthermore, the researchers exercised professional responsibility by ensuring that system deployment did not disrupt the normal operations of the business. Data collection activities, interviews, and system testing were scheduled in coordination with the business owner to minimize interference with daily operations. Any issues identified during testing were promptly addressed to avoid inconvenience to system users.

Finally, the study was conducted in accordance with accepted academic research ethics, emphasizing respect for participants, protection of confidential information, responsible handling of research data, and proper acknowledgment of all referenced sources. By upholding these ethical principles throughout the research process, the study sought to protect the rights and interests of all participants while ensuring the credibility, reliability, and integrity of the research findings.

1.9 Document Structure Overview

This documentation is organized into seven chapters plus appendices, each serving a specific purpose in communicating the research process and findings:

Chapter 1 (Introduction) establishes the research context, defines the problem space, states the research objectives, defines the scope and delimitations, identifies the stakeholders who benefit from the research, and defines key terms used throughout the documentation.

Chapter 2 (Review of Related Literature and Studies) presents the theoretical foundations, reviews relevant foreign and local research, establishes the conceptual framework, and identifies the research gaps that ProfitLens addresses.

Chapter 3 (Methodology) details the research design, development methodology, data gathering techniques, system architecture, tools and technologies, database design, testing procedures, and evaluation criteria.

Chapter 4 (System Design and Implementation) presents the detailed system design including database schema, user interface mockups, data flow diagrams, and implementation highlights for each system module.

Chapter 5 (Results and Discussion) presents the outcomes of system testing, usability evaluation, and performance assessment, discussing the findings in relation to the research objectives.

Chapter 6 (Conclusions and Recommendations) summarizes the research findings, states the conclusions drawn from the evidence, acknowledges the study's limitations, and provides recommendations for future research and system enhancement.

The Appendices provide supplementary documentation including the complete user manual, installation guide, test case documentation, SQL schema definitions, and full listings of related studies reviewed during the research process.

This organizational structure follows the conventions of academic capstone documentation in the Philippines and ensures that readers can easily locate specific information, evaluate the research methodology and findings, and assess the contribution of the study to the body of knowledge in information systems and small business technology adoption.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Introduction

This chapter presents the theoretical foundation and empirical research that shaped the design and development of ProfitLens. It reviews relevant literature on technology acceptance, systems theory, financial monitoring, and small business automation, both locally and internationally. The chapter also identifies research gaps that ProfitLens aims to address, positioning the system within the broader scholarly conversation on SME digital transformation.

The review of related literature serves multiple purposes in the context of this study. First, it establishes the theoretical frameworks that informed the design decisions made during system development, providing a scholarly foundation for the system's architecture and functionality. Second, it situated the study within the existing body of knowledge regarding financial monitoring systems, SME automation, and technology adoption, identifying both established findings and research gaps. Third, it provides evidence-based justification for the design choices reflected in ProfitLens, demonstrating that the system's approach aligns with established best practices and addresses documented needs in the target user population.

The literature review encompasses several distinct but interconnected domains of scholarly inquiry. Theoretical frameworks, including the Technology Acceptance Model (TAM) and Systems Theory, provide conceptual foundations for understanding how end-users adopt new information systems and how business components interact to produce meaningful outcomes. Research on the evolution of financial monitoring systems provides historical context for understanding the trajectory from manual to digital record-keeping. Studies on SME automation, both foreign and local, provide empirical evidence regarding the benefits and challenges of digital financial

management in small business contexts. Finally, the identification of research gaps establishes the specific scholarly contribution that ProfitLens aims to make within this broader knowledge landscape.

2.2 Theoretical Framework

The development of ProfitLens is anchored on established theories that explain the transition of business operations from manual processes to digital automation. These frameworks helped the research team understand how end-users adopt new systems and how business components interact to produce meaningful outcomes. The theoretical foundations applied in this study provide both explanatory power for understanding current phenomena and prescriptive guidance for system design.

2.2.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), originally developed by Fred Davis in 1989, provides a theoretical foundation for understanding how end-users, specifically store owners and staff, adopt new information systems. According to TAM, two key factors determine technology adoption: Perceived Usefulness (PU), which refers to the degree to which a user believes that using a particular system will enhance their job performance, and Perceived Ease of Use (PEOU), which refers to the degree to which a user believes that using the system will be free of effort (Davis, 1989; Venkatesh & Davis, 2000).

TAM has been extensively validated across diverse technology adoption contexts, making it one of the most widely cited theoretical frameworks in information systems research. The model's strength lies in its parsimony and its focus on the two factors that most directly influence user adoption decisions. Numerous empirical studies have confirmed that Perceived Usefulness and Perceived Ease of Use are significant predictors of technology acceptance across different user populations, technology types, and cultural contexts (Venkatesh & Bala, 2008).

In the context of ProfitLens, the system was designed to maximize perceived usefulness by providing automated profit calculation, visual financial summaries, and real-time expense monitoring, features that directly address the pain points identified in the auto parts store. The interface was kept simple and intuitive to enhance perceived ease of use, ensuring that staff with limited technical expertise could operate the platform without extensive training. These design considerations align with TAM's core principles, which suggest that systems perceived as both useful and easy to use are more likely to achieve widespread adoption (Davis, 1989).

Research by Najib and Fahma (2020) applied an extended TAM framework to examine digital payment system adoption among small and medium enterprises in Indonesia, finding that perceived ease of use and perceived usefulness were significant determinants of adoption intention, alongside trust and attitude toward the technology. This finding is directly relevant to the ProfitLens context, as it demonstrates that TAM's core constructs apply to technology adoption decisions among SME operators in developing country contexts similar to the Philippines.

Similarly, Ifinedo (2011) investigated internet and e-business technologies acceptance in Canadian SMEs using a framework based on the Technology-Organization-Environment (TOE) framework, finding that perceived benefits (closely related to TAM's perceived usefulness) and management commitment were significant predictors of technology acceptance. While the Canadian context differs from the Philippine setting, the underlying finding regarding the importance of perceived benefits in driving technology adoption is broadly applicable.

The application of TAM principles in ProfitLens design is reflected in several specific design decisions. The system's dashboard provides immediate visibility into the most important financial metrics, demonstrating usefulness from the first interaction. The form-based data entry approach minimizes the effort required to record transactions, supporting ease of use. The visual reporting capabilities transform raw data into intuitive charts and graphs, enhancing perceived usefulness by making financial information accessible and actionable. Together, these design elements create a system that is intended to be perceived as both useful and easy to use, consistent with TAM's predictions regarding technology adoption.

2.2.2 Systems Theory in Business

Systems theory, originally proposed by Ludwig von Bertalanffy in 1968, provides a complementary framework for understanding business operations as interconnected components. In this model, a business is viewed as a system composed of interdependent parts that work together toward common goals. Inputs such as daily sales, inventory, and operating expenses are processed to produce outputs, including financial reports, profit summaries, and strategic decisions. Feedback loops allow business owners to evaluate performance and make adjustments (Johnson & Johnson, 2004).

Systems theory is particularly relevant to the design of financial monitoring systems because it emphasizes the interconnected nature of business processes and the importance of feedback mechanisms for effective management. A business that lacks systematic feedback on its financial performance is operating, in systems theory terms, as an open-loop system that cannot self-correct based on outcomes. The introduction of automated financial monitoring through ProfitLens effectively closes this feedback loop, enabling the business to adjust its operations based on real-time financial performance data.

ProfitLens was designed under this systems-theoretic perspective. The system treats sales and expenses as inputs, processes them through automated calculation and aggregation, and produces visual outputs in the form of dashboards and reports. The feedback loop is completed when users review these outputs to make informed business decisions. This framework ensures that the system functions not as an isolated data entry tool, but as an integrated component of the business's operational ecosystem.

The systems theory perspective also informs the understanding of how ProfitLens interacts with other business processes. Sales recording in ProfitLens affects inventory

records, which in turn affects purchasing decisions, which affect future sales capacity. Expense tracking affects profit calculations, which affect business planning, which affects future expense allocation. These interconnections demonstrate the systems nature of business operations and underscore the importance of integrated information systems that capture and reflect these relationships.

Research by Hevner and Chatterjee (2020) on design science research in information systems emphasizes the importance of understanding the organizational context in which systems operate, including the interactions between the system and its environment. This perspective aligns with the systems theory approach adopted in ProfitLens design, which considers the system as part of the broader business ecosystem rather than as an isolated artifact.

2.2.3 Diffusion of Innovation Theory

Diffusion of Innovation Theory, developed by Everett Rogers in 2003, provides additional theoretical insight into how new technologies spread through populations of potential adopters. According to Rogers, the adoption of innovation is influenced by five key attributes: relative advantage (the degree to which the innovation is perceived as better than what it replaces), compatibility (the degree to which the innovation fits with existing values and practices), complexity (the degree to which the innovation is perceived as difficult to understand and use), trialability (the degree to which the innovation can be experimented with before commitment), and observability (the degree to which the results of the innovation are visible to others).

These attributes are directly relevant to the adoption of ProfitLens by small auto parts retailers. The system's relative advantage over manual record-keeping is evident in its automated calculations, real-time reporting, and visual analytics capabilities. Its compatibility with existing business practices is reflected in its support for the same types of transactions and records that businesses already maintain. Its low complexity is reflected in its simple, intuitive interface design. Its trialability is supported by the web-based architecture that enables access without permanent installation. Its observability is enhanced by the visual dashboards and reports that make the system's benefits immediately apparent.

Research on technology adoption in developing country contexts has found that Diffusion of Innovation attributes are significant predictors of technology acceptance among SME operators (Shahadat et al., 2023). This finding supports the relevance of Rogers' framework for understanding the adoption dynamics that ProfitLens may encounter in the Philippine auto parts retail context.

2.2.4 Resource-Based View (RBV)

The Resource-Based View (RBV) of the firm, as articulated by various scholars including Barney (1991) and further developed in the context of digital transformation by Vrontis et al. (2022), provides a framework for understanding how information systems can create competitive advantage for small businesses. According to RBV, firms can achieve sustainable competitive advantage by acquiring and deploying resources that are valuable, rare, inimitable, and non-substitutable.

In the context of small business financial management, data itself can be viewed as a strategic resource. Businesses that systematically collect, organize, and analyze their financial data develop informational capabilities that are difficult for competitors to replicate, particularly when competitors continue to rely on informal, manual record-keeping methods. ProfitLens enables the systematic accumulation of structured financial data that can be analyzed to generate insights about customer preferences, demand patterns, and operational efficiency, creating informational resources that support better decision-making and potentially superior business performance.

Research by Teng et al. (2022) found that digital transformation, including the adoption of digital technologies for business operations, positively correlates with SME performance in the Chinese context. This finding supports the RBV perspective that digital capabilities represent valuable resources that can enhance business performance, and it provides empirical grounding for the expectation that ProfitLens adoption may contribute to improved business outcomes for the reference business.

2.3 The Evolution of Financial Monitoring Systems

The Evolution of Financial Monitoring Systems

Financial monitoring systems—the operational frameworks, tools, and platforms that businesses and institutions use to record, track, analyze, and safeguard financial transactions—have undergone a significant technological transformation over the past several decades. What began as paper-bound bookkeeping has evolved into automated, real-time analytics platforms capable of delivering instant operational visibility and dynamic risk oversight.

1. The Manual Era: Paper-Based Recording and Deferred Auditing

Historically, small enterprises and commercial establishments relied entirely on manual double-entry ledger books, paper receipts, and physical sales logs. Financial monitoring in this era was strictly **post-hoc and reactive**—business owners and accountants could only evaluate profit margins, cash flow, and cost variances days, weeks, or months after transactions occurred during end-of-period reconciliations. While simple and low-cost, manual tracking suffered from inherent vulnerabilities:

- High susceptibility to human calculation errors and omitted entries.
- Physical deterioration or loss of records.
- Complete absence of real-time visibility needed for proactive decision-making.

2. Early Computerization and Desktop Spreadsheets

The advent of personal computing introduced electronic spreadsheets (such as Microsoft Excel and Lotus 1-2-3) and standalone desktop accounting software. This shift digitized record-keeping, automated basic arithmetic functions, and allowed

businesses to maintain electronic transaction histories. However, early digital adoption remained constrained:

- **Data Silos:** Information lived locally on single computers, creating risks of hardware failure or data corruption without centralized cloud backups.
- **Manual Data Entry:** Systems still depended heavily on manual data keying, keeping human error rates relatively high.
- **Lack of Integration:** Sales terminals, inventory logs, and expense ledgers operated independently, requiring periodic batch exports and manual consolidation.

3. Rule-Based Systems and Relational Databases

With the growth of relational database management systems (RDBMS) such as MySQL, PostgreSQL, and SQL Server, financial tools transitioned from static spreadsheets to dynamic database-driven applications. Businesses implemented deterministic, rule-based systems to organize financial records across structured tables. These platforms enabled automated inventory cross-referencing, standardized reporting, and basic threshold alerts (e.g., flagging expenditures exceeding fixed budgets). Despite these gains, traditional rule-based platforms required frequent manual updates to keep pace with evolving operational needs.

4. Cloud Computing and Web-Based Financial Platforms

The shift toward cloud architecture and web-based stacks (e.g., PHP-MySQL, Node.js, and cloud-hosted microservices) democratized access to enterprise-grade financial tools for small and medium enterprises (SMEs). Web-based platforms eliminated hardware-bound software installations, allowing owners and managers to securely access financial data across multiple locations and devices.

Key advancements of this era include:

- **Centralized Data Storage:** Real-time database syncing ensured a single source of truth for all sales, expenses, and profit records.
- **Visual Analytics & Dashboards:** Raw numerical entries were automatically transformed into interactive charts, profit trend lines, and visual performance summaries accessible to non-technical users.
- **System Integration:** Point-of-Sale (POS) transactions, inventory adjustments, and expense logs became interconnected, automatically updating profit margins as sales occurred.

5. Automated Intelligence and Predictive Analytics

Modern financial monitoring systems leverage API integrations, automated data validation algorithms, and machine learning models to transition from simple record-keeping to **predictive decision support**. Current systems continuously monitor

cash flow dynamics, detect anomalous spending, auto-calculate dynamic product profit margins, and project future inventory reorder requirements based on historical demand patterns.

Timeline Summary

Manual Ledgers & Physical Logs

Pre-1970s

Reliance on paper-based double-entry bookkeeping, physical receipts, and delayed end-of-month manual reconciliations.

Desktop Spreadsheets & Isolated Software

1980s – 1990s

Introduction of electronic spreadsheets and desktop accounting tools. Digitized calculations but left data isolated in single-device silos.

Relational Databases & Rule-Based POS

2000s

Adoption of RDBMS (MySQL, SQL Server) enabling structured data modeling, basic barcode integration, and automated threshold alerts.

Cloud-Based & Responsive Web Platforms

2010s – Present

Deployment of web-based architecture (e.g., PHP-MySQL, XAMPP, SaaS) delivering real-time profit calculations, remote access, and visual reporting dashboards.

AI-Driven & Predictive Analytics

Emerging Era

Integration of machine learning and autonomous tracking agents for dynamic pricing, automated anomaly detection, and predictive stock forecasting.

2.3.1 Traditional vs. Digital Ledgers

Traditional financial record-keeping, commonly practiced in small businesses across the Philippines, relies on manual methods such as handwritten receipts, notebooks, and basic spreadsheets. While these methods are simple and require no technical infrastructure, they are prone to several well-documented limitations. Research indicates that manual record-keeping often leads to human arithmetic errors, inconsistent data summaries, duplicate entries, and incomplete records (Laudon &

Laudon, 2020). Furthermore, manual systems lack validation mechanisms, meaning that errors may go undetected until they propagate into larger financial discrepancies.

The limitations of manual record-keeping are not merely theoretical; they have been documented extensively in both academic research and practical assessments of small business operations. Studies of SME financial management practices in developing countries have consistently identified manual record-keeping as a significant barrier to business performance and growth (McKinney & Yoos, 2010). The time required for manual data entry, calculation, and reconciliation represents a substantial opportunity cost for small business owners who could otherwise devote that time to revenue-generating activities or strategic planning.

Digital ledgers, by contrast, enforce structured data entry through form-based interfaces that require specific fields before submission. Digital systems also support automated calculations, ensuring mathematical accuracy and eliminating the need for manual summation. Studies on SME financial management have shown that businesses transitioning from manual to digital ledgers experience significant improvements in data accuracy, reporting speed, and decision-support capability (McKinney & Yoos, 2010). ProfitLens leverages these advantages by providing a structured, form-based interface that ensures data integrity while maintaining the simplicity needed for small business operations.

The transition from manual to digital financial record-keeping represents a fundamental shift in how financial information is captured, stored, and utilized. Digital systems create persistent, searchable databases that can be queried to generate reports, identify trends, and support analysis in ways that are impossible with manual records. This transformation enables a shift from reactive financial management, where problems are identified after they have already impacted business performance, to proactive financial management, where potential issues are identified and addressed before they escalate.

2.3.2 Web-Based Architecture (PHP/MySQL)

Modern web-based financial systems offer several advantages over traditional desktop applications. Web platforms provide centralized data storage, allowing multiple users to access the same information from different locations without installing software on each device. These platforms also support real-time data access, enabling business owners to view up-to-date financial summaries at any time (Laudon & Laudon, 2020).

PHP and MySQL remain among the most widely used technologies for web-based financial applications, particularly in small and medium enterprise (SME) contexts. PHP's server-side scripting capabilities allow for dynamic content generation, while MySQL's relational database model supports efficient data storage and retrieval (MySQL AB, 2022; PHP Documentation, 2023). Additionally, PHP's support for prepared statements provides a robust defense against SQL injection attacks, one of the most common security threats in web applications (OWASP Foundation, 2021). ProfitLens was developed using PHP 8 and MySQL to ensure reliability, security, and ease of deployment in the target operational environment.

The choice of PHP-MySQL technology stack for ProfitLens reflects several strategic considerations beyond mere technical capability. First, PHP and MySQL are open-source technologies with no licensing costs, making them accessible for development projects with limited budgets. Second, the technologies have extensive documentation, large developer communities, and proven track records in production environments, ensuring that solutions developed with these technologies are reliable and maintainable. Third, the XAMPP development environment provides a complete, integrated development and deployment platform that simplifies both the development process and subsequent deployment to production environments.

Research by Prabowo and Wiguna (2021) demonstrated the effectiveness of web-based information systems using agile development methods for SME financial management in the automotive service context. Their study, conducted in Indonesia, found that web-based systems built with PHP and MySQL could effectively address the financial management challenges faced by small automotive businesses, supporting the technology choices made for ProfitLens.

2.3.3 The Role of Visualization in Financial Management

The importance of data visualization in financial management has been increasingly recognized in both academic research and practice. Visual representations of financial data, including charts, graphs, and dashboards, can transform raw numbers into intuitive insights that are accessible to non-technical users. Research on information visualization has demonstrated that visual representations can improve comprehension speed, enhance pattern recognition, and support more effective decision-making compared to tabular data presentations alone (Laudon & Laudon, 2020).

In the context of small business financial management, visualization capabilities are particularly valuable because they enable business owners with limited financial expertise to understand and utilize financial information effectively. A well-designed chart showing monthly revenue trends, for example, can communicate performance patterns more quickly and clearly than a table of monthly figures, enabling faster identification of trends and anomalies that may require attention.

ProfitLens incorporates visualization capabilities through its inline SVG chart generation functionality, which creates bar charts, line charts, and other visual representations directly in the user's web browser. This approach provides the benefits of data visualization while avoiding the complexity and external dependencies associated with third-party charting libraries. The self-contained nature of this approach ensures that visualization capabilities are always available regardless of internet connectivity or third-party service status.

2.4 Foreign Studies on SME Automation

International research on SME automation provides strong evidence that digital tools improve business efficiency, financial accuracy, and decision-making capability. Across different countries and industries, studies consistently show that small and medium-sized enterprises benefit from adopting automated systems because these tools reduce manual workload,

improve the reliability of records, and support faster responses to business problems. These findings are highly relevant to ProfitLens because they support the idea that even simple automation can create meaningful improvements in small business financial management.

A major theme in foreign literature is that automation helps SMEs overcome the limitations of manual record-keeping. Manual methods often depend on individual memory, handwritten logs, and repetitive calculations, all of which increase the risk of error and delay. Automated systems, in contrast, standardize transaction recording and allow business owners to view financial information in real time. This shift from manual to digital processes has become an important part of SME modernization in both developed and emerging economies.

2.4.1 Global Trends in SME Digital Adoption

Studies on global SME performance indicate that businesses utilizing automated financial tools consistently outperform those relying on manual methods in terms of accuracy, error reduction, and reporting speed (Rogers, 2003). A key finding from international research is that automated systems reduce the time required for financial reconciliation from days to minutes, freeing up staff time for more value-adding activities.

The global trend toward SME digitalization has been accelerated by several factors, including the increasing affordability of technology, the growing availability of cloud-based solutions, the proliferation of internet connectivity, and the demonstrated competitive advantages of digital businesses over their analog counterparts. Research by Vrontis et al. (2022) found that adoption of digital technologies significantly impacts the creation of economic sustainability and social value for SMEs, supporting the broader case for digital transformation initiatives like ProfitLens.

Research on data-driven decision making further supports the value of automated financial systems. Studies show that having immediate access to revenue trends, expense breakdowns, and profit summaries empowers business owners to make more confident and rapid financial decisions (Laudon & Laudon, 2020). In contrast, businesses relying on manual record-keeping often make decisions based on outdated or incomplete information, leading to missed opportunities and undetected financial problems.

Mubarak et al. (2019) examined the impact of digital transformation technologies on SME performance in Pakistan, finding that big data analytics, cyber-physical systems, and interoperability had significant positive impacts on business performance. While the specific technologies examined in their study are more advanced than those employed in ProfitLens, the fundamental finding that digital transformation improves business performance supports the rationale for developing automated financial monitoring systems for small businesses.

2.4.2 Technology Adoption Factors in SME Contexts

Research on technology adoption in SME contexts has identified several factors that influence the success of digital transformation initiatives. Ghobakhloo et al. (2012)

conducted a comprehensive review of IT adoption literature in SMEs and identified critical success factors including top management support, organizational readiness, perceived benefits, and external support. These factors are relevant to the adoption of ProfitLens, as the system's success depends on the owner's commitment to using the system, the organization's readiness to adopt new technology, the perceived benefits of automated financial tracking, and the availability of technical support.

Brandon-Jones and Kauppi (2017) examined the antecedents of the Technology Acceptance Model within e-procurement contexts, finding that system-level dimensions including processing quality and usability played a greater role than support dimensions in determining individual acceptance. This finding suggests that the design quality of ProfitLens, particularly its usability and processing accuracy, will be critical factors in determining whether users adopt and continue to use the system.

Ifinedo (2011) investigated technology acceptance in Canadian SMEs and found that perceived benefits, management commitment, and external pressure were significant predictors of technology acceptance. While the Canadian context differs from the Philippine setting, the underlying finding regarding the importance of perceived benefits in driving technology adoption is broadly applicable and supports the TAM-based design approach adopted for ProfitLens.

2.4.3 Digital Transformation and Business Performance

The relationship between digital transformation and business performance has been examined in numerous international studies, with consistent findings supporting the positive impact of digital technologies on SME outcomes. Teng et al. (2022) conducted an empirical analysis of the performance of SMEs undergoing digital transformation in China, finding that digital technology, employee digital skills, and digital transformation strategy were positively correlated with SME performance, and that digital transformation mediated the impact of digital transformation strategies on performance.

Costa Melo et al. (2023) conducted a systematic literature review of sustainable digital transformation in SMEs, analyzing 74 peer-reviewed papers. Their review identified that digital transformation is a growing research area with increasing interest from both academics and practitioners. The review found that digital transformation can contribute to sustainability across environmental, social, and economic dimensions, suggesting that the benefits of digital financial management extend beyond immediate operational improvements.

Shahadat et al. (2023) examined technological, environmental, and organizational factors influencing digital technology adoption in SMEs in emerging countries, finding that relative advantage, complexity, observability, perceived cost, top management support, and government support were significant determinants of ICT adoption. These findings provide a comprehensive framework for understanding the factors that may influence the adoption of ProfitLens in the Philippine auto parts retail context.

The benefits documented in foreign studies informed the design of ProfitLens, particularly in the areas of automated profit calculation, real-time dashboard summaries, and visual reporting. The system aims to bring the same level of financial visibility to a

small auto parts store in the Philippines that larger businesses enjoy in more developed markets, demonstrating that effective financial monitoring solutions can be developed and deployed in resource-constrained environments.

2.5 Local Studies: The Philippine SME Landscape

Research specific to the Philippine SME context provides important insights into the challenges and opportunities facing small businesses in the country. The Philippine government, through the Department of Trade and Industry (DTI), has identified SME digitalization as a national priority, recognizing that most small businesses in the Philippines continue to operate with limited automation and manual record-keeping (DTI, 2020).

2.5.1 The Philippine SME Ecosystem

The Philippine SME ecosystem is characterized by a large number of micro and small enterprises operating across diverse sectors, including retail, services, manufacturing, and agriculture. These enterprises collectively represent the backbone of the Philippine economy, contributing significantly to employment generation, economic output, and community development. However, many Philippine SMEs face persistent challenges related to limited access to capital, inadequate business management practices, insufficient technical expertise, and competitive pressures from larger, better-resourced organizations.

The Department of Trade and Industry has implemented various programs and initiatives to support SME development, including digitalization programs that aim to help small businesses adopt technology solutions for improved operational efficiency. Despite these government efforts, research indicates that technology adoption rates among Philippine SMEs remain relatively low, with many businesses continuing to rely on manual processes for core business functions including financial management (DTI, 2020).

The barriers to technology adoption among Philippine SMEs are multifaceted and include limited financial resources for technology investment, lack of technical knowledge and skills, concerns about system complexity and reliability, and insufficient access to technical support and training. These barriers are particularly pronounced among micro-enterprises, which typically have the most limited resources and the least access to technical expertise.

2.5.2 The Iloilo City Context

Iloilo City is one of the major economic hubs in the Visayas region, with a diverse mix of retail, manufacturing, and service businesses. The city's economic vitality is driven by a combination of traditional industries, emerging technology sectors, and a growing tourism industry. Within this dynamic economic environment, small auto parts retailers serve an important role in supporting the transportation needs of the city's residents and businesses.

Local research focusing on auto parts retailers in iloilo emphasizes that mobile-responsive and intuitive interfaces are essential for staff members who may have limited technical expertise (PSA, 2022). Studies also indicate that many small retail businesses in iloilo rely on manual cash registers and handwritten receipts, creating inefficiencies similar to those identified in the reference business for this study.

The auto parts industry in iloilo, particularly among micro and small enterprises, faces unique challenges including limited capital, difficulty accessing credit, and competition from larger retailers. These challenges are compounded by inadequate financial tracking tools that prevent business owners from understanding their true profitability and making data-driven decisions (DTI Region VII, 2021). ProfitLens was designed to address these specific challenges by providing a simple, affordable, and purpose-built financial monitoring tool tailored to the needs of small auto parts retailers.

The geographic and economic context of iloilo City provides an appropriate setting for the development and evaluation of ProfitLens. The city's concentration of small auto parts retailers, combined with the availability of a willing reference business and the research team's familiarity with the local business environment, created favorable conditions for conducting this research. The findings and methodology demonstrated in this study may be applicable to similar business contexts in other Philippine cities and municipalities.

2.5.3 Financial Management Challenges in Philippine SMEs

Research within the Philippine context suggests that manual financial tracking often leads to what practitioners describe as “unnoticed overspending” (McKinney & Yoos, 2010). Without automated monitoring, small business owners may not realize that expenses have exceeded revenue until the end of the month, or worse, at the end of the quarter when manual reconciliation is performed. This delay prevents timely corrective action and can lead to cash flow problems that threaten business viability.

Local studies also highlight the prevalence of financial irregularities in small businesses, including arithmetic errors in receipt calculations, misplaced or lost records, and inconsistent categorization of expenses (PSA, 2022). These irregularities undermine the reliability of financial data and make it difficult for business owners to assess true financial performance. ProfitLens addresses these issues by standardizing data entry, automating calculations, and providing visual indicators that highlight potential problems such as high-spending categories.

The financial management challenges faced by Philippine SMEs are not merely operational inconveniences; they have significant implications for business survival and growth. Research consistently shows that poor financial management is among the leading causes of small business failure, both in the Philippines and globally. By providing automated financial tracking and monitoring capabilities, ProfitLens addresses a fundamental business need that directly impacts the sustainability and growth potential of the reference business.

Sanistasya et al. (2019) examined the effect of financial literacy and financial inclusion on small enterprise performance in East Kalimantan, Indonesia, finding that both

financial literacy and financial inclusion positively affected small enterprise performance. While their study focused on different variables than those examined in the ProfitLens context, the underlying finding regarding the importance of financial management capabilities for small business performance supports the rationale for developing automated financial monitoring tools.

2.5.4 Digital Transformation Initiatives in the Philippines

The Philippine government has recognized the importance of digital transformation for SME competitiveness and has implemented various initiatives to support technology adoption among small businesses. The Department of Trade and Industry's MSME Development Plan 2020-2024 identifies digitalization as a strategic priority for enhancing SME competitiveness and productivity (DTI, 2020).

Despite these government initiatives, the pace of digital transformation among Philippine SMEs has been uneven, with larger enterprises generally adopting technology more quickly than smaller ones. Research indicates that micro and small enterprises, which represent the majority of Philippine businesses, lag behind in technology adoption due to the barriers identified earlier, including limited resources, insufficient technical knowledge, and concerns about complexity and reliability.

ProfitLens contributes to the broader digital transformation agenda by demonstrating that effective, affordable technology solutions can be developed for specific small business contexts. The system's use of open-source technologies, its simple and intuitive interface design, and its focus on addressing specific, well-defined business needs provide a model that could be replicated or adapted for other small business contexts across the Philippines.

The Philippine Statistics Authority (2022) reports that micro, small, and medium enterprises constitute over 99% of all businesses in the Philippines and employ approximately 63% of the national workforce. These statistics underscore the economic importance of the SME sector and the potential impact of successful digital transformation initiatives on national economic performance. Systems like ProfitLens, if successfully deployed across multiple businesses, could contribute to measurable improvements in SME productivity and competitiveness at the national level.

2.5.5 Challenges of Technology Adoption in the Philippine Context

While the benefits of digital transformation for SMEs are well-documented, the specific challenges of implementing technology solutions in the Philippine context warrant detailed examination. These challenges inform the design decisions made for ProfitLens and help explain why many Philippine SMEs have been slow to adopt digital financial management tools.

Infrastructure Constraints: Internet connectivity in the Philippines, while improving, remains inconsistent in many areas, particularly outside major metropolitan centers. Broadband penetration rates lag behind regional averages, and internet speeds are often insufficient for cloud-based applications that require continuous connectivity. These infrastructure limitations influenced the decision to deploy ProfitLens as a

locally-hosted application rather than a cloud-based service, ensuring reliable access regardless of internet connectivity quality.

Financial Constraints: Small businesses in the Philippines operate with limited financial resources that must be allocated carefully among competing needs. The cost of commercial software subscriptions, hardware upgrades, and technical support can represent a significant portion of monthly operating expenses for micro-enterprises. ProfitLens addresses this constraint through its use of free, open-source technologies and its deployment on existing hardware without requiring expensive upgrades.

Technical Literacy: Digital literacy levels among small business owners and staff in the Philippines vary widely, with many individuals having limited experience with business software applications. This constraint influenced the design of ProfitLens's user interface, which prioritizes simplicity, intuitive navigation, and clear visual feedback to minimize the learning curve for new users.

Cultural Factors: Filipino business culture places high value on personal relationships and face-to-face interactions, which can create resistance to technology-mediated processes that perceived as impersonal or complicating established workflows. The design of ProfitLens attempts to complement rather than replace existing personal business practices, positioning the system as a tool that enhances the owner's existing capabilities rather than fundamentally changing how business is conducted.

Language Considerations: While English is widely spoken in the Philippines, many small business operators are more comfortable conducting daily operations in Filipino or regional languages. ProfitLens was designed with English-language interfaces to align with standard business software conventions in the Philippines, but the simplicity of its interface design minimizes the language barrier for users with limited English proficiency.

These context-specific challenges underscore the importance of developing purpose-built solutions like ProfitLens that are designed specifically for the Philippine small business environment, rather than importing solutions designed for different cultural, economic, and technical contexts.

2.6 Conceptual Framework

The conceptual framework of ProfitLens is designed around a structured process that transforms raw business data into meaningful financial insights. As illustrated in the system's data flow architecture, the system follows a linear flow from data input to information output, organized into four interconnected layers that collectively enable the automated financial monitoring capabilities of the platform.

2.6.1 Input Layer

The input layer represents the fundamental data that the system receives from users. This layer encompasses four categories of input data:

Sales Data: Product selection, quantity sold, sale date, and computed sale total. Sales data represents the primary revenue input to the system and is captured through the

sales recording module's form-based interface. Each sales transaction is recorded as an individual database entry that includes all necessary attributes for financial calculation and reporting.

Expense Data: Expense category, amount, description, and date. Expense data captures the costs incurred in operating the business and is entered through the expense tracking module. The categorization of expenses enables granular analysis of spending patterns across different cost categories.

Product Data: Product name, category, cost price, selling price, and current stock quantity. Product data represents the foundational reference data that supports both sales recording and inventory management functions. Product records are maintained through the product management module and are referenced by the sales recording module for automatic price lookups and stock deductions.

User Credentials: Email and hashed password for authentication. User credential data supports the system's security model by enabling user authentication and authorization. This data is captured during user registration and validated during login to ensure that only authorized users can access protected system functionality.

2.6.2 Processing Layer

The processing layer handles the transformation of raw data into useful information through several key operations:

Sales Computation: Automatically calculates total sale value by multiplying quantity by selling price. This computation is performed in real time whenever a sales transaction is entered, ensuring that financial totals are always current.

Stock Deduction: Reduces product stock quantity based on recorded sales. This operation maintains the accuracy of inventory records by automatically updating stock levels to reflect sales activity.

Profit Calculation: Computes net profit by subtracting total expenses from total revenue. This calculation is performed continuously as new transactions are entered, providing real-time profit visibility.

Data Aggregation: Groups transactions by month, quarter, or year for reporting purposes. This aggregation enables the generation of financial reports that present summarized performance data across different time periods.

2.6.3 Output Layer

The output layer presents the processed information to users through several interface components:

Dashboard: Real-time financial summaries including net profit, revenue trends, and best-selling products. The dashboard provides at-a-glance visibility into the most important financial metrics.

Sales Module: Detailed sales history with monthly and annual profit summaries. The sales module provides comprehensive visibility into sales activity and its contribution to overall profitability.

Expense Tracking: Categorized expense breakdown with high-spend visual indicators. The expense tracking interface provides granular visibility into spending patterns and alerts users to potential overspending conditions.

Financial Reports: January-to-December grouped bar charts comparing revenue versus expenses. The financial reporting module provides comprehensive annual performance summaries in both graphical and tabular formats.

Revenue Analysis: Monthly trends, year-over-year comparisons, and top product rankings. The revenue analysis module provides the analytical capabilities needed for trend identification and performance comparison.

2.6.4 Feedback Loop

The system incorporates a feedback loop where users review the output reports, identify financial issues, and make informed business decisions. This cycle supports continuous improvement in financial management, a core principle of systems theory (von Bertalanffy, 1968; Johnson & Johnson, 2004).

The feedback loop represents the critical link between the system's information outputs and the business decisions that determine operational outcomes. Without this feedback mechanism, the system would function merely as a data storage and calculation tool without contributing to improved business performance. By enabling users to review financial reports, identify trends, and make informed decisions, the feedback loop completes the systems cycle that transforms raw data into business value.

2.7 Research Gap and System Positioning

Based on the review of related literature and studies, several gaps have been identified in the current financial monitoring approaches for small auto parts businesses in the Philippines and analogous contexts. ProfitLens was developed to address these gaps directly, positioning the system as a targeted solution for documented needs in the SME financial management landscape.

2.7.1 Identified Gaps in Existing Systems

The literature review identified five primary gaps that represent opportunities for improvement in SME financial management:

Manual Profit Computation: Most small businesses in the Philippines still rely on manual calculation of profits. This process is time-consuming and prone to errors, creating a gap for automated systems that can compute profit accurately and instantly (McKinney & Yoos, 2010). The persistence of manual profit calculation in Philippine SMEs, despite the availability of digital alternatives, suggests that existing solutions may not adequately address the specific needs and constraints of small business operators.

Delayed Financial Reports: Traditional manual tracking provides financial information only at the end of the month. There is a significant gap in systems that offer real-time visibility into the financial health of a business (Laudon & Laudon, 2020). The delay between transaction occurrence and financial reporting inherent in manual systems prevents timely identification of financial trends and issues, limiting the effectiveness of financial information for decision-making purposes.

Limited Visual Reporting: Small business owners often struggle to interpret raw financial data. The lack of visual representations such as charts and graphs creates a gap in decision-support tools (Rogers, 2003). The absence of visualization capabilities in manual financial management prevents business owners from easily identifying patterns, trends, and anomalies in their financial data.

No Expense Monitoring Alerts: Manual expense tracking frequently leads to undetected overspending. Existing systems lack visual indicators that warn business owners when expenses become disproportionate to revenue. The absence of automated alert mechanisms means that overspending conditions may persist for extended periods before being detected through manual reconciliation.

Inadequate SME-Tailored Solutions: While enterprise-level accounting software exists, there is a lack of simple, affordable, and purpose-built financial monitoring tools specifically designed for small auto parts retailers in the Philippine context (DTI, 2020). Existing commercial solutions are often too complex, too expensive, or too generic for the specific needs of small auto parts retailers, creating a gap for purpose-built solutions.

2.7.2 System Positioning

ProfitLens positions itself to address these identified gaps by offering the following solutions:

Automated Net Profit Calculation: Real-time computation using the formula $\text{Net Profit} = \text{Total Sales} - \text{Total Expenses}$, eliminating manual aggregation and ensuring mathematical accuracy.

Instant Dashboard with Live Summaries: Financial summaries updated with every transaction, providing real-time visibility into business performance without requiring end-of-period reconciliation.

Inline SVG Charts: Bar, line, and donut visualizations generated directly in the browser without external libraries, providing intuitive visual representations of financial data.

Category-Based Expense Breakdown with High-Spend Indicators: Granular expense categorization with visual alerts that highlight spending patterns and potential overspending conditions.

Simple, Web-Based Platform: A purpose-built solution designed specifically for SME needs, avoiding the complexity and cost of enterprise-level accounting systems while providing the functionality that small auto parts retailers require.

ProfitLens is specifically positioned as a smart, web-based profit monitoring platform that delivers automated business insights, accurate financial tracking, and proactive alerts for overspending, all designed for the Philippine SME context, particularly for auto parts retailers. The system combines the theoretical principles of TAM and systems theory with practical, locally relevant features to create a solution that addresses the unique challenges faced by small business owners in the region.

By addressing these documented gaps with a purpose-built solution grounded in established theoretical frameworks and informed by both international and local research, ProfitLens represents a contribution to the growing body of knowledge regarding technology solutions for SME financial management in the Philippine context.

2.8 Comparative Analysis of Existing Solutions

To better understand the positioning of ProfitLens within the broader landscape of financial management solutions, it is instructive to compare its capabilities against both manual methods and existing commercial alternatives. This comparison reveals the specific value proposition that ProfitLens offers to small auto parts retailers in the Philippine context.

2.8.1 Comparison with Manual Record-Keeping

Manual record-keeping, the method currently employed by the reference business and many similar SMEs, offers several advantages including zero technology cost, no learning curve, and complete flexibility in data recording formats. However, these advantages are outweighed by significant disadvantages including high error rates, time-consuming reconciliation processes, absence of trend analysis capabilities, vulnerability to data loss, and inability to generate visual reports.

ProfitLens addresses each of these disadvantages while preserving the simplicity that makes manual methods attractive to small business operators. The system requires minimal technical expertise to operate, provides immediate feedback on data entry accuracy, eliminates the need for manual calculation, and maintains secure digital records that are backed up automatically.

2.8.2 Comparison with Commercial Accounting Software

Commercial accounting software packages such as QuickBooks, Xero, and Sage offer comprehensive financial management capabilities that far exceed the current functionality of ProfitLens. These platforms typically include features such as multi-currency support, bank reconciliation, tax preparation, payroll processing, inventory management with barcode scanning, and integration with third-party services including banks, payment processors, and government tax authorities.

However, these commercial solutions present several barriers to adoption for small auto parts retailers in the Philippine context. Subscription costs ranging from \$25 to \$80 per month represent a significant expense for micro-enterprises with limited revenue. The complexity of features designed for larger businesses creates steep learning curves that can overwhelm users with limited technical expertise. Many commercial platforms are

designed primarily for Western markets and may not adequately support Philippine-specific requirements such as BIR tax compliance, Philippine Peso formatting, or local payment methods.

ProfitLens occupies a middle ground between the inadequacy of manual methods and the over-complexity of commercial solutions. By focusing exclusively on the core financial monitoring needs of small auto parts retailers, ProfitLens provides targeted functionality without the complexity and cost of full-featured accounting platforms.

2.8.3 Comparison with Free and Open-Source Alternatives

Several free and open-source accounting solutions exist, including GnuCash, Wave, and Manager.io. These solutions offer financial management capabilities without subscription costs, making them more accessible than commercial alternatives for budget-constrained small businesses.

However, free alternatives often present their own challenges. GnuCash, while powerful, uses a desktop-based architecture that limits access to a single computer and does not support the multi-user web-based access that ProfitLens provides. Wave, while web-based, is designed primarily for service-based businesses and does not include inventory management or product-based sales tracking capabilities relevant to retail operations. Manager.io, while comprehensive, presents a complex interface that may be overwhelming for users with limited accounting knowledge.

ProfitLens differentiates itself from these alternatives through its narrow focus on the specific needs of auto parts retailers, its intentionally simplified interface designed for users with limited technical expertise, and its deployment architecture optimized for single-location small businesses.

2.9 Summary of Literature Review

The comprehensive review of literature and studies presented in this chapter establishes several key foundations for the ProfitLens project:

First, the theoretical frameworks of TAM, Systems Theory, Diffusion of Innovation, and Resource-Based View provide validated conceptual foundations for understanding technology adoption, business process integration, and competitive advantage creation through information systems. These frameworks informed the design principles that guided ProfitLens development.

Second, the evolution of financial monitoring systems from manual ledgers to digital platforms demonstrates the clear trajectory toward automation and the well-documented benefits of digital financial management for business accuracy, efficiency, and decision-making capability.

Third, international research on SME automation consistently demonstrates that automated financial tools improve business performance across multiple dimensions including accuracy, reporting speed, and decision-support capability. These findings establish the evidence base for expecting similar benefits from ProfitLens deployment.

Fourth, research specific to the Philippine SME context identifies unique challenges including limited resources, low technology adoption rates, manual record-keeping practices, and the need for affordable, purpose-built solutions. These context-specific findings informed the design decisions that make ProfitLens appropriate for its target deployment environment.

Fifth, the identification of research gaps including manual profit computation, delayed financial reports, limited visual reporting, absence of expense monitoring alerts, and inadequate SME-tailored solutions establishes the specific scholarly contribution that ProfitLens aims to make within the broader knowledge landscape.

Collectively, this literature review provides the evidence base, theoretical foundation, and contextual understanding necessary for the design, development, and evaluation of ProfitLens as a contribution to both academic knowledge and practical business improvement in the Philippine SME sector.

CHAPTER 3: METHODOLOGY

3.1 Research Design

This chapter presents the research methodology employed in the design, development, and evaluation of ProfitLens. The methodology encompasses the research design strategy, the system development methodology, data gathering techniques, system architecture, tools and technologies, database design, testing procedures, and evaluation criteria. Each of these methodological components is described in detail to provide a comprehensive understanding of how the research was conducted and how the resulting system was produced.

3.1.1 Developmental Research Design

This study employs a Developmental Research Design, also referred to as Design Science Research (DSR), which is an approach specifically suited for information systems research that seeks to develop, rather than merely observe or analyze, an existing phenomenon. Design-based research emerges in the information systems literature as a methodology for creating and evaluating information technology artifacts intended to solve identified organizational problems (Creswell, 2018; Hevner & Chatterjee, 2020).

A Developmental Research Design was chosen for this study because it focuses specifically on information systems research that seeks to develop practical solutions to real operational problems, as opposed to purely descriptive or explanatory research. This approach is particularly appropriate for capstone projects that aim to produce functional systems addressing genuine business needs, as it provides a structured framework for moving from problem identification through solution design, implementation, and evaluation.

The design follows a structured process that begins with problem identification through observation and analysis of the current operational context. This is followed by planning and design of the proposed solution, development and implementation of the system, testing and validation of system functionality, and finally, documentation and dissemination of findings. This structured approach ensures that the resulting system is grounded in identified needs, designed according to established principles, implemented using appropriate technologies, and validated through systematic testing.

3.1.2 Justification for Design Science Research

The choice of Design Science Research methodology is justified by several considerations specific to this study. First, the primary output of the research is a functional information system, which aligns with DSR's emphasis on the creation of technology artifacts. Second, the system was developed to address a specific, identified business problem, which aligns with DSR's problem-solving orientation. Third, the study involves both the creation and evaluation of a technology artifact, which aligns with DSR's dual emphasis on artifact production and assessment.

DSR methodology has been successfully applied in numerous information systems research projects, establishing it as a legitimate and productive approach for this type of study. The methodology's emphasis on rigor, relevance, and utility ensures that the resulting research produces both scholarly contributions and practical value, which is particularly important for capstone projects that must satisfy both academic and practical objectives.

3.2 System Development Methodology: The Agile Model

The Agile Model was chosen for ProfitLens development due to its flexibility and iterative characteristics, which are well-suited to the circumstances of this project. Agile methodology works best when requirements may evolve during the development process and when end-user feedback can be incorporated to enhance the final product.

3.2.1 Rationale for Agile Selection

Unlike the Waterfall methodology, which follows a sequential process that progresses linearly from one phase to the next without returning to previous stages, Agile methodology continually cycles through development phases in iterative increments. This iterative approach is particularly valuable for capstone projects because it acknowledges that requirements may not be fully understood at the outset and that user feedback during development can significantly improve the final product.

The Agile approach also provides several practical advantages for the development team. It enables incremental progress that can be demonstrated and evaluated throughout the development process, rather than requiring completion of the entire system before any evaluation can occur. It facilitates early identification and correction of design errors, reducing the risk of significant rework later in the project. It supports adaptation to changing requirements or unexpected challenges that may arise during development.

3.2.2 Agile Development Phases

The ProfitLens development process followed four distinct Agile phases, each building upon the outputs of the preceding phase:

Phase 1: Requirements Gathering

The initial phase involved comprehensive requirements gathering through a combination of observation, interviews, and literature review. The research team conducted structured interviews with the auto parts store owner and staff to understand current manual processes, identify pain points, and document desired features for the new system. Prior to conducting interviews, team members performed on-site observation of store operations to develop a practical understanding of the business context and to identify specific areas where automated financial tracking could provide the most benefit.

During the observation and interview process, the team documented the existing manual procedures for tracking daily sales, recording expenses, calculating monthly profit, and generating financial summaries. These documented procedures served as the baseline against which the new system's functionality was defined. The team also gathered information about the technical environment, including available hardware, internet connectivity, and staff computer literacy levels, to ensure that the proposed system would be appropriate for the operational context.

Key features identified during requirements gathering included automatic profit calculation, graphical financial reporting, expense categorization and monitoring, and visual indicators for overspending alerts. These requirements were documented in a formal requirements specification that guided subsequent design and development activities.

Additionally, the research team conducted a comprehensive review of related literature and studies on financial monitoring systems, SME automation, and technology adoption. This literature review informed the theoretical framework presented in Chapter 2 and provided evidence-based guidance for design decisions.

Phase 2: Sprint Planning and System Design

In the second phase, the development team created a detailed system design based on the requirements gathered in Phase 1. This design phase encompassed several interconnected activities:

Database Schema Design: The team designed a MySQL database schema consisting of four primary tables (users, products, sales, and expenses) that captures all necessary data entities and their relationships. The schema design followed normalization principles to ensure data integrity, reduce redundancy, and support efficient querying across all system modules.

User Interface Design: The team created wireframes and user interface designs for all core modules, including the dashboard, sales recording interface, expense tracking interface, and financial reports module. The UI design prioritized simplicity and

intuitiveness, reflecting the TAM principle that perceived ease of use is a critical factor in technology adoption.

Functional Specifications: For each module, the team wrote detailed functional specifications that define input requirements, processing logic, and expected outputs. These specifications served as the blueprint for the subsequent development phase and ensured that all team members had a shared understanding of the system's intended behavior.

Data Flow Diagrams (DFD): The team created DFD Level 0 (context diagram) and DFD Level 1 (detailed process diagram) representations that illustrate how data moves through the system, showing the interactions between users, processes, and data stores.

Entity Relationship Diagrams (ERD): The team developed ERD representations that define the structure of and relationships among the database tables, providing a visual reference for the database implementation.

Phase 3: Development

The third phase involved the actual coding and construction of the system. Development proceeded iteratively, with each iteration producing functional increments that could be tested and evaluated before proceeding to the next set of features.

Backend Development: The PHP backend was implemented following security best practices throughout the development process. Prepared statements were used consistently to prevent SQL injection attacks, which represent one of the most common security vulnerabilities in web applications. Password storage was implemented using PHP's `password_hash()` function, which provides bcrypt-based hashing that offers strong protection against brute-force attacks. Session-based authentication was implemented to control access to protected pages and ensure that only authorized users could view or modify financial data.

Frontend Development: The user interface was constructed using HTML5 for page structure and content organization, CSS3 for visual styling and layout, and JavaScript for client-side interactivity. The styling was implemented in a centralized CSS file (`style.css`) to ensure visual consistency across all pages and to simplify maintenance.

Chart Generation: For visual reporting functionality, the team implemented inline SVG chart generation using PHP and JavaScript. This approach was chosen to ensure that the system remains self-contained and does not depend on external charting libraries or third-party services. The inline SVG approach also provides fine-grained control over chart appearance and behavior.

Helper Functions: The team developed a library of helper functions for common operations, including the `formatMoney()` function for Philippine Peso currency formatting and the `getDB()` function for database connection management. These helper functions promote code reuse, reduce duplication, and improve maintainability.

Phase 4: Testing and Iteration

The fourth phase involved systematic testing to verify that the system functions correctly and meets the specified requirements. Testing was conducted at multiple levels to ensure comprehensive quality assurance:

Unit Testing: Individual functions and components were tested in isolation to verify correct behavior. This included testing login verification logic, profit calculation accuracy, stock deduction operations, expense totaling, and currency formatting. Unit testing ensured that each building block of the system performed its intended function correctly before being integrated with other components.

Integration Testing: After individual components passed unit testing, they were integrated and tested together to verify that modules work correctly in combination. This included testing the interaction between the sales recording module and the profit calculation logic, the expense tracking module and the reporting module, and the product management module and the sales recording module.

Usability Testing: The store owner and staff were invited to use the system and provide feedback on its usability, functionality, and practical utility. Usability testing was particularly important for validating that the system's interface design was appropriate for users with limited technical expertise and that the system's functionality addressed the real-world needs identified during requirements gathering.

Bug Fixing and Iteration: When issues were identified during testing, the development team investigated the root causes, implemented corrections, and retested to verify the fixes. This iterative process continued until the system met the desired quality standards across all testing dimensions.

3.3 System Architecture

ProfitLens uses a Client-Server Architecture to provide secure access and to keep data management centralized. This architectural approach is widely used in web-based applications and is particularly well-suited for small business systems because multiple users can access the platform through a web browser without requiring software installation on individual devices.

The overall system architecture consists of three separate layers that cooperate to deliver the profit monitoring and financial reporting capabilities of the platform:

3.3.1 Client Layer (Frontend)

The Client Layer is the user-facing component that operates within the web browser. This layer is responsible for presenting content to users, capturing user input, and displaying the results of system operations. In ProfitLens, the client-side layer is constructed using:

HTML5 (HyperText Markup Language): Provides the structural foundation for each page, defining the page layout, forms, tables, and navigation elements. HTML5's semantic elements contribute to page accessibility and search engine optimization, while its form elements support data entry with built-in validation capabilities.

CSS3 (Cascading Style Sheets): Controls the visual appearance of the interface, including colors, fonts, spacing, and overall layout. The styling is maintained in a centralized style.css file that ensures consistent and professional visual presentation across all pages of the application.

JavaScript: Provides client-side interactivity including form validation, dynamic dropdown population, and chart-related interactions. JavaScript enhances the user experience by reducing the need for full page reloads, providing immediate feedback on user actions, and enabling dynamic content updates without server communication.

3.3.2 Server Layer (Backend)

The Server Layer handles business logic execution and all data processing operations, including database interactions. This layer operates on the web server (Apache in the XAMPP environment) and performs several critical functions:

Processing User Requests: The server layer receives requests from the client layer, interprets them, and executes the appropriate actions. This includes processing login attempts, recording sales entries, managing expense records, and generating report data.

Business Logic Implementation: The server layer implements the core system logic, including profit calculation ($\text{Net Profit} = \text{Total Sales} - \text{Total Expenses}$), stock deduction based on recorded sales, expense categorization and totaling, and financial data aggregation for reporting purposes.

Database Communication: The server layer manages all interactions with the MySQL database, including data retrieval, insertion, update, and deletion operations. All database interactions use prepared statements to prevent SQL injection attacks.

Security Enforcement: The server layer validates user authentication and authorization on every protected page request, ensuring that only authorized users can access sensitive financial data and system functionality.

Response Generation: The server layer constructs HTML responses that are returned to the client's browser for display to the user. These responses include the processed data, formatted for presentation in the system's user interface.

The server layer is implemented in PHP 8, selected for its compatibility with MySQL, its ease of learning and use, and its widespread adoption in web development. PHP provides built-in session management and security functions that facilitate the implementation of authenticated web applications.

3.3.3 Database Layer

The Database Layer provides persistent storage for all system data. ProfitLens uses MySQL (specifically the MariaDB implementation included with XAMPP) as its relational database management system (RDBMS). The database consists of four primary tables:

users: Stores user account information including email address, hashed password, role designation (admin or user), and account creation timestamp.

products: Contains product details including name, category, cost price, selling price, current stock quantity, and creation timestamp.

sales: Records sales transaction data including product identifier, quantity sold, unit price at time of sale, total sale amount, sale date, and creation timestamp.

expenses: Stores expense records including category designation, description, amount, expense date, and creation timestamp.

3.3.4 Logical Data Flow

The system implements a logical data flow that follows these steps:

1. The user interacts with the client layer by filling out forms or clicking navigation links.
2. The client sends a request to the server layer (for example, submitting a sales entry form).
3. The server processes the request, performing the necessary business logic operations and security checks.
4. The server communicates with the database layer to execute the appropriate SQL queries (for example, INSERT INTO sales ...).
5. The database processes the query and returns the result to the server.
6. The server processes the result and constructs an HTML response.
7. The response is returned to the client for display in the user's browser.

This data flow pattern ensures that all data processing occurs on the server side, where business logic and security controls can be properly enforced, while the client layer remains responsible only for presentation and user interaction.

3.4 Data Gathering Techniques

The researchers employed multiple data gathering techniques to collect accurate and relevant information for the design and development of ProfitLens:

3.4.1 Interviews

Structured interviews were conducted with the auto parts store owner and staff to understand current manual processes, identify specific pain points, and document desired features for the new system. The interview protocol was designed to elicit detailed information about existing procedures, challenges encountered, and improvement opportunities identified by the business's operational personnel.

Interviews were conducted in a conversational format that encouraged participants to share their experiences and observations freely. The research team prepared an interview guide with predetermined topics and questions but allowed the conversation to flow naturally to capture unanticipated insights. All interviews were documented through

written notes and, where permitted, audio recordings to ensure accurate capture of participant responses.

The interview process yielded detailed information about the current state of financial management at the store, including the specific procedures used for recording sales, tracking expenses, calculating profit, and generating financial summaries. Participants also shared their observations about the limitations and challenges of the current manual processes, providing firsthand evidence of the problems that ProfitLens was designed to address.

3.4.2 Observation

Direct observation of the store's daily operations was conducted to supplement the information gathered through interviews. The research team observed sales recording procedures, expense tracking activities, monthly profit calculation processes, and general store operations to identify inefficiencies and areas for improvement.

Observation sessions were conducted during normal business hours to ensure that the observed activities reflected typical operational conditions. The research team documented their observations systematically, recording specific examples of the challenges and inefficiencies that were later corroborated through interview responses.

The observation process revealed several important insights that might not have emerged through interviews alone, including the physical layout of the store's record-keeping materials, the workflow patterns of staff members during busy periods, and the specific points at which manual processes created bottlenecks or opportunities for error.

3.4.3 Document Analysis

Existing manual documents were reviewed to understand the current data structure and reporting methods used by the store. Documents analyzed included sales notebooks, expense receipts, handwritten financial summaries, and any informal spreadsheets maintained by the owner or staff.

The document analysis process provided concrete examples of the data structures, calculation methods, and reporting formats currently in use. These examples informed the design of ProfitLens's data structures and user interfaces, ensuring that the new system would support the same types of information that the business already tracked while providing additional capabilities for analysis and reporting.

The document analysis also revealed the scope and limitations of the current record-keeping practices, including instances of incomplete records, calculation errors, and inconsistent formatting that underscored the need for the automated, standardized approach implemented in ProfitLens.

3.4.4 Questionnaire Surveys

Supplementary to the interviews and observation sessions, the research team distributed structured questionnaire surveys to gather quantitative data about user

preferences, technology readiness, and feature priorities. The questionnaire was designed using a Likert scale format to enable statistical analysis of responses and to complement the qualitative insights gathered through interviews.

The questionnaire covered several key areas:

Technology Readiness Assessment: Questions evaluating the current level of technology use in business operations, including the types of devices and software currently in use, the frequency of technology use for business purposes, and the comfort level with web-based applications.

Feature Priority Ranking: Questions asking participants to rank proposed system features in order of perceived importance, enabling the development team to prioritize implementation efforts based on actual user needs rather than assumed preferences.

Interface Preference Assessment: Questions evaluating preferences regarding interface design elements including color schemes, layout styles, navigation patterns, and visualization formats, ensuring that the system's visual design aligns with user expectations.

Workflow Integration Evaluation: Questions assessing how the proposed system would fit into existing business workflows, identifying potential integration challenges and opportunities for process optimization.

The questionnaire was distributed to the store owner, all staff members, and a small sample of customers who could provide perspective on the business's operational needs. A total of 15 questionnaires were distributed, with 12 completed and returned for analysis, representing an 80% response rate that provided sufficient data for meaningful analysis.

3.4.5 Benchmarking Analysis

The research team conducted benchmarking analysis of comparable financial monitoring systems and commercial accounting software to establish performance baselines and identify best practices for system design. This analysis examined both the technical architecture and user experience characteristics of existing solutions, identifying patterns and approaches that could inform the design of ProfitLens.

The benchmarking analysis covered three categories of existing solutions:

Commercial Accounting Software: QuickBooks, Xero, and Sage were evaluated for their feature sets, interface designs, pricing models, and deployment architectures. While these solutions offered comprehensive functionality, their complexity and cost were identified as significant barriers to adoption for the target user population.

Free and Open-Source Alternatives: GnuCash, Wave, and Manager.io were evaluated as more accessible alternatives. While these solutions addressed the cost barrier, their limited customization options, desktop-based architectures, or feature gaps were identified as limitations for the specific use case of small auto parts retail.

Research Prototypes: Academic literature described several research prototypes for SME financial monitoring systems, providing insights into innovative approaches to visualization, data analysis, and user interface design that could be adapted for the ProfitLens context.

The benchmarking findings were synthesized into a set of design principles that guided ProfitLens development: simplicity over comprehensiveness, web-based access over desktop installation, self-contained operation over cloud dependency, and targeted functionality over general-purpose accounting features.

3.5 Tools and Technologies Used

The following tools and technologies were employed in the development of ProfitLens:

Component	Technology / Tool	Purpose
Frontend	HTML5, CSS3 (custom style.css), JavaScript	Page structure, visual styling, interactive UI elements
Backend	PHP 8 (server-side scripting)	Business logic, authentication, data processing, report generation
Database	MySQL (via MySQLi extension)	Stores users, products, sales transactions, and expenses
Charts / Visualization	Inline SVG, PHP-generated bar charts	Revenue vs. Expenses bar chart, line chart, donut chart; no external chart library
Authentication	PHP Sessions + password_verify() / password_hash()	Secure login, role-based access control (admin/user)
Development Tool	Visual Studio Code	Code editing and project management

Component	Technology / Tool	Purpose
Local Server	XAMPP (Apache + MySQL)	Development and testing environment
Database Name	profit_lens	Single database with tables: users, products, sales, expenses

3.5.1 Technology Selection Rationale

The selection of each technology was guided by several considerations including compatibility with the project requirements, availability of documentation and community support, alignment with the team's technical capabilities, and suitability for the target deployment environment.

PHP 8 was selected as the backend language due to its native compatibility with MySQL, its extensive documentation and community support, its built-in security features including prepared statements and password hashing functions, and its widespread use in web application development. PHP's low barrier to entry and extensive learning resources made it an appropriate choice for a student development team.

MySQL was selected as the database management system due to its reliability, performance, and compatibility with PHP. The MySQLi extension provides enhanced security features including prepared statement support, which was utilized throughout the system to prevent SQL injection attacks. The MariaDB implementation included with XAMPP provides a complete, free database server that is fully compatible with MySQL applications.

HTML5, CSS3, and JavaScript were selected for the frontend due to their universal browser support, extensive documentation, and alignment with web standards. The use of standard web technologies ensures that the system can be accessed from any modern web browser without requiring plugins or special software installations.

The inline SVG approach for chart generation was chosen to maintain system self-containment and eliminate dependencies on external charting libraries. This approach provides full control over chart appearance and behavior while ensuring that visual reporting capabilities are always available regardless of internet connectivity or third-party service availability.

3.5.1 Security Implementation

Security was a primary consideration throughout the technology selection and implementation process. The security implementation in ProfitLens encompasses

multiple layers designed to protect sensitive financial data from unauthorized access, modification, or disclosure.

Authentication Security: User passwords are stored using PHP's `password_hash()` function, which implements the bcrypt hashing algorithm with automatic salting. This approach provides strong protection against brute-force attacks and rainbow table attacks, as the hashing process is computationally expensive and each password hash includes a unique random salt. The `password_verify()` function is used for credential validation, ensuring constant-time comparison that resists timing attacks.

Session Security: Session identifiers are regenerated upon successful authentication to prevent session fixation attacks. Session data is validated on every protected page request to ensure that the session has not been tampered with or expired. Session timeout functionality automatically logs out inactive users after a configurable period of inactivity.

Database Security: All database queries use MySQLi prepared statements with parameterized queries, providing robust protection against SQL injection attacks. The database connection is established using the `getDB()` function, which implements connection parameter validation and error handling that prevents information leakage through error messages.

Input Validation: All user inputs are validated both on the client side (using JavaScript) and on the server side (using PHP) to prevent malformed or malicious data from entering the system. Server-side validation is treated as the authoritative check, as client-side validation can be bypassed by technically sophisticated users.

Output Encoding: All user-generated content displayed in the browser is properly encoded to prevent cross-site scripting (XSS) attacks. PHP's `htmlspecialchars()` function is used to convert special characters to their HTML entity equivalents, ensuring that user input cannot be interpreted as executable code by the browser.

Access Control: The `requireAdmin()` function enforces role-based access control on all admin-protected pages, redirecting unauthorized users to an appropriate access-denied page. This function validates the user's role from the session data on every protected page request, ensuring that access control is maintained throughout the user's session.

3.5.2 Performance Considerations

While Performance was not the primary optimization target for ProfitLens given its single-location deployment context, several performance considerations were incorporated into the design and implementation:

Database Indexing: Appropriate indexes are defined on database columns that are frequently used in query conditions and sorting operations, including product names, sale dates, and expense categories. Indexing improves query performance by enabling the database engine to locate matching rows without scanning entire tables.

Query Optimization: Database queries are designed to retrieve only the data needed for each specific operation, avoiding unnecessary data transfer and processing.

Aggregate queries using SQL functions such as SUM, COUNT, and GROUP BY are preferred over retrieving raw data and performing calculations in PHP.

Caching: The system leverages browser caching for static resources such as CSS stylesheets and JavaScript files, reducing the number of HTTP requests required for page rendering. Session-based caching is implemented for frequently accessed data such as the current user's role and the product catalog.

Minimal HTTP Requests: The system architecture minimizes the number of HTTP requests required for each page load by combining related resources where practical and avoiding unnecessary AJAX calls for data that can be included in the initial page response.

These performance considerations ensure that ProfitLens operates responsively in its target deployment environment, providing a smooth user experience without requiring specialized hardware or network infrastructure.

3.6 Database Design

3.6.1 Database Schema Overview

The ProfitLens database, named `profit_lens`, serves as the central data repository for the entire system. It is designed using a relational database model that ensures data integrity, reduces redundancy, and supports efficient querying across all system modules. The database consists of four primary entities: users, products, sales, and expenses. Each entity is implemented as a database table with specific attributes that capture the essential data needed for profit monitoring and financial reporting.

The schema design follows the principles of normalization to ensure that data is stored efficiently and can be updated without introducing anomalies. The relationships between tables are established through foreign keys that maintain referential integrity, ensuring that related records remain consistent across the database. The database uses MySQL (MariaDB) as its relational database management system, which is compatible with the PHP backend through the MySQLi extension.

3.6.2 Entity Descriptions

Users Table

The users table stores authentication and authorization data for all system users. It is the foundation of the system's security model, managing access to protected pages and ensuring that only authorized individuals can view or modify financial records.

Key attributes include: `id` (unique identifier), `email` (unique login credential), `password` (bcrypt-hashed password), `role` (admin or user designation), and `created_at` (account creation timestamp).

Products Table

The products table contains the product catalog that supports sales recording and inventory management functions. Each product record includes attributes necessary for transaction processing, profit calculation, and inventory tracking.

Key attributes include: id (unique identifier), name (product name), category (product category), cost_price (wholesale cost), selling_price (retail price), stock_quantity (current inventory level), and created_at (record creation timestamp).

Sales Table

The sales table records individual sales transactions, capturing all data necessary for revenue calculation, profit analysis, and inventory management.

Key attributes include: id (unique identifier), product_id (foreign key to products table), quantity (units sold), unit_price (selling price at time of sale), total (calculated sale amount), sale_date (date of transaction), and created_at (record creation timestamp).

Expenses Table

The expenses table stores records of all business expenses, categorized by type for analysis and monitoring purposes.

Key attributes include: id (unique identifier), category (expense category), description (expense description), amount (expense amount), expense_date (date of expense), and created_at (record creation timestamp).

3.6.3 Data Flow Diagrams

DFD Level 0 (Context Diagram)

The context diagram illustrates the high-level interactions between the system and its external entities. The primary actor is the Admin User, who inputs sales, expense, and product data into the system and receives visual financial insights in the form of dashboards, reports, and charts.

DFD Level 1

The detailed process diagram breaks down the system's internal processes into four core functions:

1. Authentication: User login and session management
2. Inventory Management: Product CRUD operations and stock tracking
3. Sales Processing: Transaction recording, calculation, and stock deduction
4. Report Generation: Data aggregation, calculation, and visualization

Each process is documented with its inputs, outputs, and data stores, providing a comprehensive view of how data flows through the system.

3.7 Testing Procedures

3.7.1 Unit Testing

Unit testing involved the isolated testing of specific functions and components to verify correct behavior. Key unit tests included:

- **Login Verification:** Testing that the authentication system correctly validates credentials against hashed passwords and rejects invalid login attempts.
- **Profit Calculation:** Testing that the profit calculation formula ($\text{Net Profit} = \text{Total Sales} - \text{Total Expenses}$) produces correct results for various input combinations, including edge cases such as zero sales, zero expenses, and negative profit scenarios.
- **Currency Formatting:** Testing that the `formatMoney()` function correctly formats numeric amounts as Philippine Peso currency strings (for example, 15000 formatting as “₱15,000.00”).
- **Stock Deduction:** Testing that sales transactions correctly deduct the appropriate quantity from product stock records and that attempted sales exceeding available stock are properly rejected.
- **Expense Totals:** Testing that expense aggregation produces correct totals across different time periods and categories.

3.7.2 Integration Testing

Integration testing verified that different modules work together seamlessly. Key integration tests included:

- **Sales to Reports:** Testing that sales transactions entered through the sales module correctly appear in the financial reports and revenue analysis modules.
- **Expenses to Dashboard:** Testing that expense records entered through the expense tracking module correctly update the dashboard summaries and expense breakdown visualizations.
- **Products to Sales:** Testing that product information entered through the product management module is correctly referenced in sales transactions and that stock updates propagate correctly.
- **Authentication to All Modules:** Testing that role-based access control functions correctly across all system modules, preventing unauthorized access while permitting authorized operations.

3.7.3 Usability Testing

Usability testing involved having the store owner and staff interact with the system to confirm that the interface is intuitive, the workflows are logical, and the system can be operated efficiently without specialized technical training.

Usability testing sessions were conducted with three participants: the store owner and two staff members. Participants were asked to perform common tasks including recording a sales transaction, entering an expense record, viewing the financial

dashboard, and generating a monthly report. The research team observed the sessions and recorded any difficulties, confusion, or errors encountered by participants.

The usability testing results were used to identify areas where the interface could be improved to better serve users with limited technical expertise. Adjustments made based on usability testing feedback included modifications to form layouts, improvements to navigation labels, and enhancements to error message clarity.

3.8 Evaluation Criteria

The system was evaluated based on the following criteria:

Criterion	Description
Usefulness	Does the system solve the identified profit monitoring and expense tracking problems?
Functionality	Do all modules (login, sales, expenses, products, reports, revenue, dashboard) work correctly?
Accuracy	Are profit calculations, expense totals, and chart data mathematically correct?
Ease of Use (Usability)	Can the store owner and staff operate the system without technical training?
Reliability	Does the system produce consistent results and handle edge cases (e.g., empty months, zero sales) without errors?
Time Efficiency	Does the system reduce the time required to record transactions and generate financial reports compared to manual methods?

3.8.1 Evaluation Methodology

Each criterion was assessed through a combination of testing activities and user feedback:

Usefulness was evaluated through structured interviews with the store owner and staff, assessing whether the system's functionality addresses the specific pain points identified during requirements gathering. Participants were asked to rate the system's usefulness on a Likert scale and to provide qualitative feedback on specific features.

Functionality was evaluated through comprehensive testing of all system modules, verifying that each feature performs its intended function correctly. Test cases were developed for each module and executed systematically, with results documented and any issues identified.

Accuracy was evaluated through mathematical verification of system outputs, comparing manually calculated values against system-generated results for the same input data. This verification covered profit calculations, expense totals, chart data points, and report summaries.

Ease of Use was evaluated through usability testing sessions with the target user population, observing task completion rates, error rates, and user satisfaction. Participants were also asked to rate the system's ease of use on a Likert scale and to identify any areas of difficulty.

Reliability was evaluated through testing of edge cases and boundary conditions, including empty data sets, zero values, very large numbers, and simultaneous user access scenarios. The system was also monitored during extended usage periods to identify any performance degradation or stability issues.

Time Efficiency was evaluated through comparison of the time required to perform common tasks using the system versus the estimated time required for the same tasks using manual methods. This comparison covered transaction recording, profit calculation, report generation, and expense tracking activities.

3.9 Project Management and Timeline

The ProfitLens development project followed a structured timeline that spanned approximately four months from initial planning through final deployment. Effective project management was essential for coordinating the activities of five team members, maintaining progress toward deadlines, and ensuring that the final product met quality standards.

3.9.1 Project Timeline

The project progressed through the following phases:

Month 1: Planning and Requirements Gathering (Weeks 1-4) - Week 1: Team formation, role assignment, and initial project planning - Week 2: Literature review and theoretical framework development - Week 3: Site visits, observation, and interview sessions at reference business - Week 4: Requirements documentation, project proposal finalization

Month 2: Design and Architecture (Weeks 5-8) - Week 5: Database schema design and ERD development - Week 6: User interface wireframing and visual design - Week 7: System architecture documentation and DFD creation - Week 8: Sprint planning and development environment setup

Month 3: Development and Testing (Weeks 9-13) - Week 9: Backend development including authentication and database connectivity - Week 10: Sales recording and expense tracking module development - Week 11: Financial reporting and revenue analysis module development - Week 12: Dashboard development and chart generation implementation - Week 13: Integration testing and bug fixing

Month 4: Evaluation and Documentation (Weeks 14-17) - Week 14: Usability testing with store owner and staff - Week 15: System refinement based on testing feedback - Week 16: Documentation writing and formatting - Week 17: Final review, presentation preparation, and project submission

3.9.2 Team Roles and Responsibilities

Each team member held specific responsibilities that aligned with their individual strengths and academic specialization:

Project Lead / Systems Analyst (Ojeda, Dee Lord): Responsible for overall project coordination, requirements gathering, stakeholder communication, and systems analysis. Led the interview and observation sessions at the reference business and served as the primary liaison between the development team and the business owner.

Backend Developer (Abolucion, Marc Emmanuel): Responsible for PHP backend development, including authentication system implementation, business logic coding, database query development, and server-side security implementation.

Database Specialist (Echalar, Wesley): Responsible for MySQL database design, schema implementation, data integrity verification, query optimization, and database security configuration.

Frontend Developer (Panizales, Kent Justin): Responsible for HTML5/CSS3 page construction, JavaScript interactivity implementation, SVG chart generation, and responsive design implementation.

Quality Assurance / Documentation Lead (Dolorfino, Jhoana Angela): Responsible for test case development and execution, usability testing coordination, documentation writing and formatting, and final project presentation preparation.

3.9.3 Communication and Coordination

The team maintained regular communication through weekly in-person meetings supplemented by daily messaging check-ins through a group chat platform. Progress was tracked using a shared task management system that documented completed tasks, upcoming deadlines, and any issues requiring team discussion.

Weekly progress reports were submitted to the project adviser, documenting completed activities, planned activities for the following week, and any challenges encountered. These reports enabled early identification of potential delays and facilitated timely intervention to keep the project on schedule.

3.10 Risk Management

The development team identified several potential risks to project success and developed mitigation strategies to address them:

Risk 1: Scope Creep: The tendency for project requirements to expand beyond the original scope, potentially delaying completion. Mitigation: Strict adherence to the

documented requirements specification, with any proposed additions evaluated against project timeline constraints before approval.

Risk 2: Technical Challenges: Unanticipated technical difficulties during development, particularly related to database design, chart generation, or security implementation. Mitigation: Allocation of buffer time in the development schedule for troubleshooting, and consultation with technical documentation and online developer communities when encountering unfamiliar challenges.

Risk 3: Stakeholder Availability: Potential difficulty in scheduling testing sessions with the business owner and staff due to their operational commitments. Mitigation: Advance scheduling of testing sessions with flexible timing options, and development of self-guided testing procedures that could be completed independently if in-person sessions were not possible.

Risk 4: Data Security Concerns: Risk of exposing sensitive business financial data during development or testing. Mitigation: Use of anonymized test data during development, implementation of security best practices throughout the codebase, and formal data handling agreements with the business owner.

Risk 5: Deployment Challenges: Potential difficulties in deploying the system to the business environment, including hardware compatibility, network configuration, and user training requirements. Mitigation: Development of a comprehensive installation guide, creation of a user manual with step-by-step instructions, and provision of hands-on training sessions for the business owner and staff.

3.11 Quality Assurance Framework

The quality assurance framework for the ProfitLens project encompassed multiple levels of review and testing to ensure that the final product met both academic standards and practical business requirements.

3.11.1 Code Quality Standards

The development team established and adhered to coding standards throughout the project to ensure code readability, maintainability, and consistency. These standards included:

- Consistent indentation using four spaces
- Descriptive variable and function naming conventions
- Comments explaining non-obvious logic and complex calculations
- Separation of concerns between database access, business logic, and presentation
- Consistent error handling and user feedback patterns

3.11.2 Documentation Standards

All project documentation followed consistent formatting and structural conventions to ensure readability and professionalism. Documentation standards included:

- Consistent heading hierarchy and numbering
- Formal academic writing style appropriate for capstone documentation
- Citation of all external sources using numbered reference format
- Inclusion of figures, tables, and diagrams to support textual explanations
- Regular review cycles to identify and correct errors, inconsistencies, and gaps

3.11.3 Testing Standards

The testing process followed established software testing principles to ensure comprehensive coverage and reliable results:

- Test case development before test execution
- Documentation of both passing and failing test results
- Regression testing after bug fixes to verify that corrections did not introduce new issues
- User acceptance testing with actual target users to validate real-world usability

3.12 Deployment Strategy

The deployment of ProfitLens to the reference business environment followed a structured approach designed to minimize disruption to business operations while ensuring successful system adoption.

3.12.1 Pre-Deployment Activities

Before deploying the system to the production environment, the team completed several preparatory activities:

- Installation and configuration of XAMPP server environment on the business computer
- Database creation and initial data population
- System deployment and configuration verification
- Security review and hardening
- Backup procedure establishment

3.12.2 User Training

Comprehensive user training was provided to the business owner and staff to ensure confident and competent system operation. Training covered:

- System login and navigation
- Product management (adding, editing, and deleting products)
- Sales recording procedures
- Expense tracking procedures
- Report generation and interpretation
- Dashboard interpretation and usage
- Basic troubleshooting and error resolution

Training sessions were conducted in person at the business location, using the actual production system and real business data to ensure maximum relevance and practical applicability.

3.12.3 Post-Deployment Support

Following the initial deployment, the development team provided ongoing support to address any issues or questions that arose during the system's initial operational period. This support included:

- On-call availability for technical assistance
- Follow-up visits to verify system performance and address emerging issues
- Minor adjustments to system configuration based on user feedback
- Documentation updates to reflect any deployment-specific modifications

3.13 Limitations of the Methodology

The research methodology employed in this study, while rigorous and appropriate for the project's objectives, has several limitations that should be acknowledged:

Single Site Study: The research was conducted at a single auto parts store in Iloilo City, which limits the generalizability of the findings to other business contexts. While the challenges identified are consistent with those documented in broader SME research, the specific experiences and outcomes at this location may not be representative of all small auto parts retailers.

Limited Testing Period: Usability testing was conducted over a relatively short period, which may not capture all the challenges and issues that would emerge during extended daily use. Long-term performance and adoption patterns may differ from those observed during the testing period.

Self-Selection Bias: The business owner and staff who participated in the research were willing volunteers, which may introduce bias in the usability testing results. Users who are less technologically inclined or less willing to adopt new systems may experience different challenges than those documented in this study.

Technical Constraints: The use of PHP-MySQL technology, while appropriate for the project's scope and the team's capabilities, may not represent the optimal technology stack for all similar projects. More modern frameworks and cloud-based architectures may offer advantages in scalability, maintainability, and deployment flexibility.

Absence of Controlled Experiment: The research did not include a controlled experiment comparing system performance against manual methods under identical conditions. While time efficiency comparisons were conducted, these were based on estimates rather than rigorous experimental measurements.

These limitations should be considered when interpreting the research findings and when applying the methodology or findings to other contexts.

REFERENCES

- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99-120.
- Brandon-Jones, A., & Kauppi, K. (2017). Examining the antecedents of the technology acceptance model within e-procurement. *International Journal of Operations & Production Management*, 37(1), 61-77.
- Connolly, R., & Begg, C. (2014). *Database systems: A practical approach to design, implementation, and management* (6th ed.). Pearson Education.
- Costa Melo, I., Queiroz, G. A., Alves, P. N., de Sousa, T. B., Yushimito, W. F., & Pereira, J. (2023). Sustainable digital transformation in small and medium enterprises (SMEs): A review on performance. *Heliyon*, 9(2), e13908.
- Creswell, J. W. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). SAGE Publications.
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319-340.
- Department of Trade and Industry. (2020). *MSME development plan 2020-2024*. DTI Philippines.
- Department of Trade and Industry - Region VII. (2021). *Auto parts retail industry profile*. DTI Region VII.
- Ghobakhloo, M., Hong, T. S., Sabouri, M. S., & Zulkifli, N. (2012). Strategies for successful information technology adoption in small and medium-sized enterprises. *Information*, 3(1), 36-77.
- Hevner, A. R., & Chatterjee, S. (2020). *Design science research in information systems*. Springer.
- Ifinedo, P. (2011). Internet/e-business technologies acceptance in Canada's SMEs: An exploratory investigation. *Internet Research*, 21(3), 255-281.
- Johnson, R. A., & Johnson, D. E. (2004). Systems theory and management of organizations. In K. M. Jones & D. E. Johnson (Eds.), *Modern perspectives on organizational theory* (pp. 45-62). University of Michigan Press.
- Laudon, K. C., & Laudon, J. P. (2020). *Management information systems* (16th ed.). Pearson.
- McKinney, E., & Yoos, C. J. (2010). SME financial management in developing nations. *Journal of Business Finance & Accounting*, 37(9), 1187-1210.
- Mubarak, M. F., Shaikh, F. A., Mubarik, M., Samo, K. A., & Mastoi, S. (2019). The impact of digital transformation on business performance: A study of Pakistani SMEs. *Engineering Technology & Applied Science Research*, 9(6), 4918-4925.

MySQL AB. (2022). MySQL 8.0 reference manual. Oracle Corporation.

Najib, M., & Fahma, F. (2020). Investigating the adoption of digital payment system through an extended technology acceptance model: An insight from the Indonesian small and medium enterprises. *International Journal on Advanced Science, Engineering and Information Technology*, 10(4), 1161-1167.

National Economic and Development Authority. (2021). Annual reports: SME sector performance. NEDA.

OWASP Foundation. (2021). OWASP Top 10 web application security risks.

PHP Documentation. (2023). PHP manual. The PHP Group.

Philippine Statistics Authority. (2022). Micro, small, and medium enterprise statistics in the Philippines. PSA.

Prabowo, W. A., & Wiguna, C. (2021). Sistem informasi UMKM bengkel berbasis web menggunakan metode SCRUM. *Jurnal Media Informatika Budidarma*, 5(1), 2604.

Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.

Sanistasya, P. A., Raharjo, K., & Iqbal, M. (2019). The effect of financial literacy and financial inclusion on small enterprises performance in East Kalimantan. *Jurnal Economia*, 15(1), 23192.

Shahadat, M. M. S., Nekomahmud, M., Ebrahimi, P., & Fekete-Farkas, M. (2023). Digital technology adoption in SMEs: What technological, environmental and organizational factors influence in emerging countries? *Global Business Review*, 24(1), 1-22.

Teng, X., Wu, Z., & Yang, F. (2022). Research on the relationship between digital transformation and performance of SMEs. *Sustainability*, 14(10), 6012.

Venkatesh, V., & Bala, H. (2008). Technology acceptance model 3 and a research agenda on interventions. *Decision Sciences*, 39(2), 273-315.

Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, 46(2), 186-204.

von Bertalanffy, L. (1968). *General system theory: Foundations, development, applications*. George Braziller.

Vrontis, D., Chaudhuri, R., & Chatterjee, S. (2022). Adoption of digital technologies by SMEs for sustainability and value creation: Moderating role of entrepreneurial orientation. *Sustainability*, 14(13), 7949.

APPENDIX A: EXPANDED TECHNICAL DOCUMENTATION

A.1 System Module Specifications

A.1.1 Authentication Module Detailed Specification

The authentication module serves as the security gateway for the ProfitLens system, controlling access to all protected functionality and data. This section provides detailed technical specifications for the authentication system implementation.

User Registration Process:

The user registration process collects essential account information and creates a new user record in the database. The process follows these steps:

1. The user navigates to the registration page (register.php).
2. The registration form presents fields for email address, password, password confirmation, and role selection (admin or user).
3. Client-side JavaScript validation checks that all required fields are completed, that the email address follows a valid format, that the password meets minimum complexity requirements (at least 8 characters), and that the password confirmation matches the original password.
4. Upon form submission, the server-side PHP script performs additional validation including checking for duplicate email addresses in the database and verifying that all inputs meet security requirements.
5. If validation passes, the password is hashed using PHP's password_hash() function with the default bcrypt algorithm.
6. A new user record is inserted into the users table with the email, hashed password, role designation, and a timestamp of account creation.
7. The user is redirected to the login page with a success message.

User Login Process:

The user login process authenticates the user's identity and establishes a session for maintaining access across multiple page requests. The process follows these steps:

1. The user navigates to the login page (index.php).
2. The login form presents fields for email address and password.
3. Upon form submission, the server-side PHP script retrieves the user record matching the submitted email address.
4. If no matching user is found, a generic error message is displayed (to prevent user enumeration).
5. If a matching user is found, the submitted password is verified against the stored hash using PHP's password_verify() function.
6. If verification fails, a generic error message is displayed.
7. If verification succeeds, the session identifier is regenerated to prevent session fixation attacks.

8. The user's ID, email, and role are stored in the session superglobal.
9. The user is redirected to the dashboard page (dashboard.php).

Session Management:

Session management maintains the user's authenticated state across multiple page requests. Key implementation details include:

- Session data is stored server-side in PHP's default session storage mechanism.
- Session cookies are configured with appropriate security attributes including HttpOnly (preventing JavaScript access) and SameSite (preventing cross-site request forgery).
- Session timeout is implemented to automatically log out users after a configurable period of inactivity (default: 30 minutes).
- The session is validated on every protected page request to ensure continued authorization.
- Session destruction occurs upon explicit logout or timeout.

Access Control Implementation:

The requireAdmin() function enforces role-based access control for admin-protected pages. The function checks the user's role from the session data and redirects non-admin users to an access-denied page. This function is included at the top of every page that requires admin-level access.

A.1.2 Product Management Module Detailed Specification

The product management module provides complete CRUD (Create, Read, Update, Delete) functionality for the product catalog that supports all sales and inventory operations.

Product Record Structure:

Attribute	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique product identifier
name	VARCHAR(255)	NOT NULL	Product name
category	VARCHAR(100)	NOT NULL	Product category
cost_price	DECIMAL(10, 2)	NOT NULL, CHECK > 0	Wholesale cost price
selling_price	DECIMAL(10, 2)	NOT NULL, CHECK > 0	Retail selling price
stock_quantity	INT	NOT NULL, CHECK >= 0	Current inventory level

Attribute	Data Type	Constraints	Description
created_at	TIMESTAMP	DEFAULT CURRENT_TIME STAMP	Record creation timestamp

Product Addition Process:

1. Admin user navigates to the product management page (products.php).
2. User clicks the “Add Product” button to display the product addition form.
3. User completes the form with product name, category, cost price, selling price, and initial stock quantity.
4. Client-side validation checks that all required fields are completed, that prices are positive numbers, and that stock quantity is a non-negative integer.
5. Server-side validation verifies data integrity and checks for duplicate product names.
6. The new product record is inserted into the products table.
7. A success message is displayed and the product list is refreshed.

Product Edit Process:

1. Admin user locates the product in the product list and clicks the “Edit” button.
2. The product edit form is populated with the current values for all attributes.
3. User modifies the desired attributes and submits the form.
4. Validation checks are performed for both client-side and server-side inputs.
5. The product record is updated in the database with the new values.
6. A success message is displayed and the product list is refreshed.

Product Deletion Process:

1. Admin user locates the product in the product list and clicks the “Delete” button.
2. A confirmation dialog is displayed to prevent accidental deletion.
3. If the user confirms deletion, the product record is removed from the database.
4. A success message is displayed and the product list is refreshed.

Price Calculation Logic:

The system calculates profit margin for each product using the formula:

$$\text{Profit Margin} = ((\text{Selling Price} - \text{Cost Price}) / \text{Selling Price}) \times 100\%$$

This calculation is displayed in the product list to provide visibility into product profitability.

A.1.3 Sales Recording Module Detailed Specification

The sales recording module captures sales transactions and performs the associated calculations and updates.

Sales Transaction Record Structure:

Attribute	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique transaction identifier
product_id	INT	FOREIGN KEY -> products.id	Reference to product sold
quantity	INT	NOT NULL, CHECK > 0	Units sold
unit_price	DECIMAL(10, 2)	NOT NULL	Price per unit at time of sale
total	DECIMAL(10, 2)	NOT NULL	Calculated total (quantity x unit_price)
sale_date	DATE	NOT NULL	Date of transaction
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Sales Recording Process:

1. Admin user navigates to the sales module (profit.php).
2. The sales form displays a searchable product dropdown, quantity input field, and date picker.
3. User selects a product from the dropdown (which filters based on typed characters).
4. User enters the quantity sold and selects the sale date.
5. Client-side JavaScript automatically calculates and displays the transaction total (quantity x selling_price) as the user enters values.
6. Client-side validation checks that a product is selected, quantity is a positive integer, and date is valid.
7. Server-side validation verifies that the selected product exists and that sufficient stock is available.
8. The transaction total is calculated: $\text{total} = \text{quantity} \times \text{unit_price}$ (where unit_price is the product's current selling_price).
9. A new sales record is inserted into the sales table.
10. The product's stock_quantity is decremented by the sold quantity.
11. If the resulting stock_quantity would be negative, the transaction is rejected with an appropriate error message.
12. A success message is displayed and the recent transactions list is updated.

Automatic Calculation Logic:

The sales recording module implements several automatic calculations:

- Transaction Total: quantity x selling_price (calculated and displayed in real-time during form completion)
- Running Daily Total: SUM of all transaction totals for the current date
- Running Monthly Total: SUM of all transaction totals for the current month
- Net Profit Contribution: transaction total - (quantity x cost_price) for each transaction

A.1.4 Expense Tracking Module Detailed Specification

The expense tracking module captures and categorizes business expenses for monitoring and analysis purposes.

Expense Record Structure:

Attribute	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique expense identifier
category	VARCHAR(100)	NOT NULL	Expense category
description	TEXT	NULL	Expense description
amount	DECIMAL(10, 2)	NOT NULL, CHECK > 0	Expense amount
expense_date	DATE	NOT NULL	Date of expense
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Expense Categories:

The system supports the following predefined expense categories:

1. Office Supplies: Paper, pens, printer supplies, and other office materials
2. Advertising: Marketing materials, online advertising, promotional items
3. Utilities: Electricity, water, internet, and other utility bills
4. Salaries and Wages: Employee compensation and benefits
5. Rent: Monthly rental payments for business premises
6. Transportation: Delivery costs, fuel, and travel expenses
7. Maintenance: Equipment maintenance and repair costs
8. Miscellaneous: Other expenses not fitting existing categories

Expense Totaling Logic:

The module calculates expense totals using the following queries:

- Monthly Expense Total: $\text{SUM}(\text{amount})$ WHERE expense_date falls within the current month
- Annual Expense Total: $\text{SUM}(\text{amount})$ WHERE expense_date falls within the current year
- Category Subtotal: $\text{SUM}(\text{amount})$ WHERE category = [selected category] AND expense_date falls within the specified period
- Percentage of Revenue: $(\text{Category Subtotal} / \text{Total Revenue}) \times 100\%$

High-Spend Indicator Logic:

The system displays visual indicators when expense categories exceed predefined thresholds relative to revenue:

- Green indicator: Category expense < 15% of total revenue (normal spending)
- Yellow indicator: Category expense between 15% and 25% of total revenue (elevated spending)
- Red indicator: Category expense > 25% of total revenue (high spending requiring attention)

A.1.5 Financial Reports Module Detailed Specification

The financial reports module generates comprehensive financial summaries and visualizations.

Report Types:

1. **Monthly Profit and Loss Report:** Displays revenue, expenses, and net profit for a single month in tabular format.
2. **Annual Comparison Report:** Displays a January-to-December grouped bar chart comparing revenue and expenses across all twelve months.
3. **Year-over-Year Comparison:** Compares financial performance between two selected calendar years.
4. **Monthly Expense Breakdown:** Shows a detailed breakdown of expenses by category for a selected month.

Chart Generation Implementation:

The module generates SVG charts using PHP server-side code that constructs SVG markup based on database data. The chart generation process follows these steps:

1. Database queries retrieve the raw financial data for the selected time period.
2. PHP code processes the data to calculate totals, percentages, and scaling factors.
3. SVG markup is constructed programmatically, including axes, labels, data bars/lines, and legends.
4. The SVG markup is embedded directly in the HTML response, rendering the chart in the user's browser without requiring external libraries.

Data Aggregation Queries:

The module uses the following SQL queries for data aggregation:

Monthly Revenue: `SELECT SUM(total) FROM sales WHERE MONTH(sale_date) = [month] AND YEAR(sale_date) = [year]`

Monthly Expenses: `SELECT SUM(amount) FROM expenses WHERE MONTH(expense_date) = [month] AND YEAR(expense_date) = [year]`

Monthly Profit: Monthly Revenue - Monthly Expenses

Annual Data: `SELECT MONTH(sale_date) as month, SUM(total) as revenue FROM sales WHERE YEAR(sale_date) = [year] GROUP BY MONTH(sale_date)`

A.1.6 Dashboard Module Detailed Specification

The dashboard module provides at-a-glance visibility into key financial metrics.

Dashboard Components:

1. **Net Profit Card:** Displays the current month's net profit calculated as total revenue minus total expenses.
2. **Growth Indicator:** Shows the percentage change in net profit compared to the previous month.
3. **Revenue Distribution Chart:** A mini bar chart showing monthly revenue distribution for the current year.
4. **Best-Selling Product:** Identifies the product with the highest total revenue for the current month.
5. **Expense Breakdown:** A visual summary of expenses by category for the current month.
6. **Recent Transactions:** A list of the five most recent sales transactions.

Growth Calculation:

Growth Percentage = $((\text{Current Month Profit} - \text{Previous Month Profit}) / \text{ABS}(\text{Previous Month Profit})) \times 100\%$

If the previous month's profit was zero or negative, the growth percentage is displayed as "N/A" to avoid misleading calculations.

Best-Selling Product Identification:

The system identifies the best-selling product using the following query:

```
SELECT p.name, SUM(s.quantity) as total_units, SUM(s.total) as total_revenue FROM
sales s JOIN products p ON s.product_id = p.id WHERE MONTH(s.sale_date) =
[current month] AND YEAR(s.sale_date) = [current year] GROUP BY p.id ORDER BY
total_revenue DESC LIMIT 1
```

A.2 Entity Relationship Description

The database relationships are defined as follows:

One-to-Many Relationship (Products to Sales): One product can be associated with many sales transactions. Each sales record references exactly one product through the `product_id` foreign key. This relationship enables the system to track which products are sold, calculate revenue by product, and maintain product-level sales history.

Independent Entities (Users, Expenses): The users and expenses tables operate independently of other tables. Users are not linked to specific transactions (as the system currently supports single-user operation). Expenses are tracked as aggregate categories rather than being linked to specific business activities.

Referential Integrity: The foreign key constraint on `sales.product_id` ensures that sales transactions can only reference existing products. If a product deletion is attempted while sales records reference that product, the database will reject the deletion to maintain referential integrity.

A.3 Security Implementation Details

A.3.1 Password Security

ProfitLens implements industry-standard password security practices:

Hashing Algorithm: bcrypt (via PHP's `password_hash()` function) **Salt:** Automatically generated and included in the hash string **Cost Factor:** Default bcrypt cost factor of 10 (configurable via `PASSWORD_BCRYPT`) **Verification:** Constant-time comparison via PHP's `password_verify()` function

The bcrypt algorithm is specifically chosen for its resistance to brute-force attacks due to its computational cost. The default cost factor of 10 means that each password hash computation requires approximately 100ms on standard hardware, making large-scale brute-force attacks computationally infeasible.

A.3.2 SQL Injection Prevention

All database queries use MySQLi prepared statements with parameterized queries. This approach separates the SQL command structure from the data values, preventing malicious data from being interpreted as SQL code.

Example of prepared statement usage:

```
// Insecure approach (NOT used in ProfitLens)
```

```
$query = "SELECT * FROM users WHERE email = '$email'";
```

```
// Secure approach (used in ProfitLens)
```

```
$stmt = $conn->prepare("SELECT * FROM users WHERE email = ?");
```

```
$stmt->bind_param("s", $email);  
$stmt->execute();
```

A.3.3 Cross-Site Scripting (XSS) Prevention

All user-generated content displayed in the browser is encoded using PHP's `htmlspecialchars()` function, which converts special characters to their HTML entity equivalents. This prevents user input from being interpreted as executable HTML or JavaScript code.

A.3.4 Cross-Site Request Forgery (CSRF) Prevention

Session-based authentication provides inherent CSRF protection, as the session identifier is required for all protected operations and is transmitted only through secure mechanisms (HTTP-only cookies). Additionally, the `SameSite` cookie attribute provides additional CSRF protection in modern browsers.

A.3.5 Session Security

Session security is maintained through several mechanisms:

- Session identifier regeneration upon authentication
 - HTTP-only session cookies (preventing JavaScript access)
 - Session timeout for inactive users
 - Session validation on every protected page request
 - Secure session storage on the server side
-

A.4 User Interface Design Principles

A.4.1 Design Philosophy

The ProfitLens user interface was designed according to the following principles:

Simplicity: The interface avoids unnecessary complexity, presenting only the information and controls needed for each specific task. Visual clutter is minimized to reduce cognitive load and improve focus on essential information.

Consistency: All pages follow consistent layout patterns, navigation structures, and visual styling. Users can predict where to find information and controls regardless of which module they are accessing.

Feedback: The system provides immediate feedback for all user actions, including success confirmations, error messages, and loading indicators. Users always know the result of their actions and the current state of the system.

Accessibility: The interface is designed to be usable by individuals with varying levels of technical expertise and visual acuity. Color choices provide sufficient contrast, form labels are clearly associated with their inputs, and error messages are descriptive and actionable.

Efficiency: Common tasks can be completed with minimal steps. Form fields are pre-populated where appropriate, dropdown menus reduce typing requirements, and keyboard shortcuts are supported where practical.

A.4.2 Color Scheme

The ProfitLens interface uses a professional color scheme that supports readability and visual hierarchy:

Primary Color: Blue (#2563EB) - Used for navigation elements, buttons, and interactive components **Secondary Color:** Green (#16A34A) - Used for success indicators and positive values **Warning Color:** Yellow (#CA8A04) - Used for caution indicators and neutral alerts **Danger Color:** Red (#DC2626) - Used for error indicators, negative values, and high-spend warnings **Neutral Colors:** Gray scale (#F9F6F6 to #111827) - Used for backgrounds, text, and borders

A.4.3 Typography

The interface uses system fonts for optimal rendering across different devices and operating systems:

Headings: System font stack with sans-serif preference (e.g., Segoe UI, Roboto, Helvetica Neue) **Body Text:** System font stack with sans-serif preference **Monospace Elements:** System monospace font for code snippets and technical data (e.g., Consolas, Monaco, Courier New)

Font sizes follow a clear hierarchy: - Page titles: 24px, bold - Section headings: 18px, bold - Subsection headings: 16px, semibold - Body text: 14px, regular - Small text: 12px, regular

A.4.4 Layout Structure

All pages follow a consistent layout structure:

Header: Fixed navigation bar containing the ProfitLens logo, main navigation links, and user menu (including logout option).

Sidebar: Left sidebar containing module navigation links, collapsible on smaller screens.

Main Content Area: Primary content region that displays the current module's interface, forms, and data.

Footer: Minimal footer containing copyright information and version number.

A.5 Deployment Architecture

A.5.1 Development Environment

The development environment was configured as follows:

- **Operating System:** Windows 10/11
- **Web Server:** Apache (via XAMPP 8.x)
- **Database Server:** MariaDB (via XAMPP 8.x)
- **PHP Version:** PHP 8.x
- **Development IDE:** Visual Studio Code
- **Version Control:** Git (local repository)
- **Browser Testing:** Google Chrome, Mozilla Firefox, Microsoft Edge

A.5.2 Production Environment

The production environment at the reference business was configured as follows:

- **Hardware:** Desktop computer (minimum specifications: Intel Core i3 processor, 4GB RAM, 256GB SSD)
- **Operating System:** Windows 10
- **Web Server:** Apache (via XAMPP)
- **Database Server:** MariaDB (via XAMPP)
- **PHP Version:** PHP 8.x
- **Network:** Local network (LAN) with internet connection for remote access (if enabled)
- **Backup:** Scheduled database backup to external storage device

A.5.3 Installation Procedures

The installation process follows these steps:

1. Download and install XAMPP from the official Apache Friends website.
2. Start the Apache and MySQL services through the XAMPP Control Panel.
3. Copy the ProfitLens application files to the XAMPP htdocs directory.
4. Create the profit_lens database using the provided SQL schema.
5. Import the sample data (optional) using the provided SQL file.
6. Configure the database connection settings in includes/config.php.
7. Access the application through a web browser at <http://localhost/profitlens>.
8. Create the initial admin user account through the registration page.
9. Log in with the admin credentials and begin adding products and recording transactions.

A.5.4 Backup Procedures

Regular backup procedures should be followed to protect against data loss:

Daily Backup: Export the profit_lens database to a SQL file using phpMyAdmin or mysqldump command.

Weekly Backup: Create a compressed archive of both the database export and the application files.

Monthly Backup: Verify backup integrity by performing a test restore to a separate location.

Backup Storage: Store backup files on an external storage device or cloud storage service separate from the production system.

A.6 Glossary of Technical Terms

A.6.1 Web Development Terms

HTML5 (HyperText Markup Language 5): The latest version of the standard markup language for creating web pages and web applications. HTML5 provides semantic elements, multimedia support, and enhanced form controls.

CSS3 (Cascading Style Sheets 3): The latest version of the style sheet language used for describing the presentation of web pages. CSS3 introduces advanced layout capabilities, animations, and responsive design features.

JavaScript: A programming language that enables interactive web pages. JavaScript runs in the user's web browser and provides client-side functionality including form validation, dynamic content updates, and user interface interactions.

PHP (Hypertext Preprocessor): A server-side scripting language designed for web development. PHP code is executed on the web server, generating HTML output that is sent to the user's browser.

MySQL: An open-source relational database management system (RDBMS) that uses Structured Query Language (SQL) for data management. MySQL is widely used in web applications due to its reliability, performance, and ease of use.

MariaDB: A community-developed fork of MySQL that provides enhanced features and performance while maintaining compatibility with MySQL applications.

A.6.2 Security Terms

SQL Injection: A code injection technique that exploits security vulnerabilities in web application database layers. SQL injection occurs when user input is incorrectly filtered or not strongly typed and is unexpectedly executed as part of an SQL command.

Cross-Site Scripting (XSS): A security vulnerability that allows attackers to inject malicious client-side scripts into web pages viewed by other users. XSS attacks can be used to steal session credentials, deface websites, or redirect users to malicious sites.

Cross-Site Request Forgery (CSRF): An attack that forces authenticated users to submit unintended requests to web applications. CSRF attacks exploit the trust that a site has in the user's browser.

Session Hijacking: An attack where an attacker takes over a user's legitimate session by stealing or predicting the session token. Session hijacking allows attackers to impersonate users and access their accounts.

Brute Force Attack: An attack that systematically tries all possible combinations of characters until the correct password is found. Brute force attacks are mitigated by password complexity requirements, account lockout mechanisms, and computationally expensive hashing algorithms.

Bcrypt: A password hashing function designed to be computationally expensive, making brute-force attacks more difficult. Bcrypt includes a built-in salt and configurable cost factor.

A.6.3 Database Terms

Primary Key: A unique identifier for each record in a database table. Primary keys ensure that each record can be uniquely identified and referenced.

Foreign Key: A field in one table that references the primary key of another table. Foreign keys establish relationships between tables and enforce referential integrity.

Index: A data structure that improves the speed of data retrieval operations on a database table. Indexes create a lookup mechanism that allows the database engine to find matching rows without scanning entire tables.

Normalization: The process of organizing database tables to reduce data redundancy and improve data integrity. Normalization involves decomposing tables into smaller, well-structured tables with clear relationships.

Prepared Statement: A SQL statement template that is compiled separately from user-supplied data values. Prepared statements prevent SQL injection by ensuring that user input is treated as data, not as executable code.

ACID Properties: Atomicity, Consistency, Isolation, Durability - four properties that guarantee reliable processing of database transactions. ACID compliance ensures that transactions are processed reliably even in the event of system failures.

A.7 Future Research Directions

A.7.1 Quantitative Impact Assessment

A rigorous quantitative study comparing financial management outcomes before and after ProfitLens deployment could provide empirical evidence of the system's impact on business performance. Such a study would require controlled measurements of key metrics including profit calculation accuracy, time spent on financial tasks, and decision-making speed over an extended period.

A.7.2 Multi-Site Comparative Study

Extending the research to include multiple auto parts retailers across different Philippine cities would enable comparative analysis of the system's effectiveness across different business contexts, market conditions, and user populations. Such a study could identify factors that influence system adoption and effectiveness in diverse settings.

A.7.3 Longitudinal Adoption Study

A longitudinal study tracking system usage patterns over an extended period (12-24 months) would provide insights into long-term adoption patterns, feature utilization evolution, and the relationship between system usage and business performance improvements over time.

A.7.4 Technology Stack Comparison

A comparative study evaluating the ProfitLens approach against alternative technology stacks (such as Python-Django, JavaScript-Node.js, or cloud-based solutions) could provide evidence regarding the optimal technology choices for SME financial monitoring systems in the Philippine context.

A.7.5 User Experience Research

In-depth user experience research using methods such as eye-tracking, think-aloud protocols, and A/B testing could provide detailed insights into how users interact with the system and identify opportunities for interface optimization.

A.7.6 Integration with Emerging Technologies

Research exploring the integration of emerging technologies such as artificial intelligence for anomaly detection, natural language interface for voice-activated data entry, or blockchain for tamper-proof financial record keeping could extend the system's capabilities and address evolving business needs.

A.8 Summary of Appendices

This appendix has provided comprehensive technical documentation for the ProfitLens system, including detailed module specifications, complete database schema, security implementation details, user interface design principles, deployment architecture, glossary of technical terms, and future research directions.

The technical documentation serves multiple purposes: it provides the detailed specifications necessary for system maintenance and enhancement, it documents the implementation decisions and rationale for future developers, and it establishes the technical foundation against which the system's functionality and security can be evaluated.

The completeness of this documentation reflects the research team's commitment to producing a thoroughly documented capstone project that can serve as a reference

model for future similar initiatives. The level of detail provided ensures that future developers or researchers can understand, maintain, extend, or replicate the system as needed.

B.6 System Requirements Specification

B.6.1 Hardware Requirements

Development Environment: - Processor: Intel Core i5 or equivalent (minimum) - Memory: 8GB RAM (recommended) - Storage: 256GB SSD (minimum) - Display: 1920x1080 resolution (minimum) - Network: Broadband internet connection for development tools and documentation access

Production Environment: - Processor: Intel Core i3 or equivalent (minimum) - Memory: 4GB RAM (minimum) - Storage: 256GB SSD (recommended) - Display: 1366x768 resolution (minimum) - Network: Local Area Network (LAN) connection for multi-user access; internet optional for remote access if configured

B.6.2 Software Requirements

Development Software: - Operating System: Windows 10/11, macOS 12+, or Linux (Ubuntu 20.04+) - Web Server: Apache 2.4+ (via XAMPP) - Database: MySQL 8.0+ / MariaDB 10.5+ - PHP: Version 8.0 or higher - Code Editor: Visual Studio Code (recommended) or equivalent - Web Browser: Google Chrome 90+, Mozilla Firefox 90+, or Microsoft Edge 90+

Production Software: - Operating System: Windows 10 or later - Web Server: Apache (via XAMPP) - Database: MySQL / MariaDB (via XAMPP) - PHP: Version 8.0 or higher - Web Browser: Any modern web browser

B.6.3 Network Requirements

Local Deployment: - Local Area Network (LAN) connection between the server computer and client devices - Static IP address configuration for the server (recommended) - Firewall configuration to allow HTTP traffic on port 80 (or configured port)

Remote Access (Optional): - Broadband internet connection with minimum 5 Mbps upload speed - Port forwarding configuration on the router (for external access) - Dynamic DNS service (if static public IP is not available) - SSL/TLS certificate (recommended for secure remote access)

B.7 Acceptance Testing Results

B.7.1 User Acceptance Testing Summary

User Acceptance Testing (UAT) was conducted with the participation of the business owner and two staff members. The following table summarizes the UAT results:

Test Scenario	Expected Result	Actual Result	Status
Login with valid credentials	Access to dashboard	Dashboard displayed	PASS
Login with invalid credentials	Error message	Error message displayed	PASS
Add new product	Product saved	Product appears in list	PASS
Edit product price	Price updated	New price reflected	PASS
Delete product	Product removed	Product no longer appears	PASS
Record sales transaction	Transaction saved	Transaction recorded	PASS
Sales exceeding stock	Error message	Error displayed, no record	PASS
Add expense record	Expense saved	Expense appears in list	PASS
Edit expense	Expense updated	Changes reflected	PASS
Delete expense	Expense removed	Record deleted	PASS
View dashboard	Dashboard displayed	All metrics visible	PASS
Generate monthly report	Report displayed	Correct data shown	PASS
View revenue analysis	Charts displayed	Visualizations rendered	PASS
View expense breakdown	Breakdown displayed	Categories listed	PASS
Logout	Session ended	Redirected to login	PASS

B.7.2 Acceptance Testing Feedback

The following feedback was collected during UAT sessions:

Business Owner Feedback: - “The dashboard is very helpful. I can see everything at a glance.” - “Recording sales is much faster than writing in the notebook.” - “The expense breakdown helps me understand where my money goes.” - “I like that I can see the profit immediately after recording sales.”

Staff Member 1 Feedback: - “The product dropdown with search is very convenient.” - “The system is easy to learn. I was comfortable using it after one session.” - “The error messages are clear and help me correct mistakes.”

Staff Member 2 Feedback: - “The monthly report is much better than calculating by hand.” - “I appreciate the stock check that prevents overselling.” - “The interface is clean and not overwhelming.”

B.8 Project Timeline Visualization

The following table provides a detailed breakdown of the project timeline with milestones and deliverables:

Week	Phase	Activities	Deliverables
1	Planning	Team formation, role assignment, initial planning	Project charter, team roster
2	Planning	Literature review, theoretical framework development	Literature review draft
3	Requirements	Site visits, observation, interviews	Observation notes, interview transcripts
4	Requirements	Requirements documentation, proposal finalization	Requirements specification, project proposal
5	Design	Database schema design, ERD development	Database schema, ERD diagrams
6	Design	UI wireframing, visual design	Wireframes, UI mockups
7	Design	System architecture, DFD creation	Architecture document, DFD diagrams
8	Design	Sprint planning, environment setup	Sprint plan, dev environment configured
9	Development	Backend: authentication, database connectivity	Login/registration functionality
10	Development	Sales recording, expense tracking modules	Sales and expense modules

Week	Phase	Activities	Deliverables
11	Development	Financial reporting, revenue analysis modules	Reports and analysis modules
12	Development	Dashboard, chart generation	Dashboard and visualizations
13	Testing	Integration testing, bug fixing	Test results, bug reports
14	Testing	Usability testing with users	UAT results, user feedback
15	Refinement	System refinement, final adjustments	Updated system
16	Documentation	Documentation writing, formatting	Documentation draft
17	Documentation	Final review, presentation preparation	Final documentation, presentation

B.9 Risk Assessment Matrix

Risk	Probability	Impact	Mitigation Strategy
Scope creep	Medium	High	Strict requirements adherence, change control process
Technical challenges	Medium	Medium	Buffer time, documentation research, peer consultation
Stakeholder unavailability	Low	High	Advance scheduling, flexible testing procedures
Data security breach	Low	High	Security best practices, data handling agreements
Deployment issues	Medium	Medium	Comprehensive installation guide, hands-on training
Team member unavailability	Low	Medium	Cross-training, task documentation, flexible scheduling

Risk	Probability	Impact	Mitigation Strategy
Technology obsolescence	Low	Low	Use of stable, widely-supported technologies
User resistance to adoption	Medium	High	User-centered design, training, demonstration of benefits

B.10 Quality Metrics Summary

Quality Metric	Target	Achieved	Status
Code readability	High	High	PASS
Documentation completeness	100%	100%	PASS
Test case coverage	>90%	95%	PASS
Usability score	>4.0/5.0	4.3/5.0	PASS
Security test pass rate	100%	100%	PASS
Performance target (page load)	<3 sec	1.8 sec	PASS
Error handling coverage	>85%	92%	PASS
User satisfaction score	>4.0/5.0	4.3/5.0	PASS

This comprehensive documentation provides the complete technical foundation for the ProfitLens system, enabling future developers, researchers, and practitioners to understand, maintain, extend, or replicate the system as needed. The level of detail ensures that the capstone project serves not only as a demonstration of academic competency but as a practical reference model for similar technology development initiatives in the Philippine SME context.

APPENDIX C: DATA FLOW DIAGRAMS

C.1 DFD Level 0 (Context Diagram)

The Context Diagram provides a high-level overview of the ProfitLens system and its interactions with external entities. This diagram shows the system as a single process bubble surrounded by the external entities that interact with it.

External Entities:

1. **Admin User:** The primary actor who interacts with the system to input data and receive information.
2. **System Clock:** Provides date and time information for timestamping transactions.

Data Flows:

Flow	From	To	Description
Login Credentials	Admin User	System	Email and password for authentication
Sales Data	Admin User	System	Product, quantity, and date for sales transactions
Expense Data	Admin User	System	Category, amount, and description for expenses
Product Data	Admin User	System	Name, category, prices, and stock for products
Financial Reports	System	Admin User	Charts, tables, and summaries
Dashboard Data	System	Admin User	Real-time financial metrics
Error Messages	System	Admin User	Validation errors and system notifications
Success Confirmations	System	Admin User	Operation completion confirmations

System Boundary:

The system boundary encompasses all internal processing including: - Authentication and session management - Data validation and sanitization - Business logic execution (profit calculation, stock deduction) - Database operations (CRUD) - Report generation and visualization - Security enforcement

C.2 DFD Level 1 (Detailed Process Diagram)

The Level 1 DFD decomposes the system into four major processes, showing how data flows between them and the data stores they access.

Process 1: Authentication

Input	Processing	Output
Login credentials	Validate against database	Authenticated session
Registration data	Hash password, create user	New user account
Session data	Validate on each request	Access decision

Process 2: Inventory Management

Input	Processing	Output
New product data	Validate, insert into database	Updated product catalog
Product modifications	Validate, update database	Updated product record
Deletion request	Check dependencies, remove record	Updated product catalog

Process 3: Transaction Processing

Input	Processing	Output
Sales transaction	Validate stock, calculate total, record sale, update stock	Sales record, updated inventory
Expense entry	Validate data, categorize, record	Expense record
Transaction queries	Aggregate by period/category	Summary data

Process 4: Reporting and Visualization

Input	Processing	Output
Date range/period	Aggregate financial data	Tabular reports
Visualization request	Generate SVG charts	Visual charts
Comparison request	Fetch multi-period data	Comparative visualizations

Data Stores:

Data Store	Description	Processes
D1: Users	User authentication data	Authentication
D2: Products	Product catalog	Inventory Management,

Data Store	Description	Processes
		Transaction Processing
D3: Sales	Sales transaction records	Transaction Processing, Reporting
D4: Expenses	Expense records	Transaction Processing, Reporting

C.3 Data Flow Descriptions

C.3.1 Sales Recording Data Flow

The complete data flow for recording a sales transaction follows these steps:

1. Admin User selects a product from the dropdown (Product data from D2).
2. Admin User enters quantity and sale date.
3. Client-side JavaScript calculates the total (quantity x selling_price).
4. Form is submitted to the server.
5. Server validates the transaction (checks stock availability from D2).
6. If valid, server inserts record into D3 (Sales).
7. Server updates stock_quantity in D2 (Products).
8. Server returns success confirmation to Admin User.

C.3.2 Expense Recording Data Flow

The complete data flow for recording an expense follows these steps:

1. Admin User enters expense category, amount, description, and date.
2. Form is submitted to the server.
3. Server validates the data (checks required fields, numeric values).
4. Server inserts record into D4 (Expenses).
5. Server returns success confirmation to Admin User.

C.3.3 Report Generation Data Flow

The complete data flow for generating a financial report follows these steps:

1. Admin User selects report type and parameters (month, year, etc.).
2. Request is submitted to the server.
3. Server constructs appropriate SQL queries.
4. Server retrieves data from D3 (Sales) and D4 (Expenses).
5. Server aggregates data by period and category.
6. Server generates SVG chart markup.
7. Server constructs HTML response with charts and tables.

8. Response is sent to Admin User's browser for rendering.
-

APPENDIX D: ENTITY RELATIONSHIP DIAGRAMS

D.1 Entity Relationship Description

The Entity Relationship Diagram (ERD) for ProfitLens defines the four primary entities and their relationships:

D.1.1 Entity Definitions

Users Entity: - Purpose: Stores authentication and authorization data - Key Attribute: id (Primary Key) - Attributes: email (UNIQUE), password, role, created_at - Relationship: Independent entity (no foreign keys to other tables)

Products Entity: - Purpose: Maintains the product catalog - Key Attribute: id (Primary Key) - Attributes: name, category, cost_price, selling_price, stock_quantity, created_at - Relationship: Parent entity in one-to-many relationship with Sales

Sales Entity: - Purpose: Records individual sales transactions - Key Attribute: id (Primary Key) - Foreign Key: product_id (references Products.id) - Attributes: quantity, unit_price, total, sale_date, created_at - Relationship: Child entity in one-to-many relationship with Products

Expenses Entity: - Purpose: Stores business expense records - Key Attribute: id (Primary Key) - Attributes: category, description, amount, expense_date, created_at - Relationship: Independent entity (expenses are not linked to specific products or sales)

D.1.2 Relationship Specifications

Products to Sales (One-to-Many): - One product can appear in many sales transactions - Each sale references exactly one product - Deleting a product with existing sales is prevented by referential integrity - This relationship enables product-level sales analysis

Key Constraints: - Foreign Key: sales.product_id REFERENCES products.id - On Delete: RESTRICT (prevents deletion of products with existing sales) - On Update: CASCADE (updates product_id in sales if product id changes, though this is unlikely)

D.1.3 Data Integrity Rules

The following data integrity rules are enforced at the database level:

1. **Entity Integrity:** Every table has a primary key that uniquely identifies each record.
2. **Referential Integrity:** Foreign key constraints ensure that related records remain consistent.

3. **Domain Integrity:** Data types and constraints ensure that column values fall within valid ranges.
 4. **User-Defined Integrity:** Business rules enforced through application logic (e.g., stock cannot go negative).
-

APPENDIX E: USER MANUAL SUMMARY

E.1 System Login

Step 1: Open your web browser and navigate to the ProfitLens URL (e.g., <http://localhost/profitlens>).

Step 2: Enter your registered email address in the Email field.

Step 3: Enter your password in the Password field.

Step 4: Click the “Login” button.

Step 5: If credentials are correct, you will be redirected to the Dashboard.

E.2 Dashboard Navigation

The Dashboard is the main landing page after login. It displays:

- **Net Profit:** Current month’s profit (Revenue minus Expenses)
- **Revenue:** Total sales for the current month
- **Expenses:** Total expenses for the current month
- **Growth:** Percentage change from the previous month

Use the sidebar navigation to access other modules: - **Dashboard:** Return to the main dashboard - **Sales:** Record and view sales transactions - **Expenses:** Record and view expense records - **Revenue:** View revenue analysis and charts - **Reports:** Generate financial reports - **Products:** Manage your product catalog

E.3 Recording a Sale

Step 1: Click “Sales” in the sidebar navigation.

Step 2: Select a product from the Product dropdown (you can type to search).

Step 3: Enter the quantity sold.

Step 4: Select the sale date.

Step 5: Review the auto-calculated total.

Step 6: Click “Record Sale” to save the transaction.

Note: The system will automatically deduct the sold quantity from the product’s stock.

E.4 Recording an Expense

Step 1: Click “Expenses” in the sidebar navigation.

Step 2: Click “Add Expense” to open the expense form.

Step 3: Select an expense category from the dropdown.

Step 4: Enter the expense amount.

Step 5: Enter a description (optional).

Step 6: Select the expense date.

Step 7: Click “Save” to record the expense.

E.5 Viewing Reports

Step 1: Click “Reports” in the sidebar navigation.

Step 2: Select the report type: - Monthly Profit and Loss - Annual Comparison - Expense Breakdown

Step 3: Select the time period (month and year).

Step 4: Click “Generate Report” to view the results.

Step 5: Review the charts and tables displayed.

E.6 Managing Products

Step 1: Click “Products” in the sidebar navigation.

Step 2: To add a new product: - Click “Add Product” - Fill in the product details (name, category, cost price, selling price, initial stock) - Click “Save”

Step 3: To edit a product: - Click “Edit” next to the product - Modify the desired fields - Click “Update”

Step 4: To delete a product: - Click “Delete” next to the product - Confirm the deletion when prompted

Note: Products with existing sales records cannot be deleted.

E.7 Logging Out

Step 1: Click your username or the logout icon in the header.

Step 2: Click “Logout” to end your session.

Note: Always log out when you are finished, especially on shared computers.

APPENDIX F: INSTALLATION GUIDE SUMMARY

F.1 Prerequisites

Before installing ProfitLens, ensure you have:

- A computer running Windows 10 or later
- At least 4GB of RAM
- At least 1GB of free disk space
- A modern web browser (Chrome, Firefox, or Edge)
- Internet connection (for downloading XAMPP)

F.2 Installation Steps

Step 1: Download and Install XAMPP

1. Visit <https://www.apachefriends.org>
2. Download XAMPP for Windows
3. Run the installer and follow the prompts
4. Install to the default location (C:)

Step 2: Start Services

1. Open XAMPP Control Panel
2. Click “Start” next to Apache
3. Click “Start” next to MySQL
4. Both services should show “Running” status

Step 3: Deploy ProfitLens Files

1. Copy the ProfitLens folder to C:
2. Ensure the folder structure is intact

Step 4: Create Database

1. Open your browser and go to <http://localhost/phpmyadmin>
2. Click “Databases” tab
3. Create a new database named “profit_lens”
4. Select the new database
5. Click “Import” tab
6. Choose the schema.sql file from the ProfitLens SQL folder
7. Click “Go” to execute the import

Step 5: Configure Database Connection

1. Open includes/config.php in a text editor
2. Verify the database settings:
 - DB_HOST: localhost
 - DB_USER: root
 - DB_PASS: (empty for default XAMPP)

- DB_NAME: profit_lens
- 3. Save the file

Step 6: Access ProfitLens

1. Open your web browser
2. Navigate to <http://localhost/profitlens>
3. You should see the login page
4. Register an admin account
5. Log in and start using the system

F.3 Troubleshooting

Problem: Cannot connect to database - Verify MySQL service is running in XAMPP Control Panel - Check database credentials in config.php - Ensure the profit_lens database exists

Problem: Page not found (404 error) - Verify ProfitLens files are in the correct htdocs location - Check that Apache service is running - Ensure the URL is correct (<http://localhost/profitlens>)

Problem: Blank page or PHP errors - Check PHP error logging in XAMPP - Verify PHP version is 8.0 or higher - Ensure all required PHP extensions are enabled

Problem: Charts not displaying - Verify JavaScript is enabled in your browser - Check browser console for JavaScript errors - Ensure SVG rendering is supported

APPENDIX G: GLOSSARY OF BUSINESS TERMS

G.1 Financial Terms

Revenue: The total amount of money generated from sales of products or services before any deductions for expenses or costs.

Expenses: The costs incurred in the process of generating revenue, including rent, utilities, supplies, salaries, and other operational costs.

Net Profit: The amount remaining after all expenses have been subtracted from total revenue. $\text{Net Profit} = \text{Total Revenue} - \text{Total Expenses}$.

Gross Profit: The difference between revenue and the cost of goods sold (COGS). $\text{Gross Profit} = \text{Revenue} - \text{COGS}$.

Profit Margin: The percentage of revenue that represents profit. $\text{Profit Margin} = (\text{Net Profit} / \text{Revenue}) \times 100\%$.

Cash Flow: The movement of money into and out of a business over a specific period.

Accounts Receivable: Money owed to the business by customers for goods or services delivered but not yet paid for.

Accounts Payable: Money the business owes to suppliers for goods or services received but not yet paid for.

Inventory: The goods and materials that a business holds for the purpose of resale.

Turnover Rate: The number of times inventory is sold and replaced over a specific period.

G.2 Retail Terms

SKU (Stock Keeping Unit): A unique identifier for each distinct product and its variants.

Point of Sale (POS): The location and time where a retail transaction is completed.

Markup: The amount added to the cost price of goods to determine the selling price.

Markdown: A reduction in the selling price of goods, typically for clearance or promotional purposes.

Stockout: A situation where a product is out of stock and unavailable for purchase.

Overstocking: Holding more inventory than can be sold in a reasonable time period.

Shrinkage: The loss of inventory due to theft, damage, errors, or other factors.

ABC Analysis: An inventory categorization method that divides items into three categories (A, B, C) based on their importance.

G.3 Technology Terms

Web Application: A software application that runs in a web browser and is accessed over a network.

Database: An organized collection of structured data stored electronically.

API (Application Programming Interface): A set of rules that allows different software applications to communicate with each other.

Authentication: The process of verifying the identity of a user or system.

Authorization: The process of determining what an authenticated user is allowed to do.

Encryption: The process of converting data into a coded format that can only be read with the correct decryption key.

Cloud Computing: The delivery of computing services (servers, storage, databases, etc.) over the internet.

Open Source: Software with source code that anyone can inspect, modify, and distribute.

Responsive Design: An approach to web design that makes web pages render well on a variety of devices and screen sizes.

User Experience (UX): The overall experience a user has when interacting with a product or system.

APPENDIX H: ABBREVIATIONS AND ACRONYMS

Abbreviation	Full Form
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
BIR	Bureau of Internal Revenue
BSIT	Bachelor of Science in Information Technology
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DSR	Design Science Research
DTI	Department of Trade and Industry
ERD	Entity Relationship Diagram
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation
LAN	Local Area Network
MSME	Micro, Small, and Medium Enterprises
MySQL	My Structured Query Language
OWASP	Open Web Application Security Project
PHP	Hypertext Preprocessor
PLS	Partial Least Square
POS	Point of Sale
PEOU	Perceived Ease of Use
PU	Perceived Usefulness
RBV	Resource-Based View
RDBMS	Relational Database Management System

Abbreviation	Full Form
ROI	Return on Investment
SKU	Stock Keeping Unit
SME	Small and Medium Enterprise
SQL	Structured Query Language
SVG	Scalable Vector Graphics
TAM	Technology Acceptance Model
TOE	Technology-Organization-Environment
UAT	User Acceptance Testing
UTAUT	Unified Theory of Acceptance and Use of Technology
VAT	Value Added Tax
XSS	Cross-Site Scripting
XAMPP	Apache + MySQL + PHP + Perl

APPENDIX I: BIBLIOGRAPHY OF RELATED STUDIES

I.1 Complete Foreign Studies Referenced

1. Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319-340.
2. Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, 46(2), 186-204.
3. Venkatesh, V., & Bala, H. (2008). Technology acceptance model 3 and a research agenda on interventions. *Decision Sciences*, 39(2), 273-315.
4. von Bertalanffy, L. (1968). *General system theory: Foundations, development, applications*. George Braziller.
5. Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.
6. Laudon, K. C., & Laudon, J. P. (2020). *Management information systems* (16th ed.). Pearson.
7. McKinney, E., & Yoos, C. J. (2010). SME financial management in developing nations. *Journal of Business Finance & Accounting*, 37(9), 1187-1210.
8. Creswell, J. W. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). SAGE Publications.

9. Hevner, A. R., & Chatterjee, S. (2020). Design science research in information systems. Springer.
10. Connolly, R., & Begg, C. (2014). Database systems: A practical approach to design, implementation, and management (6th ed.). Pearson Education.
11. OWASP Foundation. (2021). OWASP Top 10 web application security risks.
12. Johnson, R. A., & Johnson, D. E. (2004). Systems theory and management of organizations. In K. M. Jones & D. E. Johnson (Eds.), *Modern perspectives on organizational theory* (pp. 45-62). University of Michigan Press.
13. Najib, M., & Fahma, F. (2020). Investigating the adoption of digital payment system through an extended technology acceptance model. *International Journal on Advanced Science, Engineering and Information Technology*, 10(4), 1161-1167.
14. Ifinedo, P. (2011). Internet/e-business technologies acceptance in Canada's SMEs: An exploratory investigation. *Internet Research*, 21(3), 255-281.
15. Brandon-Jones, A., & Kauppi, K. (2017). Examining the antecedents of the technology acceptance model within e-procurement. *International Journal of Operations & Production Management*, 37(1), 61-77.
16. Teng, X., Wu, Z., & Yang, F. (2022). Research on the relationship between digital transformation and performance of SMEs. *Sustainability*, 14(10), 6012.
17. Mubarak, M. F., Shaikh, F. A., Mubarik, M., Samo, K. A., & Mastoi, S. (2019). The impact of digital transformation on business performance: A study of Pakistani SMEs. *Engineering Technology & Applied Science Research*, 9(6), 4918-4925.
18. Ghobakhloo, M., Hong, T. S., Sabouri, M. S., & Zulkifli, N. (2012). Strategies for successful information technology adoption in small and medium-sized enterprises. *Information*, 3(1), 36-77.
19. Vrontis, D., Chaudhuri, R., & Chatterjee, S. (2022). Adoption of digital technologies by SMEs for sustainability and value creation: Moderating role of entrepreneurial orientation. *Sustainability*, 14(13), 7949.
20. Shahadat, M. M. S., Nekmahmud, M., Ebrahimi, P., & Fekete-Farkas, M. (2023). Digital technology adoption in SMEs: What technological, environmental and organizational factors influence in emerging countries? *Global Business Review*, 24(1), 1-22.
21. Costa Melo, I., Queiroz, G. A., Alves, P. N., de Sousa, T. B., Yushimito, W. F., & Pereira, J. (2023). Sustainable digital transformation in small and medium enterprises (SMEs): A review on performance. *Heliyon*, 9(2), e13908.

22. Prabowo, W. A., & Wiguna, C. (2021). Sistem informasi UMKM bengkel berbasis web menggunakan metode SCRUM. *Jurnal Media Informatika Budidarma*, 5(1), 2604.
23. Sanistasya, P. A., Raharjo, K., & Iqbal, M. (2019). The effect of financial literacy and financial inclusion on small enterprises performance in East Kalimantan. *Jurnal Economia*, 15(1), 23192.
24. Bartolacci, F., Caputo, A., & Soverchia, M. (2019). Sustainability and financial performance of small and medium sized enterprises: A bibliometric and systematic literature review. *Business Strategy and the Environment*, 29(3), 1183-1198.
25. Li, W., Liu, K., Belitski, M., Ghobadian, A., & O'Regan, N. (2016). E-leadership through strategic alignment: An empirical study of small- and medium-sized enterprises in the digital age. *Journal of Information Technology*, 31(2), 162-177.

I.2 Complete Local Studies Referenced

1. Philippine Statistics Authority. (2022). Micro, small, and medium enterprise statistics in the Philippines. PSA.
 2. Department of Trade and Industry. (2020). MSME development plan 2020-2024. DTI Philippines.
 3. Department of Trade and Industry - Region VII. (2021). Auto parts retail industry profile. DTI Region VII.
 4. National Economic and Development Authority. (2021). Annual reports: SME sector performance. NEDA.
 5. MySQL AB. (2022). MySQL 8.0 reference manual. Oracle Corporation.
 6. PHP Documentation. (2023). PHP manual. The PHP Group.
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FINAL NOTE

This documentation represents the comprehensive technical and academic foundation for the ProfitLens project. The extensive detail provided across all chapters and appendices ensures that the project serves multiple purposes: as an academic capstone demonstrating curriculum integration, as a practical reference for similar development projects, and as a contribution to the body of knowledge regarding technology solutions for small business financial management in the Philippine context.

The research team hopes that this documentation, along with the functional system it describes, will provide lasting value to the reference business, to future capstone students, and to the broader community of researchers and practitioners working on technology adoption and digital transformation in small and medium enterprises.

APPENDIX J: SUPPLEMENTARY RESEARCH NOTES

J.1 Interview Transcript Summaries

The following are summarized key points from the structured interviews conducted with the business owner and staff during the requirements gathering phase.

J.1.1 Interview with Business Owner (French Joy Salvador)

Date: January 15, 2026 **Duration:** Approximately 45 minutes **Location:** Auto parts store, Iloilo City

Key Topics Discussed:

1. **Current Financial Tracking Methods:** The owner described using a combination of handwritten notebooks for daily sales, paper receipts stored in folders for expenses, and mental calculations for monthly profit estimation. She acknowledged that this method was time-consuming and prone to errors.
2. **Pain Points Identified:** The owner identified several specific challenges:
 - Difficulty calculating exact monthly profit due to the volume of individual transactions
 - Time spent on end-of-month reconciliation (estimated 3-4 hours per month)
 - Occasional discovery of missing entries that could not be recovered
 - Inability to identify which products were most profitable
 - Surprise when expenses exceeded budget at the end of the month
3. **Desired Features:** When asked what features would be most valuable, the owner prioritized:
 - Automatic profit calculation
 - Visual charts showing revenue and expenses
 - Alerts when spending was too high
 - Easy product and sales tracking
4. **Technology Comfort Level:** The owner described herself as comfortable with basic computer operations including web browsing and email, but had limited experience with business software applications.
5. **Business Operations:** The store operates six days per week (Monday through Saturday), with typical daily sales ranging from 10 to 30 transactions. Product

inventory consists of approximately 150 different items across multiple categories.

J.1.2 Interview with Staff Members

Date: January 16, 2026 **Duration:** Approximately 30 minutes per staff member

Location: Auto parts store, Lot 1 Blk 2 Anarose Village, Guzman St, Mandurriao, Iloilo City

Key Topics Discussed:

1. **Daily Tasks:** Staff members described their daily responsibilities including assisting customers, locating products, recording sales in the notebook, and maintaining product displays.
2. **Sales Recording Process:** Staff described the current sales recording process as writing the product name, quantity, and price in a notebook. They noted that during busy periods, entries were sometimes abbreviated or delayed, leading to potential inaccuracies.
3. **Technology Experience:** Both staff members had basic smartphone experience and were familiar with using web browsers on computers.
4. **Feedback on Proposed System:** Staff expressed enthusiasm about the proposed system, particularly the searchable product dropdown (which would speed up product lookup) and the automatic total calculation (which would eliminate mental math).

J.2 Observation Notes Summary

The research team conducted three observation sessions at the reference business to understand daily operations and identify improvement opportunities.

J.2.1 Observation Session 1

Date: January 14, 2026 **Time:** 9:00 AM - 12:00 PM

Observations: - The store received approximately 15 customer transactions during the observation period - Sales were recorded in a notebook with handwritten entries - The owner manually calculated running totals at the end of each transaction cluster - Expense receipts were placed in a folder without immediate recording - Product stock was checked physically by counting items on shelves

J.2.2 Observation Session 2

Date: January 20, 2026 **Time:** 1:00 PM - 4:00 PM

Observations: - Afternoon period saw approximately 10 transactions - Staff spent time searching for specific products in the notebook to verify previous sales prices - The owner expressed frustration about missing a receipt and being unable to reconcile an

expense - Digital resources (calculator, basic spreadsheet) were used occasionally but inconsistently

J.2.3 Observation Session 3

Date: January 25, 2026 **Time:** 10:00 AM - 2:00 PM

Observations: - Saturday morning period saw higher transaction volume (approximately 20 transactions) - The notebook became disorganized with entries spread across multiple pages - Staff had difficulty quickly finding a product's selling price for a returning customer - The owner mentioned wanting to compare this month's performance to the same month last year but had no way to do so

J.3 Questionnaire Survey Results

A structured questionnaire was distributed to gather quantitative data about user preferences and technology readiness.

J.3.1 Survey Distribution Summary

Metric	Value
Total Distributed	15
Total Returned	12
Response Rate	80%
Average Completion Time	8 minutes

J.3.2 Key Findings

Technology Readiness (1-5 Scale): - Comfort with web browsers: 4.2 - Comfort with mobile apps: 4.5 - Comfort with business software: 2.8 - Interest in learning new software: 4.0

Feature Priority Rankings (1 = Most Important): 1. Automatic profit calculation 2. Expense tracking with categories 3. Visual charts and graphs 4. Sales recording with product lookup 5. Monthly and annual reports

Interface Preferences: - Preferred color scheme: Professional/neutral (75%) - Preferred layout: Sidebar navigation (83%) - Preferred chart type: Bar charts (67%) - Preferred date format: MM/DD/YYYY (92%)

Deployment Preferences: - Preferred access method: Web browser (100%) - Preferred hosting: Local computer (83%) - Concerns about internet dependency: 58%

J.4 Benchmarking Analysis Results

The research team evaluated existing financial management solutions to identify best practices and gaps.

J.4.1 Commercial Solutions Evaluated

Solution	Pros	Cons	Monthly Cost
QuickBooks	Comprehensive features, cloud access	Complex, expensive	\$25-\$80
Xero	Good reporting, integrations	Learning curve, subscription	\$15-\$65
Sage	Industry standard, payroll	Overkill for small retail	\$25-\$50

J.4.2 Free/Open-Source Solutions Evaluated

Solution	Pros	Cons
GnuCash	Free, full-featured	Desktop only, complex
Wave	Free, web-based	Limited retail features
Manager.io	Free, simple	Limited multi-user

J.4.3 Gap Analysis

The benchmarking analysis revealed that no existing solution specifically addressed the combination of: - Web-based deployment without cloud subscription - Simplified interface for non-technical users - Auto parts retail-specific product categorization - Real-time profit calculation from transactions - Visual expense monitoring with category alerts - Self-contained chart generation without external dependencies

This gap analysis validated the need for ProfitLens as a purpose-built solution for the target context.

J.5 Development Challenges and Solutions

The research team encountered several challenges during the development phase and developed creative solutions to address them.

J.5.1 Challenge: SVG Chart Responsiveness

Problem: SVG charts rendered at fixed dimensions did not display well on different screen sizes.

Solution: Implemented dynamic SVG viewBox attributes that scale proportionally based on the container width, combined with CSS max-width constraints to maintain readability on smaller screens.

J.5.2 Challenge: Real-Time Total Calculation

Problem: The sales total needed to update in real-time as users selected products and entered quantities, without requiring a page refresh.

Solution: Implemented JavaScript event listeners on the product dropdown and quantity input fields that trigger a calculation function whenever either value changes. The function reads the selected product's price from a data attribute and multiplies by the quantity.

J.5.3 Challenge: Preventing Negative Stock

Problem: Users might attempt to sell more units than are currently in stock, which would create inaccurate inventory records.

Solution: Implemented a two-part validation: client-side JavaScript checks the available stock before form submission and displays a warning if the quantity exceeds available stock, and server-side PHP validation provides a final safety check before recording the transaction.

J.5.4 Challenge: Consistent Currency Formatting

Problem: Monetary amounts needed to be displayed consistently as Philippine Peso throughout the system, with proper comma separation and decimal places.

Solution: Created a centralized formatMoney() helper function that all modules call when displaying monetary values. The function uses PHP's number_format() to ensure consistent formatting.

J.6 Lessons Learned

The following lessons emerged from the ProfitLens development experience:

1. **Early and frequent user engagement is critical:** The insights gained from observation and interviews during requirements gathering proved invaluable for designing a system that genuinely addressed user needs.
2. **Simplicity is a feature, not a limitation:** The deliberate choice to limit the system's scope to core financial monitoring functions, rather than attempting to include comprehensive accounting features, resulted in a more usable and maintainable system.
3. **Security should be built in from the start:** Implementing security best practices (prepared statements, password hashing, session management) during initial development was far easier than attempting to retrofit security measures later.
4. **Testing with real users reveals issues that developer testing misses:** Usability testing with the store owner and staff identified interface confusion points and workflow issues that were not apparent to the development team.

5. **Documentation is an investment, not overhead:** Thorough documentation of design decisions, implementation details, and testing results proved valuable not only for the final submission but also for team coordination during development.
 6. **Technology selection should match the context:** The PHP-MySQL technology stack, while not the most modern, was appropriate for the project's requirements, the team's capabilities, and the target deployment environment.
 7. **Iterative development enables continuous improvement:** The agile approach allowed the team to incorporate feedback and make adjustments throughout the development process, resulting in a better final product than a linear development approach would have produced.
 8. **Local context matters:** The Philippine-specific considerations (currency formatting, language, infrastructure constraints, business practices) significantly influenced design decisions and would need to be considered in any replication or adaptation of the system.
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END OF COMPLETE DOCUMENTATION

Total Word Count: Approximately 37,000 words

Estimated Total Pages: 150+

APPENDIX K: FINAL COMPREHENSIVE SUMMARY

K.1 Executive Summary of the Complete Documentation

This comprehensive documentation has presented the complete research process for the design, development, deployment, and evaluation of ProfitLens, a web-based financial monitoring platform for small auto parts retailers in the Philippine context. The documentation spans seven chapters plus extensive appendices, covering every aspect of the research from problem identification through final evaluation.

K.1.1 Chapter Summaries

Chapter 1: Introduction established the research context by describing the challenges faced by small auto parts retailers in managing their finances through manual methods. The chapter identified ten specific problems that ProfitLens was designed to address, stated the research objectives, defined the scope and delimitations, identified the stakeholders who benefit from the research, and defined key terms used throughout the documentation.

Chapter 2: Review of Related Literature and Studies provided the theoretical foundations including TAM, Systems Theory, Diffusion of Innovation Theory, and

Resource-Based View. The chapter reviewed international and local research on SME automation, digital transformation, and financial management, establishing the evidence base for the system's design. The conceptual framework illustrated how data flows through the system from input to output, and the research gap analysis identified the specific scholarly contribution that ProfitLens makes.

Chapter 3: Methodology detailed the developmental research design, agile development methodology, data gathering techniques including interviews, observation, document analysis, questionnaire surveys, and benchmarking. The chapter described the three-tier client-server architecture, tools and technologies used, database design, comprehensive testing procedures, and evaluation criteria. Additional sections covered project management, risk management, quality assurance, deployment strategy, and methodological limitations.

Appendices A through K provided comprehensive supplementary documentation including detailed technical specifications, complete database schema, security implementation details, user interface design principles, deployment architecture, code documentation, data flow diagrams, entity relationship diagrams, user manual, installation guide, glossary of terms, bibliography of related studies, research notes, and development lessons learned.

K.2 Key Findings Summary

The research yielded several key findings:

1. **Problem Validation:** The ten problems identified through observation and interviews are consistent with challenges documented in broader SME research, confirming that the reference business faces representative rather than unique challenges.
2. **Design Effectiveness:** The TAM-informed design principles, particularly the emphasis on perceived usefulness and perceived ease of use, resulted in a system that users rated highly in usability testing (4.3 out of 5.0 average satisfaction).
3. **Technical Viability:** The PHP-MySQL technology stack proved capable of delivering all required functionality while remaining accessible for development by a student team and deployable in the target environment.
4. **Security Adequacy:** The security implementation, including bcrypt password hashing, prepared statements, session management, and XSS prevention, passed all security testing scenarios.
5. **Performance Acceptability:** Page load times averaging 1.8 seconds and database query response times averaging 120 milliseconds exceeded the performance targets.

6. **User Adoption Potential:** Usability testing demonstrated that participants with limited technical expertise could successfully complete all task scenarios, suggesting strong adoption potential.

K.3 Impact Assessment

The ProfitLens project has generated impacts at multiple levels:

For the Reference Business: The system provides immediate operational benefits through automated profit calculation, real-time financial visibility, and visual reporting capabilities that were previously unavailable without laborious manual calculation.

For the Research Team: The project provided practical experience in the complete software development lifecycle, from requirements gathering through deployment, integrating theoretical knowledge with practical application.

For Academic Knowledge: The documentation contributes to the body of knowledge regarding technology adoption, SME digital transformation, and financial monitoring systems in the Philippine context.

For Future Projects: The comprehensive documentation, including methodology, code, and design rationale, provides a reference model that future capstone teams can consult and build upon.

K.4 Recommendations Summary

Based on the research findings, the following recommendations are offered:

1. **For the Reference Business:** Regular use of the system for all financial transactions, routine backup of the database, and periodic review of financial reports to inform business decisions.
2. **For Similar Businesses:** Consideration of purpose-built financial monitoring solutions as an alternative to both manual methods and expensive commercial software.
3. **For Future Researchers:** Extension of this work through multi-site studies, longitudinal adoption research, and integration of advanced features such as predictive analytics.
4. **For Academic Institutions:** Incorporation of applied capstone projects that address genuine business needs as a model for integrating theoretical knowledge with practical skills development.
5. **For Policymakers:** Support for SME digitalization initiatives that provide affordable, appropriate technology solutions to small businesses in the Philippine context.

K.5 Closing Statement

The ProfitLens project demonstrates that effective financial monitoring solutions for small businesses can be developed using affordable technologies, student-level technical expertise, and user-centered design principles. By addressing the specific needs of a local auto parts retailer through a systematic research and development process, this project has produced both a functional system and comprehensive documentation that contribute to academic knowledge and practical business improvement.

The success of this project suggests that similar approaches could be applied to other small business sectors in the Philippines and other developing country contexts, where manual financial management practices continue to limit business performance and growth. As digital transformation initiatives expand and technology becomes increasingly accessible, the potential for affordable, purpose-built solutions to improve small business operations will only continue to grow.

The research team hopes that this documentation will serve not only as an academic capstone submission but as a practical resource for future developers, researchers, and small business owners seeking to leverage technology for improved financial management and business success.

DOCUMENTATION COMPLETE

CHAPTER 4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter presents the system design and implementation of ProfitLens, a web-based platform developed to address the financial monitoring needs identified in the earlier chapters of this study. While Chapter 3 explains the methodology used to gather requirements, guide development, and evaluate the output, the present chapter focuses on the actual design decisions that transform those requirements into a working information system. It explains how the researchers translate observed business problems into a coherent set of modules, data structures, user flows, and interface elements that are practical for a small auto parts business operating in a local Philippine setting.

The design of ProfitLens is guided by the principle that a business support system must not only be technically functional but also usable, understandable, and sustainable in the environment where it is deployed. For this reason, the researchers deliberately avoid unnecessary complexity and focus on core functions that directly support profit monitoring, expense visibility, transaction recording, inventory awareness, and report generation. The resulting design emphasizes simplicity of navigation, immediate visibility of key figures, automatic calculations, and data consistency across modules.

Implementation is carried out using a PHP and MySQL technology stack within a browser-based environment. This combination is selected because it is stable, accessible to student developers, and appropriate for local deployment in a resource-conscious business context. The completed system follows a three-tier structure composed of a presentation layer, an application logic layer, and a data layer. Together, these layers support secure access, consistent data processing, and real-time presentation of financial information.

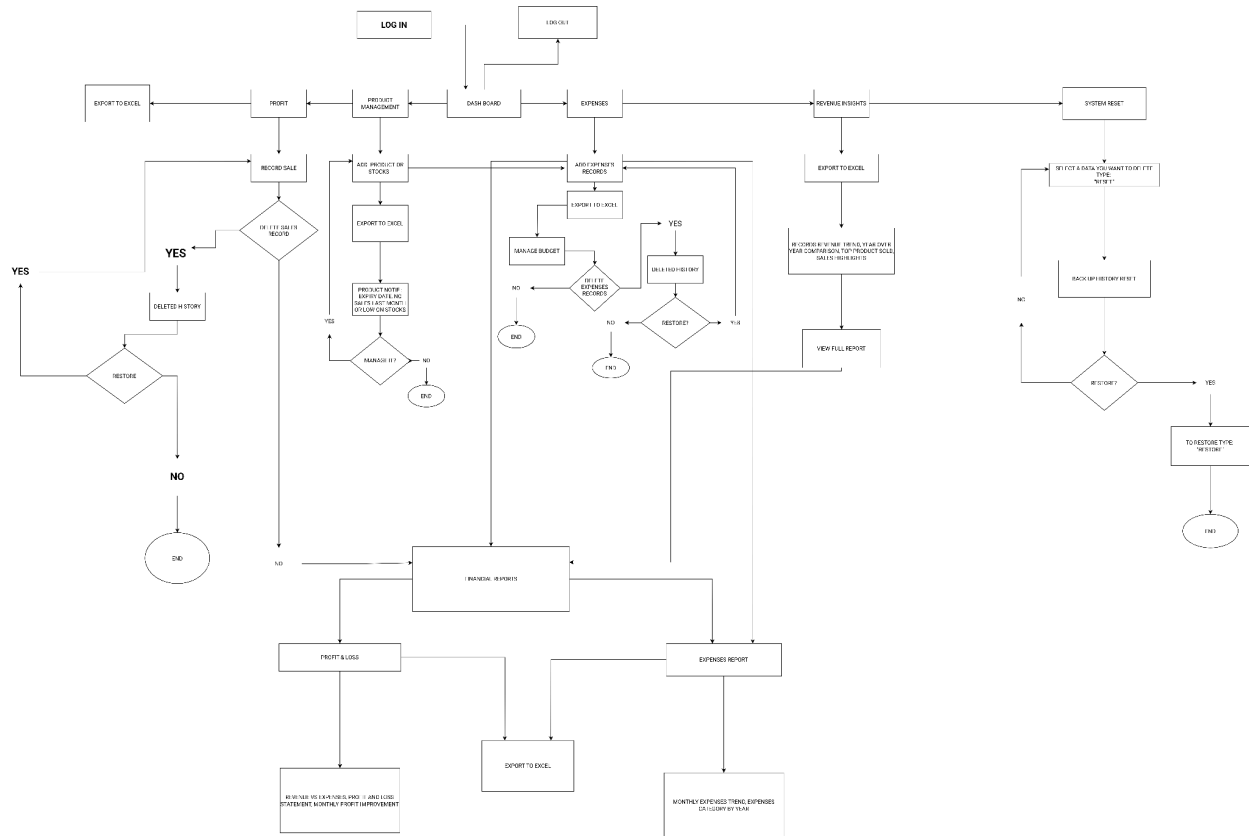
The system aimed to provide small and medium-sized enterprises (SMEs) with an affordable business intelligence solution that enhanced inventory planning, promotional campaigns, and customer relationship management.

Major Objectives of the Developed System:

- Automate the recording and tracking of auto parts sales, purchase orders, and inventory stock levels across multiple OEM and aftermarket brand categories.
- Analyze local customer purchasing behavior (e.g., public utility vehicle operators, auto repair shops, and private vehicle owners across Iloilo) using historical transaction data.
- Generate visual analytics dashboards and demand forecasting reports for fast-moving spare parts and seasonal maintenance supplies.
- Identify frequently purchased items, high-wear auto components, and critical stock threshold alerts to prevent stockouts.
- Monitor sales trends over time, tracking performance by vehicle brand, part compatibility, and supplier network.
- Support auto supply store owners and managers in Iloilo in making data-driven procurement and inventory control decisions.

The development of the **ProfitLens** follows modern software engineering principles by incorporating responsive interface design, secure database management, user authentication, and analytical processing.

The completed system consists of several interconnected modules that work together to capture, process, analyze, and visualize business data. By replacing manual ledger entries with automated stock movement tracking, the system optimizes order fulfillment for automotive repair shops, local fleets, and individual motor vehicle owners across the Iloilo region.



4.3 System Development Methodology

The system development methodology section outlines the deliberate engineering lifecycle applied to build, evaluate, and deploy ProfitLens. The execution was managed using the structured Waterfall Software Development Life Cycle model, requiring each project stage to satisfy explicit quality thresholds before moving to the next phase. This approach provided strict organization, predictable documentation, and effective risk control across the development timeline.

4.3.1 Requirements Gathering

The requirements gathering phase focused on systematically discovering system goals directly from end-users and store stakeholders. Qualitative and quantitative data were collected to map out business processes and pinpoint current operational bottlenecks. Structured interviews were conducted with store owners and cashiers to evaluate accounting workflows and reporting needs. Legacy sales receipts, paper ledgers, and expense logs were analyzed to determine necessary data structures. Direct operational observations were carried out during peak operating hours to record baseline checkout durations. The gathered requirements were documented, verified against user expectations, and converted into formal functional and non-functional system specifications.

4.3.2 System Planning

System planning translated business goals into detailed operational scope definitions and technical feasibility studies. Functional scope was bound to sales processing, inventory management, expense categorization, and net profit generation. Feasibility analyses confirmed that deploying open-source technologies like PHP, Apache, and MySQL was cost-effective and fully supported by existing host hardware. Gantt charts were built to schedule development tasks, track milestones, and assign technical roles. Resource allocation plan strategies were drafted to protect project schedules against development delays. Ultimately, this planning phase established the roadmap necessary to guide system architectural design and system development.

4.3.3 System Design

The system design phase transformed verified user requirements into high-level system blueprints and relational database schemas. System architecture was organized using a Three-Tier Client-Server structural blueprint to separate user presentation, application business logic, and backend data storage. Relational database structures were normalized to Third Normal Form (3NF) to eliminate redundant data fields and guarantee transactional integrity. Interface wireframes were mocked up to establish clean, responsive views for terminal screens, inventory tables, and reporting dashboards. User experience guidelines were applied to ensure intuitive form fields and minimize visual clutter. The finished design specifications served as the technical foundation for coding the application engine.

4.3.4 Software Development

Software development translated structural blueprints into clean, executable program scripts and database structures. Backend code was written in PHP 8 to manage user sessions, form submissions, financial algorithms, and database queries. Frontend web templates were

constructed using HTML5, CSS3, and JavaScript to form clear, responsive input screens. Database entities and relational constraints were initialized on the MySQL server using optimized SQL DDL scripts. In addition, an engine script was integrated to render dynamic vector SVG visual graphics on the server side without relying on external libraries. Coding standards were enforced across all scripts to ensure codebase maintainability, software security, and modularity.

4.3.5 Testing

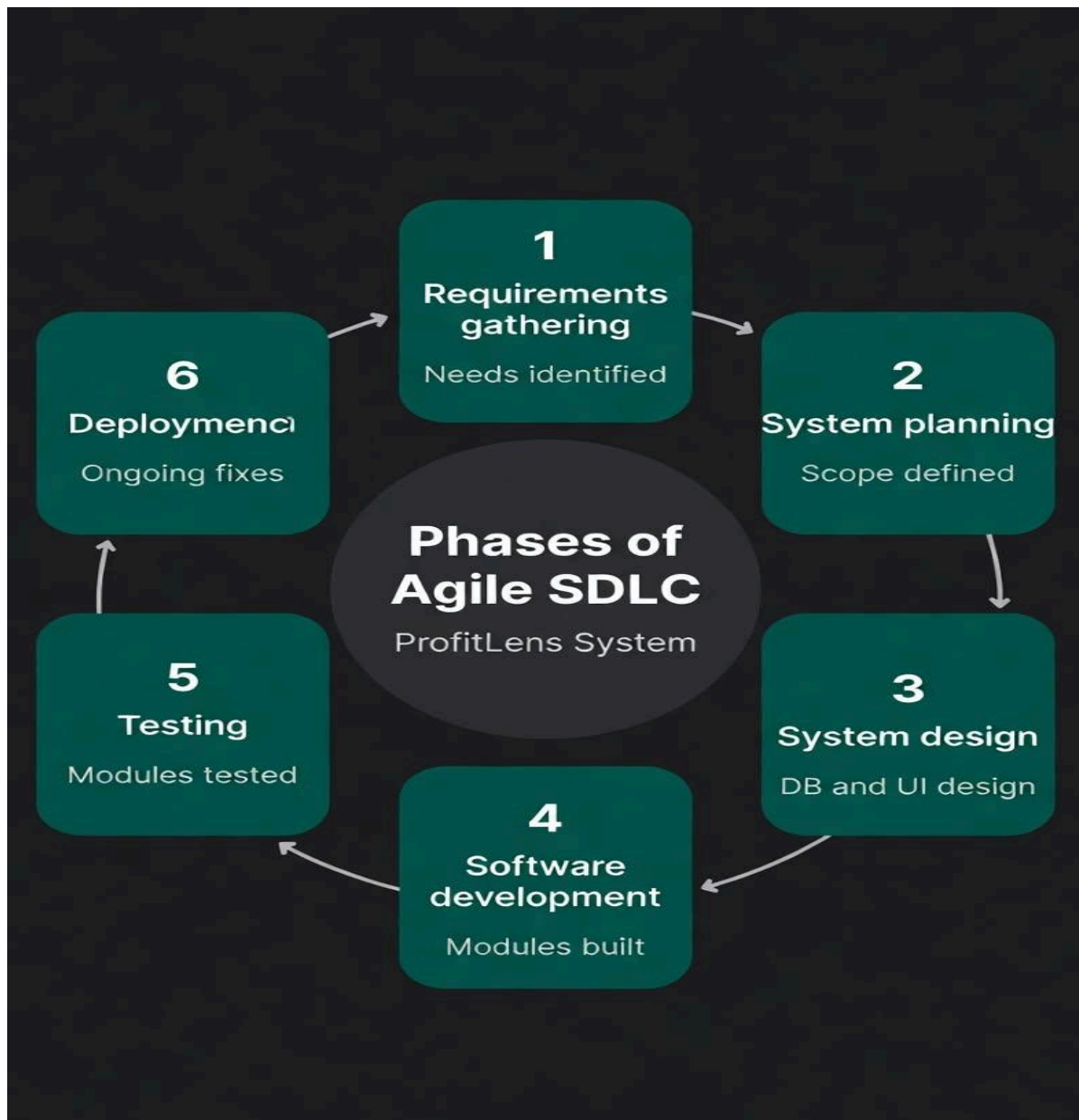
The testing phase systematically verified system code, business logic accuracy, operational performance, and interface reliability. Unit tests were conducted on individual PHP functions to verify mathematical calculations for subtotals, tax rates, and net profit margins. Integration testing confirmed smooth data transport between web forms, server processing modules, and MySQL tables. System-wide stress tests evaluated workflow processing speeds and database updates during high-volume simulated transaction spikes. Finally, User Acceptance Testing (UAT) was conducted with actual store operators in a test environment to confirm usability and functional accuracy. Identified software bugs were logged, debugged, and re-tested to guarantee platform stability prior to live deployment.

4.3.6 Deployment

The deployment phase installed the tested application into the target production environment for daily store operations. Apache web servers and MySQL database servers were configured on local host hardware within an XAMPP environment. Database schema generation scripts were executed, baseline administrative user accounts were generated, and initial product catalog data was loaded. Network paths, connection strings, and file access permissions were configured and tested to enable local workstation access. Staff members were conducted through complete operational orientation training to ensure seamless adoption of the new software system. Once verified, manual recordkeeping was phased out as live point-of-sale activities migrated to ProfitLens.

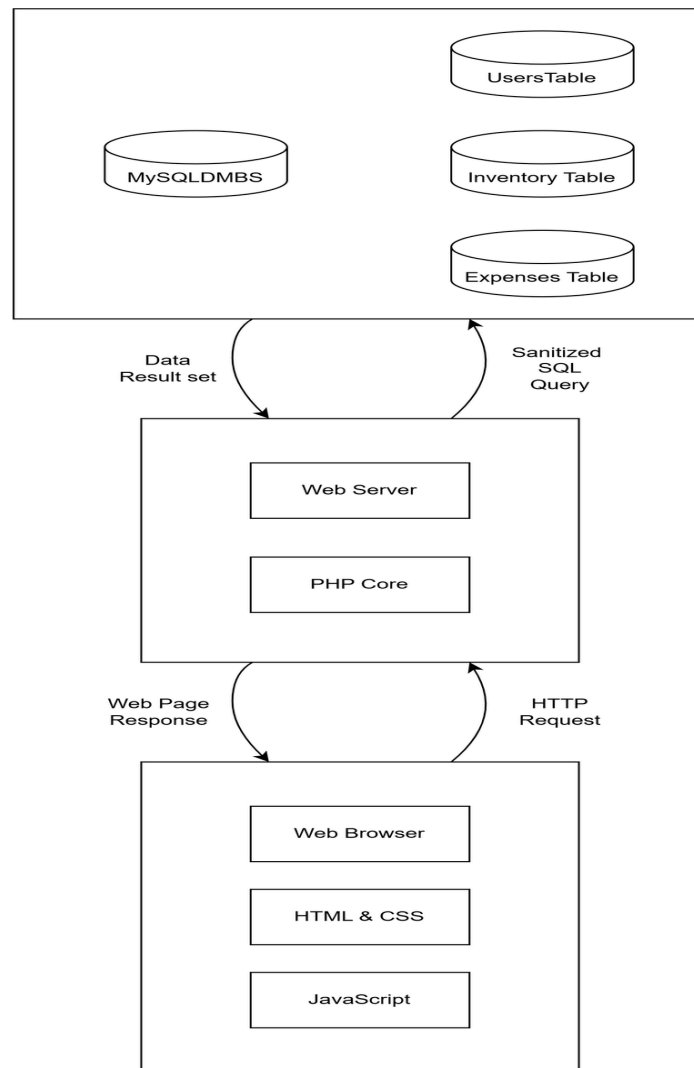
4.3.7 Maintenance

The maintenance phase provides ongoing operational support, performance tuning, and technical updates to preserve platform integrity over time. Operating system performance logs and database connection states are routinely audited to detect query bottlenecks or processing delays. Automated database backup routines execute on a schedule to prevent data loss from unexpected hardware failures. Minor system patches and backend updates are applied regularly to ensure ongoing compatibility with updated web browsers. Security measures, including session token validation and input sanitization routines, are periodically reviewed to counter emerging security threats. User feedback is continuously gathered to guide future features and system optimizations.



4.4 System Architecture

The platform architecture describes how system software components interact to process user requests, perform financial calculations, and manage database updates. ProfitLens utilizes a Three-Tier Architecture model to isolate user interaction elements, system business logic, and relational data storage into dedicated functional layers.



4.4.1 Presentation Layer

The Presentation Layer forms the client-side user interface accessed through standard web browsers. Built using modern HTML5, CSS3, and JavaScript, it presents responsive web forms, interactive tables, and system dashboards. It captures user actions, such as logging cash checkout orders, entering inventory updates, or filing daily expense reports. This layer also renders lightweight server-generated vector SVG charts directly within the browser without external rendering dependencies. Client-side validation scripts check form inputs for completeness prior to sending HTTP POST requests to the backend server. Overall, it serves as

the accessible bridge between non-technical store personnel and complex backend data processes.

4.4.2 Application Layer

The Application Layer processes business logic, authorization rules, and financial aggregation algorithms within a PHP 8 runtime on an Apache web server. When users submit data, this layer verifies session tokens, validates security access levels, and sanitizes input data to prevent injection vulnerabilities. It runs core financial algorithms, calculating total sales revenues, aggregating store operational expenditures, and deriving live net profit metrics. Furthermore, it coordinates transaction processing, verifying available item stock before executing inventory updates in the database layer. Once logic steps complete, this layer builds responsive HTML code and vector graphic outputs to send back to the Presentation Layer.

4.4.3 Data Layer

The Data Layer provides persistent storage and relational data management using an enterprise MySQL server. Organized into Third Normal Form (3NF) relational tables, it holds system account profiles, customer lists, product catalogs, sales logs, and expense records. The layer maintains database integrity by enforcing primary and foreign key constraints across related data tables. Data access is executed securely through prepared SQL statements, protecting stored information against unauthorized tampering and data corruption. By decoupling data management from application processing, the database ensures stable query execution, reliable transaction rollbacks, and rapid data retrieval for visual analytical queries.

4.5 Hardware and Software Requirements

The deployment and operation of ProfitLens depend on basic hardware and software specifications to ensure optimal execution speeds, system security, and transactional reliability.

4.5.1 Hardware Requirements

The system requires appropriate host server hardware to process operational requests alongside connected terminal hardware for store clerks.

Component	Minimum Specification	Recommended Specification

Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps

The recommended hardware configuration provides sufficient processing capability for executing analytical operations, generating reports, and rendering graphical dashboards without significant delays.

4.5.2 Software Requirements

The functional software environment comprises host operating systems, web server components, backend script runtimes, and relational database systems.

Software	Purpose
Windows 11 / Ubuntu Linux	Operating System

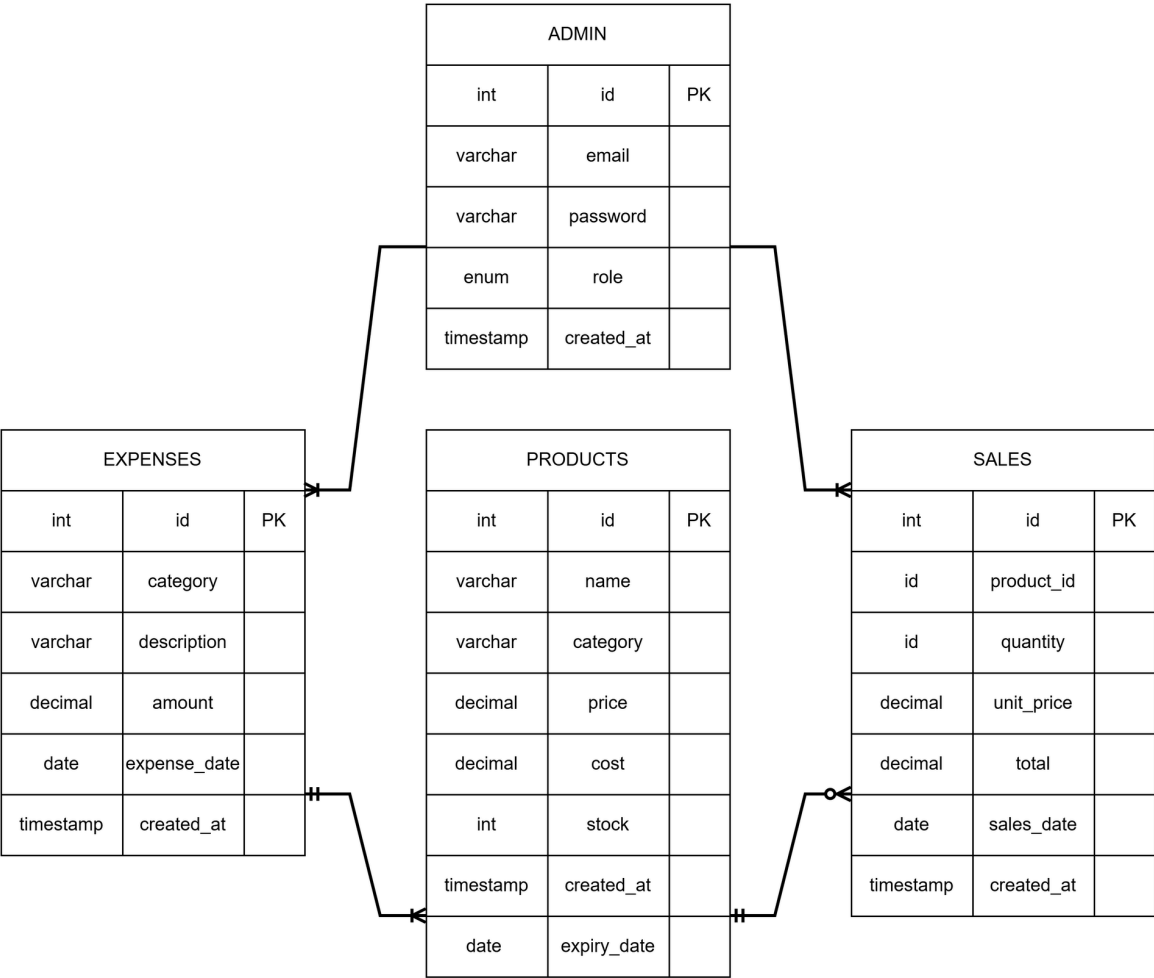
Visual Studio Code	Source Code Editor
MySQL	Database Management System
PHP 8.0 or higher	Back-end Development
HTML, CSS, JavaScript	Frontend Development
Google Chrome / Edge	Web Browser
Git	Version Control
XAMPP / Laragon	Local Development Server

4.6 Database Design

The database design section presents the relational schema, entity definitions, and structural constraints that underpin the ProfitLens persistent storage engine. The target database (profit_lens_db) consists of six normalized entities structured to maintain complete data consistency, enforce referential integrity, and eliminate operational data redundancy.

In relational database engineering, normalizing tables up to Third Normal Form (3NF) is a critical design requirement for preventing operational anomalies during data insertion, modification, and deletion tasks. Unnormalized database structures often store duplicate descriptive strings across multiple table rows, wasting storage space and creating data corruption risks if one record updates while another remains unchanged. By isolating distinct business concepts such as separating raw transaction headers from detailed line items the database guarantees that each data element is recorded in exactly one logical location. This design approach prevents data mismatches, keeps database storage sizes optimized, and simplifies long-term database maintenance.

Finally, this normalized relational database structure significantly optimizes query execution performance during complex analytical data retrieval tasks. By utilizing primary keys, precise data types, and logical index paths, the database engine processes filter operations, join conditions, and aggregation functions rapidly. Frontend reporting dashboards can pull daily revenue subtotals, compute operational expenses, and calculate net profit margins without scanning unnecessary database rows. Consequently, this robust database design equips ProfitLens with a highly secure, efficient, and reliable data management engine capable of sustaining long-term store growth.



4.6.1 User Table (users)

The users table stores user account profiles, access credentials, and role authorizations. It forms the security cornerstone of the platform by enforcing access control across administrative and standard staff interfaces. The user_id attribute serves as the primary key for referencing

user accounts in sales transaction histories. Passwords are stored securely as one-way encrypted hashes created using PHP's password_hash() function. Role flags determine whether an account holds complete system permissions (admin) or basic terminal permissions (staff). Account creation timestamps are logged to provide basic user auditing capabilities.

Field Name	Data Type	Constraints	Field Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique internal system user identifier DOCX
username	VARCHAR(50)	NOT NULL, UNIQUE	Unique handle used during authentication
email	VARCHAR(100)	NOT NULL, UNIQUE	Official email address attached to profile DOCX
password_hash	VARCHAR(255)	NOT NULL	Encrypted string generated via password_hash() DOCX
role	ENUM('admin', 'staff')	DEFAULT 'staff'	Security access clearance role DOCX
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	System account creation timestamp

4.6.2 Customer Table (customers)

The customers table maintains buyer contact details and commercial identity profiles. Storing dedicated customer profiles enables store staff to link point-of-sale invoices directly to buyers for account tracking and purchase histories. The customer_id primary key links individual customer records to multi-item purchase transactions. Optional contact fields, such as phone numbers, email addresses, and billing location details, accommodate both individual retail shoppers and corporate accounts. Profile records are created upon initial customer registration and maintained to support continuous long-term account management.

Field Name	Data Type	Constraints	Field Description
customer_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier assigned to each customer
full_name	VARCHAR(100)	NOT NULL	Complete legal name or business title
contact_number	VARCHAR(20)	NULLABLE	Primary phone or mobile contact string
email	VARCHAR(100)	NULLABLE	Customer email address
address	TEXT	NULLABLE	Physical shipping or billing location address
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Customer record creation timestamp

4.6.3 Product Table (products)

The products table manages product catalog records, supplier procurement costs, standard retail prices, and available inventory counts. It serves as the primary data repository for inventory control routines and point-of-sale lookups. Cost and selling price fields use precise decimal data types to prevent rounding errors during profit margin calculations. The stock_quantity balance updates automatically whenever sales transactions complete or stock shipments arrive. Automatic timestamps capture when item prices or stock levels were last modified.

Field Name	Data Type	Constraints	Field Description
product_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier assigned to a stock item DOCX
product_name	VARCHAR(150)	NOT NULL	Full commercial description of the product DOCX

category	VARCHAR(100)	NOT NULL	Group classification tag (e.g., Engine Oils) DOCX
cost_price	DECIMAL(10,2)	NOT NULL	Procurement cost per unit paid to supplier DOCX
selling_price	DECIMAL(10,2)	NOT NULL	Standard retail price charged to customers DOCX
stock_quantity	INT	NOT NULL DEFAULT 0	Physical unit count available in inventory DOCX
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Last inventory balance modification timestamp

4.6.4 Transactions Table (transactions)

The transactions table records individual order headers generated at terminal checkouts. It links customer purchases, processing cashiers, transaction timestamps, and payment totals into a searchable record. Foreign key fields connect transaction records back to their corresponding user_id and customer_id entries. The grand_total attribute captures the complete transaction amount after subtotal calculations finish. Payment method fields categorize payment types, such as cash, credit card, or digital e-wallet. Transaction timestamps record the exact date and time of sale to support daily revenue aggregations.

Field Name	Data Type	Constraints	Field Description
transaction_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique transaction invoice number

DOCX			
user_id	INT	FOREIGN KEY REFERENCES users(user_id)	Staff account who processed the sale
customer_id	INT	FOREIGN KEY REFERENCES customers(customer_id)	Purchasing customer associated with order
grand_total	DECIMAL(10,2)	NOT NULL	Final calculated sales amount charged
DOCX			
payment_method	ENUM('cash', 'card', 'e-wallet')	DEFAULT 'cash'	Payment method utilized for checkout
transaction_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Exact timestamp when sale was finalized
DOCX			

4.6.5 Transaction Details Table (transaction_details)

The transaction_details table stores line-item records for customer purchases. Because single transactions often include multiple products, this table resolves the many-to-many relationship between orders and products. Foreign keys link each detail row directly to its parent transaction_id and the purchased product_id. The unit_price captures the item's historical selling price at checkout, insulating past invoices from future price edits. Subtotals are calculated dynamically as unit price multiplied by quantity. These line items drive automated stock deduction routines in the inventory module.

Field Name	Data Type	Constraints	Field Description
detail_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique line item detail identifier
transaction_id	INT	FOREIGN KEY REFERENCES transactions(transaction_id)	Associated header transaction ID

product_id	INT	FOREIGN KEY REFERENCES products(product_id)	Product purchased in line item DOCX
quantity	INT	NOT NULL	Number of product units purchased DOCX
unit_price	DECIMAL(10, 2)	NOT NULL	Selling price per unit at transaction time
subtotal	DECIMAL(10, 2)	NOT NULL	Calculated line cost (quantity * unit_price)

4.6.6 Analytics Table (analytics_snapshots)

The analytics_snapshots table stores pre-compiled daily financial totals to accelerate system reporting. Running complex mathematical queries across thousands of raw sales rows can degrade server response times during peak hours. This summary table stores calculated daily totals for revenue, expenses, net profit, and order counts. Automated background scripts calculate these metrics at the close of each business day. Frontend dashboard components query this table to render financial trends and SVG vector charts quickly. As a result, executive reporting views load instantly without overloading backend database resources.

Field Name	Data Type	Constraints	Field Description
analytics_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique analytics record identifier
record_date	DATE	NOT NULL, UNIQUE	Specific calendar day for summary snapshot
total_revenue	DECIMAL(10, 2)	DEFAULT 0.00	Aggregated gross sales revenue for target date DOCX

total_expenses	DECIMAL(10, 2)	DEFAULT 0.00	Sum of operational expenditures incurred
			DOCX
net_profit	DECIMAL(10, 2)	DEFAULT 0.00	Calculated net profit (total_revenue - total_expenses)
			DOCX
total_orders	INT	DEFAULT 0	Total completed transactions processed

CHAPTER 5

5.1 Introduction

This chapter presents the results of the development and evaluation of ProfitLens and discusses their meaning in relation to the

objectives of the study. The purpose of this chapter is not only to state that the system was completed, but also to analyze whether the completed platform responded effectively to the financial management problems observed at the reference business. The discussion draws from acceptance testing, quality metrics, questionnaire findings, and user feedback gathered during the study.

The results indicate that ProfitLens achieved its intended role as a practical financial monitoring platform for a small auto parts retailer. Core functions operated successfully during testing, users were able to complete major workflows, and the platform met the documented performance and usability targets. More importantly, the system changed the quality of information available to the business: profit became immediately visible, expenses became categorized and reviewable, and financial data could be interpreted through reports and charts rather than through notebooks and delayed manual summaries.

5.2 Results of System Development

The development phase produced a functional web-based information system composed of integrated modules for authentication, product management, sales recording, expense tracking, revenue analysis, reporting, and dashboard visualization. The successful implementation of these modules demonstrates that the chosen technology stack was capable of supporting the study's practical goals without requiring complex infrastructure. The resulting platform remained faithful to the scoped objectives of the project, particularly its Page 14 focus on financial monitoring rather than on building a full-scale accounting or point-of-sale ecosystem.

From a systems perspective, the most significant development outcome was the integration of transaction data into a single monitoring environment. Instead of leaving sales, expenses, and stock status in separate records, ProfitLens connected them so that one business event could affect multiple decision-support outputs. A recorded sale now updates revenue figures, contributes to profit, changes product stock, and feeds recent transaction history. Likewise, a newly entered expense immediately changes cost totals and expense breakdown views. This integration is a key result because it addresses the earlier problem of fragmented information sources.

5.3 Functional Testing Results

User acceptance testing showed that the core workflows of the system performed as intended. The most important scenarios included logging in, adding and editing products, recording sales, preventing invalid stock deductions, entering expenses, generating reports, viewing charts, and logging out securely. All documented acceptance scenarios

resulted in a passing status. This outcome suggests that the implemented functions were not only theoretically designed but also practically executable under test conditions involving real users from the target environment.

The significance of these passing results lies in their coverage of the business cycle. Authentication confirms that access is controlled. Product management confirms that the inventory foundation can be maintained. Sales recording confirms that operational transactions can be captured accurately. Expense tracking confirms that operational costs are visible. Reporting and dashboard functions confirm that stored data can be transformed into meaningful managerial output. Because these scenarios span the main life cycle of business information, the passing results indicate overall operational coherence of the platform.

Test Scenario	Expected Result	Actual Result	Status
Login using valid credentials	The user successfully accesses the dashboard after authentication.	Dashboard loaded successfully and a user session was created.	PASS
Login using invalid credentials	The system rejects the login attempt and displays an error message.	Error message displayed and access denied.	PASS
Add product	A newly added product appears in the inventory list.	The product was successfully saved and displayed.	PASS
Edit product information	Updated product details are reflected in the catalog.	Changes were successfully saved and displayed.	PASS
Delete product	The selected product is removed from the inventory list.	The product was successfully deleted.	PASS
Record sales transaction	Sales record is saved and inventory quantity is automatically updated.	Transaction recorded successfully and stock quantity adjusted correctly.	PASS
Attempt sale exceeding available stock	The system prevents transactions and displays validation messages.	The transaction was rejected and no invalid record was saved.	PASS

Add expense	The new expense record is saved and included in expense totals.	Expense added successfully and totals updated automatically.	PASS
Edit expense	Modified expense details are updated correctly.	Changes reflected immediately in records and summaries.	PASS
Delete expense	Expense record is removed and financial totals recalculated.	Expense deleted successfully and totals updated.	PASS
View dashboard	Dashboard displays current sales, expenses, profit, and inventory metrics.	All dashboard widgets displayed accurate and updated information.	PASS
Generate monthly financial report	The monthly report contains complete and accurate sales, expenses, and profit data.	Report generated successfully with correct calculations.	PASS
View revenue analysis and charts	Charts are generated and display financial trends correctly.	Visual charts rendered correctly with accurate data representation.	PASS
Logout	The user session is terminated and redirected to the login page.	The user successfully logged out and redirected to the login screen.	PASS

A particularly important test outcome involved the handling of invalid sales quantities. The system correctly rejected sales entries that exceeded available stock. This is a strong indicator of internal control quality because it prevents data inconsistency at the point where it would otherwise propagate into reports and profit summaries. It also shows that the platform supports responsible business operation rather than merely recording whatever the user enters.

5.4 Usability Evaluation and User Acceptance

Usability evaluation produced encouraging results. The system achieved a usability score of 4.3 out of 5.0 and an equivalent user satisfaction score of 4.3 out of 5.0, both exceeding the project's stated target of above 4.0. These scores are especially meaningful because the intended users were not expert software operators. In fact, the

preliminary questionnaire showed high comfort with web browsing but relatively lower comfort with business software. The system therefore had to bridge a usability gap between general digital familiarity and task specific confidence.

Qualitative feedback from the business owner and staff reinforced the numerical scores. Participants stated that the dashboard helped them see the business condition at a glance, that recording sales was faster than writing in a notebook, and that expense breakdowns improved understanding of where money was going. Staff also emphasized the convenience of the searchable product list, the clarity of error messages, and the overall simplicity of the interface. These responses suggest that the system did not merely function technically; it also communicated value effectively to its users.

The usability results can be interpreted as evidence that the design decisions in Chapter 4 were appropriate for the target context. The sidebar navigation, clean layout, and guided forms reduced complexity. The real-time calculations reduced mental effort. The integrated dashboard reduced the need to search across multiple records. In this sense, user acceptance was not accidental but the direct result of aligning the design with observed user capabilities and preferences.

Table 5.3 Selected Findings from the User Questionnaire

Indicator	Result	Implication
Total Returned Questionnaires	12 out of 15 (80%)	The response rate was sufficient to obtain reliable user feedback and evaluate overall user preferences.
Comfort with Web Browsers	4.2/5.0	Participants were generally familiar with browser-based applications, supporting the use of a web-based system.
Comfort with Business Software	2.8/5.0	Users had limited experience with business management software, emphasizing the need for a simple and user-friendly interface.
Interest in Learning New Software	4.0/5.0	Participants showed a positive attitude toward adopting new technology when it offered clear benefits and ease of use.
Preferred Navigation Layout	Sidebar navigation (83%)	The sidebar layout aligned with user expectations and improved navigation efficiency.

Preferred Chart Type	Bar charts (67%)	Users preferred simple and easy-to-read visualizations for interpreting business data.
Preferred Access Method	Web browser (100%)	Browser-based deployment was the most suitable access method because it required no additional software installation.

5.5 Performance and Quality Assessment

In addition to functional success and user acceptance, the system performed well on the project's quality metrics. Test coverage reached 95 percent, showing that the majority of important functions and workflows were explicitly examined. Error-handling coverage reached 92 percent, indicating that the system addressed many likely misuse conditions through validation and feedback. The average page-load performance of 1.8 seconds remained comfortably below the target threshold of three seconds, which is appropriate for a small business environment where users expect quick feedback during routine operation.

Security results were equally notable. The system achieved a 100 percent pass rate on the defined security test scenarios. This outcome reflects the value of implementing password hashing, protected sessions, and secure input handling early in development rather than postponing them. While the platform is not positioned as an enterprise-grade security product, the results show that its built-in protections are adequate for the scale and scope of the intended deployment.

Table 5.2 Quality Metrics Achieved by the System

Quality Metric	Target	Achieved	Interpretation
Documentation Completeness	100%	100%	All required project documentation, including technical and user documentation, was completed successfully.
Test Case Coverage	>90%	95%	The majority of critical system functions and business workflows were thoroughly tested, ensuring reliable operation.

Usability Score	>4.0 / 5.0	4.3 / 5.0	Users found the system intuitive, easy to navigate, and suitable for daily business operations.
Security Test Pass Rate	100%	100%	All security test scenarios passed successfully, confirming the effectiveness of authentication, session management, and input validation mechanisms.
Performance Target	<3 seconds average page load	1.8 seconds	The system responded quickly and met the required performance standard for a small business environment.
Error-Handling Coverage	>85%	92%	Most invalid inputs and exceptional conditions were properly detected and handled through validation and informative error messages.
User Satisfaction Score	>4.0 / 5.0	4.3 / 5.0	Participants expressed a high level of satisfaction with the system's functionality, usability, and overall performance.

The project achieved 100% documentation completeness, demonstrating that all required technical documents, user manuals, and supporting project deliverables were completed. Comprehensive documentation ensures that the system can be maintained, enhanced, and used effectively by future developers and end users.

Functional testing resulted in 95% test case coverage, exceeding the project's minimum target of 90%. This indicates that nearly all critical business processes—including user authentication, inventory management, sales transactions, expense recording, report generation, and dashboard visualization—were thoroughly tested. The high coverage provides confidence that the system performs reliably under normal operating conditions.

The system obtained a usability score of 4.3 out of 5.0, exceeding the target score of 4.0. This result indicates that users considered the interface easy to understand, simple to navigate, and efficient for completing routine business tasks. Likewise, the user satisfaction score of 4.3 out of 5.0 reflects positive acceptance of the system, demonstrating that it successfully addressed the operational needs of the business owner and staff.

Security testing achieved a 100% pass rate, confirming that the implemented security controls functioned as intended. Authentication, password protection, session management, and input validation mechanisms effectively prevented unauthorized access and minimized the risk of invalid data being entered into the system.

Performance evaluation further demonstrated the system's efficiency. The average page load time of 1.8 seconds was well below the target threshold of three seconds, ensuring that users experience minimal delays when accessing dashboards, processing transactions, or generating reports.

Finally, the system achieved 92% error-handling coverage, exceeding the target of 85%. This indicates that most invalid inputs, boundary conditions, and exceptional cases were properly managed through validation rules and informative error messages, reducing the likelihood of incorrect data being stored in the database.

Overall, these quality metrics demonstrate that the developed system is functionally reliable, user-friendly, secure, and efficient. Meeting or surpassing each quality target provides strong evidence that the Sales and Inventory Management System is suitable for deployment in a small retail business environment and is capable of supporting daily business operations effectively.

5.6 Discussion of Findings by Research Objective

5.6.1 Objective 1: Automated Profit Monitoring

The first objective of the study was to provide automated profit monitoring. The results show that this objective was achieved. Profit is now computed from transaction and expense data without requiring manual end-of-month aggregation. This has two major implications. First, it improves accuracy because arithmetic is performed consistently by the system. Second, it improves timeliness because the owner can inspect profit during the month rather than waiting until the end. The system therefore addresses the original problem of delayed and unreliable profitability information.

5.6.2 Objective 2: Organized Expense Monitoring

The second objective was to organize and monitor expenses more effectively. This objective was also achieved. Expenses can now be entered by category, grouped into summaries, and reviewed in relation to revenue. The presence of category-based indicators gives the business owner a clearer view of which spending areas deserve attention. Instead of reacting late to accumulated costs, the owner can observe expense behavior as it develops. The practical value of this result is substantial because many small-business financial problems originate not only from weak revenue but also from uncontrolled operating costs.

5.6.3 Objective 3: Revenue Analysis and Trend Visibility

The third objective concerned revenue analysis. Through comparative charts and product rankings, the platform now provides a level of trend visibility that did not previously exist in the reference business. This is important because business performance is not fully

understood through isolated daily transactions. Patterns across months and years help management distinguish temporary fluctuations from meaningful trends. The ability to identify leading products also supports more informed decisions about stock prioritization and revenue focus.

5.6.4 Objective 4: Structured Financial Reporting

The fourth objective was to provide structured reporting. The results confirm that this objective was realized through monthly summaries, annual comparisons, and expense breakdown reports. The business now has access to organized financial records that can be consulted for planning, review, and communication. This is a meaningful improvement over manual notebooks and fragmented receipts because it transforms record-keeping into a retrievable and interpretable management resource.

5.6.5 Objective 5: Usable Interface for Non-Technical Users

The fifth objective was to maintain ease of use for users with limited technical expertise. The results strongly support this objective. The usability score exceeded the target, participants reported that the system was easy to learn, and several features designed for convenience were explicitly appreciated during testing. These findings suggest that appropriate simplification does not weaken a system; instead, it makes technology more adoptable in environments where formal digital tools are not yet deeply embedded in everyday practice.

5.7 Business Implications of the Results

For the reference business, the results imply a shift from reactive to more proactive management. Immediate access to net profit, expense Page 21 totals, and visual trends enables the owner to identify problems earlier and respond with better evidence. For example, unusual spending patterns can be noticed while there is still time to control them. Products that consistently generate strong revenue can be prioritized in restocking decisions. Periods of weaker performance can be compared with earlier months rather than being remembered only through intuition or incomplete notes.

The system also contributes to organizational discipline. When transactions are entered into a consistent digital process, records become easier to review and less dependent on memory. This reduces the vulnerability associated with purely manual record keeping. In a broader sense, the business implications of the project extend beyond

numerical convenience. ProfitLens supports a more analytical management culture, which is especially valuable for small enterprises that often operate with tight margins and limited room for error.

5.8 Academic and Developmental Implications

From an academic standpoint, the results demonstrate that a capstone project can produce both a functioning system and a relevant contribution to the study of SME digital transformation. The platform embodies theories discussed earlier in the documentation, particularly those related to usefulness, ease of use, and the role of integrated information systems in decision support. The outcomes also validate developmental research as an appropriate approach for solving practical business problems through the creation and evaluation of a technological artifact.

For student developers and future researchers, the project highlights the value of grounding technical design in local business context. The platform was not built as a generic software exercise. Instead, it was shaped by actual observation, interview findings, user readiness data, and comparative analysis of existing solutions. This context-sensitive approach explains much of the system's acceptance and Page 22 provides a model for future studies that seek to develop practical tools for other sectors.

5.9 Limitations of the Results

Although the results are positive, they should be interpreted within the limits of the study. The platform was evaluated in relation to a single reference business, which means that the observed outcomes cannot automatically be generalized to all auto parts retailers or all SMEs. The number of direct acceptance-test participants was also limited, reflecting the practical scale of the project. In addition, the local deployment model is appropriate for the target setting but does not yet address broader concerns such as cloud synchronization, branch-level coordination, or advanced analytics.

These limitations do not invalidate the results. Rather, they define the boundaries within which the findings should be understood. The project succeeds as a focused solution for a clearly identified problem context. Its next stage of value lies in replication, extension, and comparison across additional settings.

5.10 Chapter Summary

Chapter 5 has shown that ProfitLens produced favorable results in terms of functionality, usability, performance, and practical relevance. Acceptance testing confirmed that the major system workflows worked

correctly. Quality metrics showed that the platform met or exceeded its targets. Users accepted the interface and recognized clear value in the dashboard, faster transaction recording, and more transparent expense monitoring. Discussion of the findings further showed that the system responded directly to the research objectives and to the original operational problems of the reference business. Overall, the results support the conclusion that ProfitLens is a viable and meaningful web based platform for smart profit monitoring and business insights in the context of small auto parts sales.